

Security Guide

AOS Security 7.3

August 27, 2026

Contents

Audience & Purpose.....	6
Nutanix Security Infrastructure.....	7
Security Configuration Management Automation Implementation.....	7
RHEL 8 STIG Implementation in Nutanix Controller VM.....	7
Generating a STIG Compliance Report.....	8
Viewing Security Updates.....	8
Security Management Using Prism Central.....	9
Identity and Access Management.....	9
Identity and Access Management Features.....	9
Identity and Access Management Requirements.....	10
Identity and Access Management Considerations.....	10
Identity Provider Configuration in Prism Central.....	11
Configuring Active Directory Authentication in Prism Central.....	13
Configuring OpenLDAP Authentication in Prism Central.....	16
Configuring a SAML-Based Identity Provider in Prism Central.....	18
Downloading the Prism Central Metadata File.....	20
Common Access Card Authentication with Client Chain Certificate in Prism Central.....	20
Configuring Common Access Card Authentication in Prism Central.....	21
Enabling Certificate Revocation Checking using Online Certificate Status Protocol.....	22
Enabling Certificate Revocation Checking using Certificate Revocation List.....	22
Updating ADFS When Using SAML Authentication.....	23
Local User Account in Prism Central.....	24
Creating a Local User Account in Prism Central.....	24
Editing a Local User Account in Prism Central.....	26
Changing a Local User Account Password in Prism Central.....	26
Disabling Login Access to a Local User Account in Prism Central.....	27
Enabling Login Access to a Local User Account in Prism Central.....	27
Resetting a Local User Account Password Using nCLI.....	28
Service Accounts.....	29
Configuring a Service Account in Prism Central.....	29
Managing Service Account Keys.....	30
Roles Management.....	31
Built-in Roles List.....	32
Creating a Custom Role.....	34
Creating a Custom Role from an Existing Role.....	36
Adding an Authorization Policy to a Role.....	38
Duplicating a Built-in Role.....	38
Duplicating a Custom Role.....	39
Deleting a Role.....	39
Updating a Custom Role.....	40
Viewing Role Permissions.....	40
Cluster or Category Independent Entities.....	40
Changes to Registered Operations and Custom Role Migration in Prism Central pc.7.3.....	42
Authorization Policies.....	53
Creating an Authorization Policy for Full Access.....	53

Creating an Authorization Policy for Configurable Access.....	54
Editing an Authorization Policy.....	56
Duplicating an Authorization Policy.....	56
Deleting an Authorization Policy.....	56
Deleting a User or User Group.....	57
Configuring Your Profile.....	58
Updating Your Profile.....	58
Changing Your Password.....	58
Centralized Password Management Using System Accounts.....	59
Changing System Account Passwords.....	60
SSL Certificate Management in Prism Central.....	60
Generating a Self-Signed SSL Certificate with Subject Alternative Name.....	61
Generating a Certificate Signing Request with Subject Alternative Name for Submission to Certificate Authority.....	63
Verifying the Certificate Generation Request.....	66
Troubleshooting the Certificate Generation Request.....	66
Importing a Self-Signed SSL Certificate in Prism Central.....	67
Importing a CA-Signed SSL Certificate in Prism Central.....	69
Regenerating a Self-Signed Certificate in Prism Central.....	71
Supported Key Configurations for SSL Certificates.....	71
Cluster Lockdown in Prism Central.....	72
Configuring Cluster Lockdown in Prism Central.....	72
Deleting a User SSH Key in Prism Central.....	73
Security Policies using Nutanix Flow.....	73
Data-in-Transit Encryption.....	74
Enabling Data-in-Transit Encryption.....	74
Disabling Data-in-Transit Encryption.....	74
Securing AHV VMs with a Virtual Trusted Platform Module.....	75
Virtual Trusted Platform Module Requirements.....	75
Virtual Trusted Platform Module Limitations.....	75
Configuring a VM with a Virtual Trusted Platform Module.....	76
Enabling a Virtual Trusted Platform Module on an Existing VM.....	77
Disabling a Virtual Trusted Platform Module.....	77
Security Dashboard.....	78
Security Dashboard Requirements.....	79
Security Dashboard Widget in Prism Central Dashboard.....	79
Security Dashboard Wizard.....	80
Accessing the Security Dashboard.....	80
Security Dashboard Details Page.....	81
Manually Refresh the Security Dashboard.....	84
Manually Upgrade the Security Dashboard Using LCM.....	84
Enabling Security Dashboard on Dark Sites.....	85
Portal Proxy Connection in Prism Central.....	86
Configuring the Portal Proxy Connection in Prism Central.....	86
Updating the Portal Proxy Connection in Prism Central.....	87
Disabling the Portal Proxy Connection in Prism Central.....	87

Security Management Using Prism Element..... 88

Authentication in Prism Element.....	88
Configuring Active Directory Authentication in Prism Element.....	88
Configuring OpenLDAP Authentication in Prism Element.....	91
Common Access Card Authentication with Client Chain Certificate in Prism Element.....	93
Configuring Common Access Card Authentication in Prism Element.....	94
Authentication Best Practices.....	95
Emergency Local Account Usage.....	95

Modifying the Default CVM Password.....	95
Setting Admin Session Timeout.....	96
Role Mapping.....	97
Configuring a Role Mapping.....	97
Editing a Role Mapping.....	98
Deleting a Role Mapping.....	99
Local User Account in Prism Element.....	99
Creating a Local User Account in Prism Element.....	99
Backup Admin Role Capabilities.....	101
Editing a Local User Account in Prism Element.....	101
Deleting a Local User Account in Prism Element.....	101
Disabling Login Access to a Local User Account in Prism Element.....	102
Resetting a Local User Account Password Using nCLI.....	102
Configuring Your Profile.....	103
Updating Your Profile.....	103
Changing Your Password.....	104
Exporting an SSL Certificate for Third-Party Backup Applications.....	104
SSL Certificate Management in Prism Element.....	105
Generating a Self-signed SSL Certificate with Subject Alternative Name.....	106
Generating a Certificate Signing Request with Subject Alternative Name for Submission to Certificate Authority.....	108
Verifying the Certificate Generation Request.....	111
Troubleshooting the Certificate Generation Request.....	111
Importing a Self-Signed SSL Certificate in Prism Element.....	112
Importing CA-Signed SSL Certificate in Prism Element.....	114
Regenerating Self-Signed Certificate in Prism Element.....	116
Supported Key Configurations for SSL Certificates.....	116
Cluster Lockdown in Prism Element.....	117
Configuring Cluster Lockdown in Prism Element.....	117
Deleting a User SSH Key in Prism Element.....	118
Data-at-Rest Encryption.....	118
Data-at-Rest Encryption (SEDs).....	120
Data-at-Rest Encryption (Software Only).....	129
Switching from SED-EKM to Software-LKM.....	144
Configuring Dual Encryption.....	144
Backing up Keys.....	145
Taking a Consolidated Backup of Keys (Prism Central).....	145
Importing Keys.....	146
Data-at-Rest Encryption with Linux Unified Key Setup.....	147
Internationalization (i18n).....	147
Log Forwarding.....	148
Documenting the Log Fingerprint.....	148
Admin Account Password Retry Lockout.....	149
Portal Proxy Connection in Prism Element.....	149
Configuring the Portal Proxy Connection in Prism Element.....	149
Updating the Portal Proxy Connection in Prism Element.....	150
Disabling the Portal Proxy Connection in Prism Element.....	150
Hardening Instructions Using nCLI.....	151
AHV Security Hardening.....	151
CVM Security Hardening.....	153
PCVM Security Hardening.....	156
IP Set Based Firewall.....	158

Eliminate the Default Passwords During a Cluster Creation.....	159
Removing the Default SSH Password of a CVM.....	159
Enabling SSH Access to a CVM Using a Public Key.....	159
Enabling Cluster Lockdown Mode.....	160
Firewall Best Practices.....	161
Securing Traffic Through Network Segmentation.....	162
Traffic Types In a Segmented Network.....	163
Segmented and Unsegmented Networks.....	163
Supported Environment.....	169
RDMA Requirements for Network Segmentation.....	170
Prerequisites.....	170
Limitations.....	170
Cluster Services That Support Traffic Isolation.....	171
Unsupported Configurations for Network Segmentation.....	171
Troubleshooting Tips.....	171
Prepare the Network on an AHV Host.....	171
Configuring the Network on Existing Nodes.....	172
Configuring the Network on New Nodes.....	172
Network Segmentation for Traffic Types- Backplane and Management.....	173
Isolate the Backplane Traffic Logically Using VLAN-Based Segmentation.....	174
Isolating the Backplane Traffic Logically on an Existing Cluster.....	174
Physically Isolate the Backplane Traffic on an Existing Cluster.....	175
Enabling Physical Backplane Segmentation on Hyper-V Using CLI.....	181
Enabling Backplane Network Segmentation on a Mixed Hypervisor Cluster.....	182
Backplane IP Pool.....	183
Configuring Backplane IP Pool.....	183
Reconfiguring the Backplane Network.....	184
Update Backplane Port Groups.....	185
Disabling Network Segmentation on an ESXi and Hyper-V Clusters.....	186
Disabling Network Segmentation on an AHV Cluster.....	187
Service-Specific Traffic Isolation.....	189
Isolating Service-Specific Traffic.....	189
Modifying Network Segmentation Configured for a Service.....	190
Disabling Network Segmentation Configured for a Service.....	191
Deleting a vNIC Configured for a Service.....	191
Service-Specific Settings and Configurations.....	192
Remote Direct Memory Access over Converged Ethernet.....	196
Priority-Based Flow Control without Zero-Touch RoCE Specifications.....	198
NIC Compatibility Matrix for RDMA Features.....	198
Isolating the Backplane Traffic on an Existing RDMA Cluster.....	200
Support for iSCSI Extensions for RDMA.....	201
Customizing IP Addresses for each CVM and Host.....	208
Network Segmentation during Cluster Expansion.....	209
Network Segmentation-Related Changes During an AOS Upgrade.....	210
Accessing a List of Open Source Software Running on a Cluster.....	211
Copyright.....	212

AUDIENCE & PURPOSE

Understand the intended audience and scope of the Nutanix Security Guide.

This document is written for security-minded people who are responsible for architecting, managing, and supporting infrastructures, especially those who want to address security without adding more human resources or additional processes to their datacenters.

The Nutanix Security Guide offers an overview of the security development life cycle (SecDL) and host of security features that Nutanix supports. It also demonstrates how Nutanix complies with security regulations to streamline infrastructure security management. In addition, the guide covers the technical requirements that are site-specific or compliance standards that users must adhere to that are not enabled by default.

From AOS 6.8 version onward, the underlying operating system upgrades to RHEL 8, which does not support TCP Wrapper. Therefore, AOS version 6.8 or later versions do not include TCP Wrapper. For any assistance in configuring an alternative to TCP Wrapper in AOS 6.8 or later version, contact Nutanix Support.

Note: The hardening of the guest OS or any applications running on top of the Nutanix infrastructure is beyond the scope of this guide. Nutanix recommends that you refer to the documentation of the products that you deployed in your Nutanix environment.

NUTANIX SECURITY INFRASTRUCTURE

Nutanix implements a holistic approach to security with a secure platform, extensive automation, and a robust partner ecosystem.

Nutanix integrates security into every stage of product development through its security development life cycle (SecDL), rather than treating it as an afterthought. Nutanix makes the SecDL a foundational part of product design. The strong pervasive culture and processes built around security harden the enterprise cloud platform and eliminate zero-day vulnerabilities. The efficient one-click operations and self-healing security models enable automation to maintain security in an always-on hyperconverged solution.

Nutanix conforms to RHEL 8 Security Technical Implementation Guides (STIGs) that use machine-readable code to automate compliance against rigorous common standards, because traditional manual configuration and checks cannot keep up with the ever-growing list of security requirements. Nutanix Security Configuration Management Automation (SCMA) allows you to quickly and continually assess and remediate your platform to ensure that it meets or exceeds all regulatory requirements. Nutanix standardized the security profile of the Controller VM to a security compliance baseline that meets or exceeds the standard high-governance requirements.

In United States, vendors most commonly use the following references to build products according to the following set of technical requirements:

- The National Institute of Standards and Technology Special Publications Security and Privacy Controls for Federal Information Systems and Organizations (NIST 800.53)
- The US Department of Defense Information Systems Agency (DISA) Security Technical Implementation Guides (STIG)

Security Configuration Management Automation Implementation

The Nutanix platform and all its products leverage the Security Configuration Management Automation (SCMA) framework to continuously inspect services for deviations from established security policies.

For both Nutanix storage and AHV, Nutanix uses SCMA to check multiple security entities, including those defined in relevant STIGs (Security Technical Implementation Guides). The system automatically reports log inconsistencies and reverts them to the baseline. The SCMA has the lowest system priority within the virtual storage controller, ensuring that security checks do not interfere with platform performance.

Note: You can modify only the SCMA schedule. The system runs the AIDE schedule on a fixed weekly interval. For more information on how to change the SCMA schedule for AHV or Controller VM (CVM), see the [Security Hardening](#) on page 82 or [Hardening Instructions Using nCLI](#) on page 151.

RHEL 8 STIG Implementation in Nutanix Controller VM

Nutanix Controller VMs take advantage of SaltStack and Security Configuration Management Automation (SCMA) to self-heal any deviation from the security baseline configuration.

If any component of the system deviates from the security baseline, the component is set back to the supported security settings without any intervention. Nutanix set up the Controller VM to support STIG compliance with the RHEL 8 STIG as published by DISA. STIG rules cover various security configurations, including the boot loader, packages, file system, boot process, services, file ownership, authentication, kernel, and logging.

For example, STIG rules for authentication prohibit direct root login, lock system accounts other than root, enforce several password maintenance details, cautiously configure SSH, enable screen-locking, configure user shell defaults, and display warning banners.

For more information on Nutanix STIGs, see [Security Dashboard STIG Guidance Reference](#) and [Security Dashboard](#) on page 78.

Generating a STIG Compliance Report

You can leverage the OpenSCAP report to ensure that the Controller VM is STIG compliant.

About this task

To generate a STIG compliance report, follow these steps:

Procedure

1. SSH into any of the Controller VMs (CVMs) as a **nutanix** user:

```
$ ssh nutanix@cvm_ip_address
```

2. Create an output directory for the STIG report:

```
nutanix@cvm$ mkdir /tmp/stig_report
```

3. Generate the STIG compliance report:

```
nutanix@cvm$ sudo python3 -B /home/nutanix/config/security/stig_scripts/  
stig_scan_driver.py /home/nutanix/config/security/stig_scripts/  
U_RHEL_8_V1R11_STIG_SCAP_1-2_Benchmark.xml /tmp/stig_report/
```

The STIG compliance report is generated in the /tmp/stig_report directory.

Note: Depending on the size of your environment, the STIG report generation script runs for approximately 60 to 90 minutes to completion.

Viewing Security Updates

Nutanix Security Advisories provide detailed information on the available security fixes and updates, including the vulnerability description and affected product or version.

About this task

To view the list of security advisories, follow these steps:

Procedure

1. Log in to [Nutanix Support Portal](#).
2. From the [Entities Menu](#), select **Security > Advisories**.
The system displays the list of security advisories.

SECURITY MANAGEMENT USING PRISM CENTRAL

Prism Central provides several mechanisms and features to enforce security of your multicluster environment.

Security in Prism Central is managed through a comprehensive identity and access management (IAM) framework that spans multiple clusters. IAM supports local accounts, Active Directory, OpenLDAP, SAML identity providers, and common access card (CAC) authentication. You can configure roles, authorization policies, and service accounts with precise permissions. Additional capabilities include session timeouts, certificate-based authentication, and automatic role migration during upgrades, ensuring consistent and controlled access across the environment.

Identity and Access Management

Manage fine-grained access control in Prism Central using identity and access management (IAM).

IAM in Prism Central provides authentication and authorization to control access at a granular level. IAM includes the following components:

- **Identity:** A user or user group that requires access. You can configure authentication for local users, directory services and a wide selection of identity providers including SAML-based identity providers. For more information, see [Identity Provider Configuration in Prism Central](#) on page 11.
- **Entity:** A resource such as a virtual machine or disk that the identity interacts with.
- **Operation:** An action performed on an entity, such as creating a VM.
- **Role:** A set of permissions that define what operations an identity can perform. For more information, see [Roles Management](#) on page 31.
- **Authorization Policy:** A rule that enforces granular role-based access control (RBAC) by binding identities, roles, and entities. For more information, see [Authorization Policies](#) on page 53.

Prism Central enables IAM by default. When you enable Microservices Infrastructure either manually or by default, Prism Central automatically enables IAM. For more information, see [Microservices Infrastructure](#).

Identity and Access Management Features

Understand identity and access management (IAM) features in Prism Central:

- **Highly Scalable Architecture:** Based on Kubernetes open source platform, IAM uses independent pods for authentication, authorization, and data storage or replication, and supports rolling upgrades to ensure zero downtime.
- **Secure by Design:** IAM uses mutual TLS (mTLS) to secure communication between components. The Microservices Infrastructure in Prism Central provisions the certificates.

- **Support for Multiple SAML Identity Providers (IDPs):** IAM supports SAML-based authentication with the following IDPs:
 - Microsoft Active Directory Federation Services (ADFS)
 - Microsoft Entra ID (formerly Azure Active Directory)

Note: IAM supports only SAML-based authentication with Microsoft Entra ID. IAM does not support LDAP-based authentication with Microsoft Entra ID.

- Okta
- PingOne
- Shibboleth
- Keycloak

Users can log in only from the Prism Central web console. IAM does not support IDP-initiated authentication workflows from an identity provider (IDP) web page or portal.

Nutanix supports SAML 2.0 compliant IDPs.

Identity and Access Management Requirements

Understand software and infrastructure requirements of identity and access management (IAM) feature:

- For IAM specific minimum software support and requirements, see the [Prism Central release notes](#).
- For Prism Central, Prism Element Clusters, and Microservices infrastructure requirements, see the [Microservices Infrastructure](#) in *Prism Central Infrastructure Guide*.

Identity and Access Management Considerations

Understand the considerations of the identity and access management (IAM) feature:

- After you upgrade to pc.2022.9 or later, log in to Prism Central for the first time using your Nutanix local account default admin credentials. You can use Active Directory (AD) account for subsequent logins.
- IAM automatically migrates existing authentication and authorization settings, including Common Access Card (CAC) configurations.
- IAM does not support signed single logout (SLO), leading to an error on the SAML Microsoft Active Directory Federation Services (ADFS) page when you logout of Prism Central.
- If you previously enabled Microservices Infrastructure and IAM, both the services remain enabled after upgrading Prism Central, provided the cluster meets all requirements. Contact Nutanix Support for custom configurations.

Note: After upgrade to pc.2022.9 if the Security Assertion Markup Language (SAML) IDP is configured, you need to download the Prism Central metadata and re-configure the SAML IDP to recognize Prism Central as the service provider. See [Updating ADFS When Using SAML Authentication](#) on page 23 to create the required rules for ADFS.

- Maximum user session lifetime is eight hours with a 15-minute idle timeout. After 15 minutes of idle timeout, IAM logs out the user and requires re-authentication.
- IAM supports client authentication only when CAC authentication is enabled. Ensure port 9441 is open in the firewall for CAC client authentication.
- During CAC authentication, certificate checks can fail if certificates are invalid or the Online Certificate Status Protocol (OCSP) server is unreachable. As a result, the browser can show a generic error before identity and access management (IAM) processes the request.

- IAM is available only on on-premise Prism Central deployments running on AOS clusters with AHV or ESXi. Other hypervisors are not supported.
- You cannot delete local or directory users from the Prism Central web console. However, you can disable and re-enable local or directory users.

Note: Use nCLI to delete Prism Central users or user groups. For more information, see [Deleting a User or User Group](#) on page 57.

- In directory services such as Active Directory or OpenLDAP, IAM group-based authentication and authorization depend on the group's full distinguished name (DN). Changing the DN by renaming or moving the group causes the system to treat it as a different group, which can result in role mapping and access issues.
- When you change the scope in an authorization policy to include or exclude individual entities, it takes approximately five minutes for the filtered items to update and reflect on the entity page. During this time, the entities might display incorrectly.
- After you upgrade to pc.2024.1 or later, IAM migrates built-in roles user admin and viewer to new authorization policies with similar permissions. If users were assigned these roles, IAM automatically migrates the role mapping configuration as new authorization policies.

Note: IAM automatically migrates the existing role mapping data into equivalent authorization policies, that are renamed using the format `Role_Mapping_<Role_Name>_<timestamp>`. These migrated policies are available on the [Authorization Policies](#) page in Prism Central.

- The **Role Details** page view does not display the associated authorization policies and users.
- In an authorization policy configuration, if you select **All Entity Types** for the entity filter, scope of the authorization policy is limited to specific clusters or categories. The authorization policy does not grant access to Prism Central entities that do not have cluster-scoped attributes. To include such entities, explicitly allow access for those entities in the authorization policy configuration.

For more information on list of entities that do not support cluster or category-based scoping, see [Cluster or Category Independent Entities](#) on page 40.

- Starting from pc.7.3, IAM operations include several changes. For more information, see [Changes to Registered Operations and Custom Role Migration in Prism Central pc.7.3](#) on page 42.

Identity Provider Configuration in Prism Central

Prism Central supports the following user authentication methods:

- Active Directory
- OpenLDAP
- SAML identity provider
- Client Access Card (CAC)

Your Prism Central login experience depends on the configured identity provider. For example, if you configure both local user account and AD authentication, the Prism Central login page defaults to AD authentication.

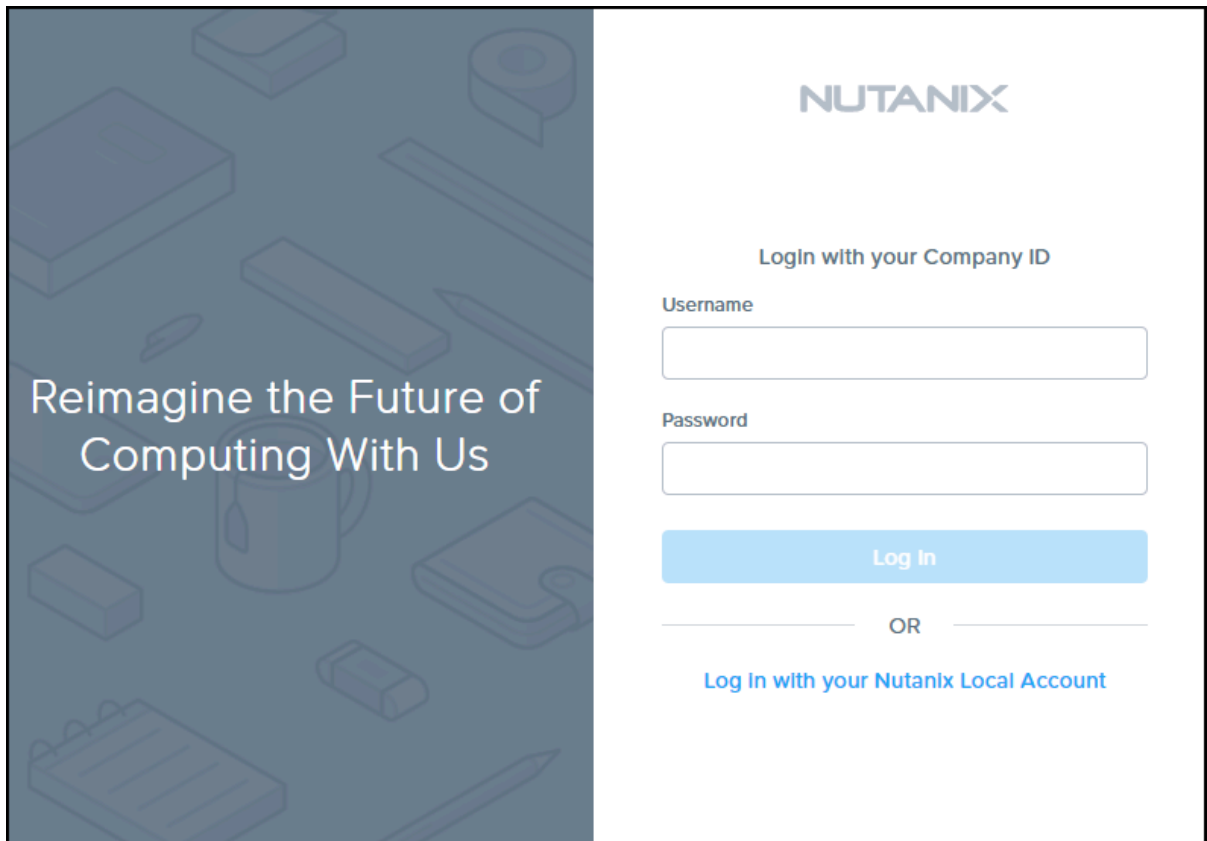


Figure 1: Sample Prism Central Login Page with Active Directory And Local User Authentication

For example, if you configure multiple identity providers such as SAML authentication (Shibboleth), AD (AD2), and local user account, the Prism Central login page displays login fields for each.

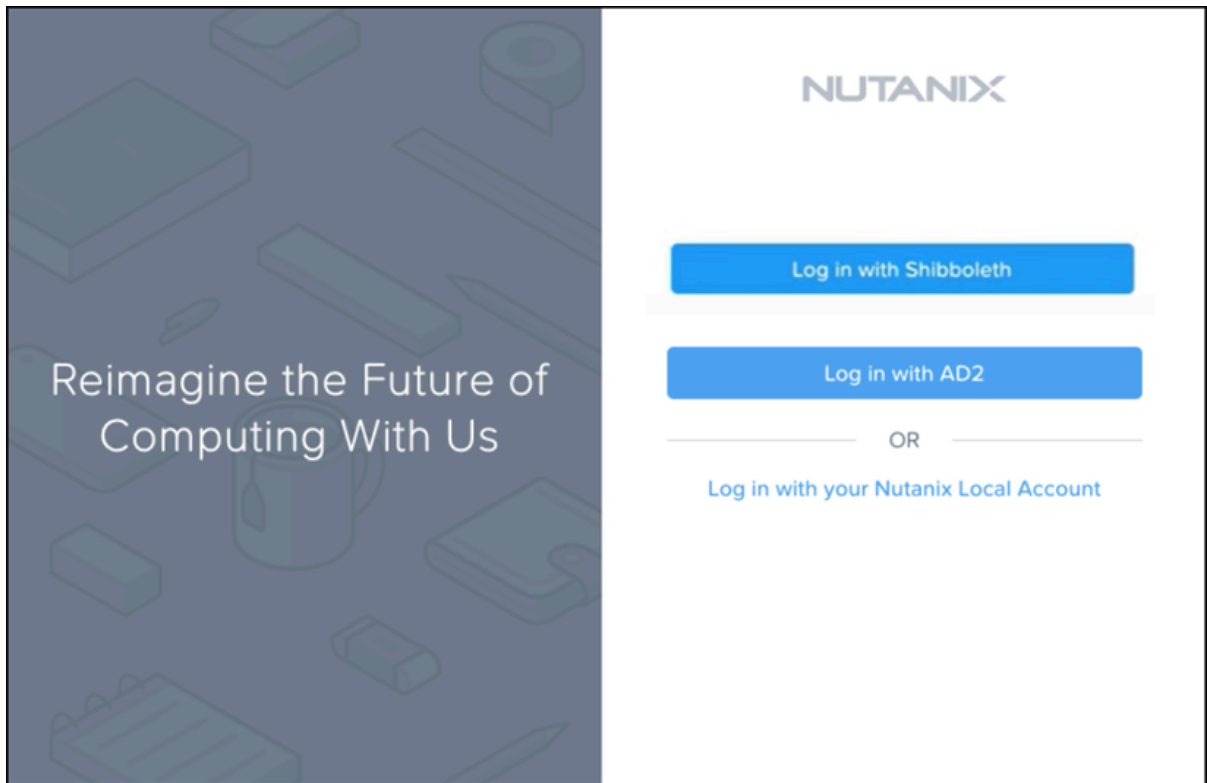


Figure 2: Sample Prism Central Login Page with SAML Identity Provider, Active Directory , and Local User Authentication

Configuring Active Directory Authentication in Prism Central

Configure Active Directory (AD) as the identity provider in Prism Central to enable AD credential-based login for users.

Before you begin

Before you configure AD in Prism Central, consider the following points:

- The service account for AD must have full read permissions on the directory service.
- Users with the User must change password at next logon attribute enabled cannot authenticate to Prism Central. Ensure that users change their password on a domain workstation before logging into Prism Central.
- If SSL is enabled on the AD server, ensure that Prism Central's firewall allows access to the appropriate port.
- Use of the Protected Users group is not supported for Prism Central authentication. For more information, see *Guidance about how to configure protected accounts* on Microsoft documentation website.
- Prism Central supports AD with LDAP v2 on Windows Server 2012 R2, Windows Server 2016, and Windows Server 2019.

Important: Prism Central does not support insecure SSLv2 and SSLv3 ciphers. To prevent the SSL fallback situation and being denied access to Prism Central, configure the following settings in your browser:

- Disable SSLv2 and SSLv3
- Enable TLS

About this task

To configure AD in Prism Central, follow these steps:

Procedure

1. Log in to Prism Central as an administrator.
2. From the [Application Switcher Function](#), select the **Admin Center** application and from the navigation bar, select **IAM**.
3. Select the **IdP Configuration** tab.
4. Click **Add Identity Provider > Active Directory/OpenLDAP**.

The **Configure Directory** window appears.

5. In the **Configure Directory** window, enter the following information:

- **Directory Type:** From the dropdown list, select **Active Directory**.
- **Name:** Enter a name to identify the directory service on Prism Central.
- **Domain:** Enter the domain or sub-domain of the directory service.

For example nutanix.com

- **Directory URL:** Enter the URL address of the directory in `ldap://host:ldap_port_num` format.

host: The host value is either the IP address or fully qualified domain name.

ldap_port_num: The default LDAP port number is 389. Nutanix also supports ports 636 for LDAPS, 3268 and 3269 for LDAP/S global catalog servers.

Examples:

- Default LDAP (Port 389), when the configuration is single domain, single forest, and not using SSL:

`ldap://ldap.example.com` or `ldap://10.1.4.111:389`

- LDAPS (Port 636), when the configuration is single domain, single forest, and using SSL:

`ldaps://ldap.example.com` or `ldaps://10.1.4.111:636`

LDAPS requires that all AD Domain Controllers have properly installed SSL certificates.

- LDAP/S Global Catalog non-SSL (Port 3268), when the configuration is multiple domain, single forest, and not using SSL:

`ldap://globalcatalog.example.com:3268` or `ldap://10.1.4.111:3268`

- LDAP/S Global Catalog SSL/TLS encrypted (Port 3269), when the configuration is multiple domain, single forest, and using SSL:

`ldaps://globalcatalog.example.com:3269` or `ldaps://10.1.4.111:3269`

When the directory URL is set to the domain FQDN, LDAP authentication can fail if any directory server becomes unavailable. For high availability (HA), configure an LDAP HA cluster and specify the cluster virtual IP (VIP) address in the directory URL instead of the domain FQDN. For more information, see [KB-12568](#).

When constructing your LDAP/S URL to use a global catalog server, ensure that the domain control IP address or name belongs to a global catalog server within the domain that you are configuring. Otherwise, queries over port 3268 or 3269 might fail.

Cross-forest trust between multiple AD forests is not supported.

LDAPS support does not require custom certificates or certificate trust import.

For the complete list of required ports, see [Port Reference](#).

- **Search Type:** From the dropdown list, select either **Non Recursive (Default)** or **Recursive**.

When users are present in the first level group, select **Non Recursive (Default)**.

When you have multi level group assignments, select **Recursive**. A recursive search performs a multi level or nested search during authentication, which might cause slowness. If you experience slowness, select **Non-Recursive (Default)**.

- **Service Account Username:** Enter the service account user name in `user_name@domain.com` format.

Note: A service account is created to run only a particular service or application with the credentials specified for the account. According to the requirement of the service or application, the administrator can limit access

to the service account. The service account is under the Managed Service Accounts in the AD server. An application or service uses the service account to interact with the operating system.

- **Service Account Password:** Enter the service account password.

Note: Be sure to update the service account credentials if the password changes or you use a different service account. User authentication and authorization requests fail if the service account password is updated in Active Directory but not in Prism Central.

6. Click **Save**.

What to do next

By default, permissions are not granted to the AD users. To grant permissions, you must create an authorization policy. For more information, see [Authorization Policies](#) on page 53.

Configuring OpenLDAP Authentication in Prism Central

Configure OpenLDAP as the identity provider in Prism Central to enable OpenLDAP credential-based login for users.

Before you begin

Before you configure OpenLDAP in Prism Central, consider the following points:

- The service account for OpenLDAP must have full read permission on the directory service.
- Prism Central uses a service account to query OpenLDAP directories for user information and does not currently support certificate-based authentication with the OpenLDAP directory.
- The values for the **User Object Class**, **User Search Base**, **Username Attribute**, **Group Object Class**, **Group Search Base**, **Group Member Attribute**, and **Group Member Attribute** fields depend on your OpenLDAP configuration.
- Prism Central supports OpenLDAP with LDAP v2.

Important: Prism Central does not support insecure SSLv2 and SSLv3 ciphers. To prevent the SSL fallback situation and being denied access to Prism Central, configure the following settings in your browser:

- Disable SSLv2 and SSLv3
- Enable TLS

About this task

To configure OpenLDAP in Prism Central, follow these steps:

Procedure

1. Log in to Prism Central as an administrator.
2. From the [Application Switcher Function](#), select the **Admin Center** application and from the navigation bar, select **IAM**.
3. Select the **IdP Configuration** tab.
4. Click **Add Identity Provider > Active Directory/OpenLDAP**.

The **Configure Directory** window appears.

5. In the **Configure Directory** window, enter the following information:

- **Directory Type:** From the dropdown list, select **OpenLDAP**.
- **Name:** Enter a name to identify the directory service on Prism Central.
- **Domain:** Enter the domain or sub-domain of the directory service.
For example nutanix.com
- **Directory URL:** Enter the URL address of the directory in ldap://host:ldap_port_num format.
host: The host value is either the IP address or fully qualified domain name.
ldap_port_num: The default LDAP port number is 389. Nutanix also supports ports 636 for LDAPS.

Examples:

- Default LDAP Port (389), when the configuration is single domain, single forest, and not using SSL:
`ldap://ldap.example.com or ldap://10.1.4.111:389`
- LDAPS (Port 636), when the configuration is single domain, single forest, and using SSL:
`ldaps://ldap.example.com or ldaps://10.1.4.111:636`

When the directory URL is set to the domain FQDN, LDAP authentication can fail if any directory server becomes unavailable. For high availability (HA), configure an LDAP HA cluster and specify the cluster virtual IP (VIP) address in the directory URL instead of the domain FQDN. For more information, see [KB-12568](#).

LDAPS support does not require custom certificates or certificate trust import.

For the complete list of required ports, see [Port Reference](#).

- **Search Type:** From the dropdown list, select either **Non Recursive(Default)** or **Recursive**.

When users are present in the first level group, you can select **Non Recursive(Default)**.

When you have multi level group assignments, you can select **Recursive**. A recursive search performs a multi level or nested search during authentication, which might cause slowness. If you experience slowness, select **Non-Recursive (Default)**.

- **User Object Class:** Enter the object class of users in the directory service.
For example: user, person, inetOrgPerson, organizationalPerson, or posixAccount
- **User Search Base:** Enter the base Distinguished Name (DN) to search for users.
The base DN must include the domain components, each represented in the format of *dc=domain_component*, separated by commas.

For example, if your domain name is nutanix.com, then base DN includes:

`ou=users,dc=nutanix,dc=com`

`cn=users,dc=nutanix,dc=com`

- **Username Attribute:** Enter the attribute of the user object class which uniquely identifies a user in the directory.
For example: uid
- **Group Object Class:** Enter the object class of groups in the directory service.
For example: posixGroup or groupOfNames
- **Group Search Base:** Enter the base DN to search for user groups.

The base DN must include the domain components, each represented in the format of *dc=domain_component*, separated by commas.

For example, if your domain name is nutanix.com, then base DN includes:

cn=groups,dc=nutanix,dc=com

ou=groups,dc=nutanix,dc=com

- **Group Member Attribute:** Enter the attribute of the group object that holds the user association.

For example: member or memberUid

- **Group Member Attribute Value:** Enter the attribute of the user object that is used to configure the membership in the group object.

For example: uid

- **Service Account Username:** Enter the service account user name in cn=username, dc=example, dc=com format.

For example cn=username, dc=nutanix, dc=com

Note: A service account is created to run only a particular service or application with the credentials specified for the account. According to the requirement of the service or application, the administrator can limit access to the service account. The service account is under the Managed Service Accounts in the OpenLDAP server. An application or service uses the service account to interact with the operating system.

- **Service Account Password:** Enter the service account password.

Note: Be sure to update the service account credentials if the service account password changes or a different service account is used. User authentication and authorization requests fail if the service account password is updated in OpenLDAP but not in Prism Central.

6. Click **Save**.

What to do next

By default, permissions are not granted to the OpenLDAP users. To grant permissions, you must create an authorization policy. For more information, see [Authorization Policies](#) on page 53.

Configuring a SAML-Based Identity Provider in Prism Central

Configure a SAML-based identity provider in Prism Central.

Before you begin

Before you configure a SAML-based identity provider in Prism Central, consider the following points:

- You must download the identity provider's XML metadata file from their website and upload it to Prism Central when configuring SAML authentication.
- An identity provider (typically a server or other computer) is the system that provides authentication through a SAML request. There are various implementations that can provide authentication services in line with the SAML standard.
- You can specify other tested standard-compliant IDPs in addition to Microsoft Entra ID (formerly Azure Active Directory). For more information, see the [Prism Central Release Notes](#).
- Prism Central supports SAML 2.0-compliant identity providers for authentication. To enforce additional security, you can configure multi-factor authentication (MFA) within your identity provider. When you configure MFA, the identity provider issues a SAML token after authentication, which provides single sign-on (SSO) access to Prism Central.

- You must configure your identity provider to return the **NameID** attribute in a SAML response. Prism Central uses the **NameID** attribute for role mapping.
- Prism Central only supports redirect binding for identity provider sign on URLs. You must configure and include the redirect binding in identity provider's metadata.

About this task

To configure a SAML-based identity provider in Prism Central, follow these steps:

Procedure

1. Log in to Prism Central as an administrator.
2. From the [Application Switcher Function](#), select the **Admin Center** application and from the navigation bar, select **IAM**.
3. Select the **IdP Configuration** tab.
4. Click **Add Identity Provider > SAML Identity Provider**.
5. In the **Configure Identity Provider** window, enter the following information:

- **Configuration Name:** Enter a name for the identity provider.
- **Username Attribute:** Enter a username attribute of the identity provider configuration.
- **Email Attribute:** Enter an email attribute of the identity provider configuration.
- **Redirect URL (Optional):** Enter the post-login redirection URL. This value is required for SAML integration in the following cases:
 - The Prism Central instance is deployed behind a reverse proxy.
 - The hostname of the Prism Central instance changes, such as during a transition from an IP address to a fully qualified domain name (FQDN).

Note: Note: After configuring or updating the redirect URL, you must download the updated metadata file from the Prism Central instance and import it into your identity provider to re-establish trust. For more information on downloading metadata, see [Downloading the Prism Central Metadata File](#) on page 20.

- **Group Attribute Name (Optional):** Enter the group attribute name.

Note: Ensure that the name matches the group attribute name provided in the IDP configuration.

- **Group Attribute Delimiter (Optional):** Enter a delimiter that must be used when multiple groups are selected for the group attribute.
- **Import Metadata:** Upload the metadata file that contains the identity provider information.

Note: For information about configuring Entra ID integration from both Prism Central and Entra ID Administration, see [KB-16526](#).

6. Click **Save**.

What to do next

- You must download the Prism Central metadata file and configure your SAML-based identity provider to recognize Prism Central as the service provider. For more information, see [Downloading the Prism Central Metadata File](#) on page 20.
- By default, permissions are not granted to the SAML IDP users. To grant permissions, you must create an authorization policy. For more information, see [Authorization Policies](#) on page 53.

Downloading the Prism Central Metadata File

Download the Prism Central metadata file to configure the callback URL in the SAML-based identity provider's website.

About this task

To download the Prism Central metadata file, follow these steps:

Procedure

1. Log in to Prism Central as an administrator.
2. From the [Application Switcher Function](#), select the **Admin Center** application and from the navigation bar, select **IAM**.
3. Select the **IdP Configuration** tab.
4. Select the identity provider checkbox from the list.
5. Click **Actions** and select **Download Metadata**.
A metadata file that describes the Prism Central attributes is downloaded in XML format.
6. On the identity provider's website, upload the metadata file.
7. (Optional) If your identity provider asks you to manually enter the Prism Central callback URL, you can find it in the `AssertionConsumerService` tag with attribute `Location` in the downloaded XML metadata file.

The following sample XML metadata file shows `AssertionConsumerService` tag with attribute `Location`:

```
<md:AssertionConsumerService Binding="urn:oasis:names:tc:SAML:2.0:bindings:HTTP-POST"
  Location="https://10.46.153.235:9440/api/iam/authn/callback"
  index="1" />
```

Common Access Card Authentication with Client Chain Certificate in Prism Central

Prism Central uses client chain certificate authentication to enable the Common Access Card (CAC) login. Client chain certificate authentication is a certificate-based authentication mechanism where the user is prompted to select a certificate for login instead of password.

Client chain certificate authentication provides enhanced security by requiring users to present a digital certificate instead of a password, reducing the risk of password-based attacks. In a one-way authentication process, Prism Central presents a certificate, and the user's browser verifies it. When the client chain certificate authentication is enabled, the process becomes a two-way authentication. During authentication, Prism Central also verifies the user's identity with a valid digital certificate to ensure that only authorized users can access the cluster. A user must provide a valid certificate when accessing Prism Central.

The CAC authentication has the following workflow:

- **Certificate Installation:** A user installs the certificate on their local machine.

Note: The Certificate Authority (CA) must be the same for both the client chain certificate and the certificate on the local machine.

- **Certificate Presentation and Chain Verification:** When the user accesses Prism Central, the system prompts them to select their CAC certificate. Prism Central verifies the presented certificate against its trusted CA chain. The chain establishes trust by linking the client certificate to a trusted root CA. This chain verification step confirms the certificate's validity.
- **EDIPI Extraction and Account Verification:** Prism Central extracts the Electronically Data Interchange Personal Identifier (EDIPI) from the validated CAC certificate. It then queries Active Directory (AD) to verify that the user has a valid and active account associated with this EDIPI.
- **Login:** Prism Central logs in the user based on their validated certificate and verified Active Directory account.

Before you configure the CAC, consider the following points:

- Prism Central supports AD as directory service for CAC. OpenLDAP is not supported.
- When you authenticate Prism Central with client chain certificate, the **Subject name** field must be present. The subject name should match the UserPrincipalName (UPN) in the AD. The UPN is a username with domain address. For example user1@nutanix.com.
- If you map a Prism role to a CAC user and not to an AD group or organizational unit to which the user belongs, specify the EDIPI UPN or UPN of that user in the role mapping. A user who presents a CAC with a valid certificate is mapped to a role and taken directly to the web console home page. The web console login page is not displayed.
- By default, users with an admin role can log in to the Prism Central console using CAC authentication. For other LDAP user roles, ensure the correct role mappings or role assignments are configured to enable CAC authentication.
- Client chain certificate must be PEM encoded.
- After logging in to Prism Central using CAC authentication, log out by closing your browser. For added security, consider clearing your browser's cookies.
- Nutanix recommends configuring a certificate revocation checking (OCSP or CRL), which IAMv2 uses to perform a secondary check, verifying certificates are not marked invalid during certificate validation.
- During CAC authentication, certificate checks can fail if certificates are invalid or the Online Certificate Status Protocol (OCSP) server is unreachable. As a result, the browser can show a generic error before identity and access management (IAM) processes the request.

Configuring Common Access Card Authentication in Prism Central

Configure Common Access Card (CAC) authentication in Prism Central.

Before you begin

Ensure that port 9441 is open in your firewall. After enabling the CAC authentication, the CAC login redirects the browser to use port 9441.

About this task

To configure the CAC authentication in Prism Central, follow these steps:

Procedure

1. Log in to Prism Central as an administrator.

2. From the [Application Switcher Function](#), select the **Admin Center** application and from the navigation bar, select **IAM**.
3. Select the **IdP Configuration** tab.
4. Click **Add Identity Provider > Common Access Card**.
The **Configure Client Chain Certificate** window appears.
5. Click **Upload** and select a client chain certificate from your local machine.

Note: Client chain certificate must be PEM encoded.
6. Select the **Enable CAC Authentication** checkbox.
7. From the **Select Active Directory** dropdown list, select the Active Directory.
8. Click **Save**.

Enabling Certificate Revocation Checking using Online Certificate Status Protocol

Nutanix recommends Online Certificate Status Protocol (OCSP) as the standard method for checking certificate revocation in client chain certificate authentication.

About this task

To enable OCSP for client chain certificate authentication, follow these steps:

Procedure

1. SSH into any of the Controller VMs (CVMs) as a **nutanix** user:

```
$ ssh nutanix@cvm_ip_address
```

2. Set the OCSP responder URL:

```
nutanix@cvm$ ncli authconfig set-certificate-revocation set-ocsp-responder=ocsp-url
```

ocsp-url: Specify the location of the OCSP responder.

3. Verify that OCSP checking is enabled:

```
nutanix@cvm$ ncli authconfig get-client-authentication-config
```

Output:

```
Auth Config Status: true
File Name: ca.cert.pem
OCSP Responder URI: http://ocsp-responder-url
```

Enabling Certificate Revocation Checking using Certificate Revocation List

Enable the Certificate Revocation List (CRL) certificate revocation checking method.

About this task

Note: Nutanix recommends Online Certificate Status Protocol (OSCP) as the standard method for checking certificate revocation in client authentication.

To enable certificate revocation checking using CRL for client authentication, follow these steps:

Procedure

1. SSH into any of the Controller VMs (CVMs) as a `nutanix` user:

```
$ ssh nutanix@cvm_ip_address
```

2. Specify all the CRLs required for certificate validation:

```
nutanix@cvm$ ncli authconfig set-certificate-revocation set-crl-uri=uri-1,uri-2 /  
set-crl-refresh-interval=refresh-interval-in-seconds set-crl-expiration-  
interval=expiration-interval-in-seconds
```

where:

- ***set-crl-refresh-interval***: Specify the CRL refresh interval in seconds to periodically update the CRLs. The CRL refresh interval is common for the entire list of CRL distribution points. The default value is 86,400 seconds (1 day).
- ***set-crl-expiration-interval***: Specify the duration in seconds for which the periodically updated CRLs are cached in-memory. The CRLs expire after the specified duration if a particular CRL distribution point is not reachable. The CRL expiration interval is common for the entire list of CRL distribution points. The default value is 604,800 seconds (7 days).

Note: The URIs must be percent-encoded and comma-separated.

Updating ADFS When Using SAML Authentication

With Nutanix IAM, to maintain compatibility with new and existing IDP/SAML authentication configurations, update your Active Directory Federated Services (ADFS) configuration - specifically the Prism Central Relying Party Trust settings. For these configurations, you are using SAML as the open standard for exchanging authentication and authorization data between ADFS as the identity provider (IDP) and Prism Central as the service provider. See the Microsoft Active Directory Federation Services documentation for details.

About this task

In your ADFS Server configuration, update the Prism Central Relying Party Trust settings by creating claim rules to send the selected LDAP attribute as the SAML NameID in email address format. For example, map the User Principal Name to NameID in the SAML assertion claims.

As an example, this topic uses UPN as the LDAP attribute to map. You could also map the email address attribute to NameID. See the *Microsoft Active Directory Federation Services* documentation for details about creating a claims aware Relying Party Trust and claims rules.

Procedure

1. In the Relying Party Trust for Prism Central, configure a claims issuance policy with two rules.
 - a. One rule based on the **Send LDAP Attributes as Claims** template.
 - b. One rule based on the **Transform an Incoming Claim** template
2. For the rule using the **Send LDAP Attributes as Claims** template, select the **LDAP Attribute** as **User-Principal-Name** and set **Outgoing Claim Type** to **UPN**.

For User group configuration using the **Send LDAP Attributes as Claims** template, select the **LDAP Attribute** as **Token-Groups - Unqualified-Names** and set **Outgoing Claim Type** to **Group**.

3. For the rule using the **Transform an Incoming Claim** template:
 - a. Set **Incoming claim type** to **UPN**.
 - b. Set the **Outgoing claim type** to **Name ID**.
 - c. Set the **Outgoing name ID format** to **Email**.
 - d. Select **Pass through all claim values**.

Local User Account in Prism Central

Prism Central allows you to create and manage local user accounts.

During the cluster deployment process, an **admin** user account is created by default. However, you can create additional local user accounts by granting specific permissions for accessing Prism Central.

Local user accounts provide individual user access to Prism Central. To enable an application or service to perform automated tasks in your Nutanix environment, configure a service account instead. For more information, see [Service Accounts](#) on page 29.

Local User Account Considerations

Before creating a local user account, consider the following points:

- A local user account cannot SSH into CVM and PCVM. SSH access to CVM and PCVM is limited only to the built-in **nutanix** and **admin** user accounts.
- By default, no permissions are granted to a local user account. After creating a local user account, you must specify the custom or built-in roles and permissions.
- A local user account does not have password expiration policy.
- The delete option is not available for a local user account in the Prism Central web console. However, you can delete a local user account using nuclei CLI commands. For more information, see [Deleting a User or User Group](#) on page 57.
- When you enable Prism Self Service feature, the system automatically uses Active Directory as the directory service.
- Changing the Prism Central admin password does not affect the cluster registration.

Creating a Local User Account in Prism Central

You can create a local user account in Prism Central.

About this task

To create a local user account in Prism Central, follow these steps:

Procedure

1. Log in to Prism Central as an administrator.
2. From the [Application Switcher Function](#), select the **Admin Center** application and from the navigation bar, select **IAM**.
3. Select the **Identities** tab.

4. Click the **+ Add Local User** and enter the following information:

- **First Name:** Enter a first name.
- **Last Name:** Enter a last name.
- **Email:** Enter a valid user email address in the format `user@domain.top-level-domain`.

The `user` and `domain` can contain the following characters:

- Letters (a-z, A-Z)
- Digits (0-9)
- Special characters: `.`, `_`, and `-`

The `top-level-domain` must be two to five characters long.

- **Username:** Enter a user name.

The username can contain only the following characters:

- Letters (a-z, A-Z)
- Digits (0-9)
- Special characters: `-`, `_`, `.`, and `@`
- Non-ASCII (international) characters

- **Password:** Enter a password.

Ensure that your password meets the following requirements:

- At least eight characters long
- At least one lowercase letter
- At least one uppercase letter
- At least one digit
- At least one special character (allowed special characters are: "#\$%&'()*+,-./:;<=>@[\\]^_`{|}~!\\")
- At least four characters different from the old password
- Must not be among the last five passwords
- Must not have more than two consecutive occurrences of a character
- Must contain at least one character from each of the following four character classes: uppercase letters, lowercase letters, digits, and special characters
- Must not contain the words "nutanix", "ntnx", "password", or any simple dictionary words.

- **Language:** From the dropdown list, select the language setting for the user.

For example, if you select **zh-CN**, the user interface displays in Simplified Chinese when the user logs in Prism Central.

- Turn on the **User Enabled** toggle to enable the login access to the local user account.

5. Click **Save**.

Editing a Local User Account in Prism Central

You can edit the local user account settings in Prism Central.

About this task

To edit the local user account, follow these steps:

Procedure

1. Log in to Prism Central as an administrator.
2. From the [Application Switcher Function](#), select the **Admin Center** application and from the navigation bar, select **IAM**.
3. Select the **Identities** tab.
4. Select the checkbox associated with the local user account.
5. Select **Actions > Edit**.
6. In the **Edit Local User** window, edit the fields as needed and click **Update**.

Changing a Local User Account Password in Prism Central

You can change the local user account password in Prism Central.

About this task

To change the local user account password, follow these steps:

Procedure

1. Log in to Prism Central as an administrator.
2. From the [Application Switcher Function](#), select the **Admin Center** application and from the navigation bar, select **IAM**.
3. Select the **Identities** tab.
4. Select the checkbox associated with the local user account.
5. Select **Actions > Change Password**.

6. In the **Edit Password** window, enter the following information:

- **Password:** Enter the new password.
- **Re-Enter Password:** Re-enter the new password.

Ensure that your password meets the following requirements:

- At least eight characters long
- At least one lowercase letter
- At least one uppercase letter
- At least one digit
- At least one special character (allowed special characters are: "\$%&'()*+,-./:;<=>@[\\]^_`{|}~!\\")
- At least four characters different from the old password
- Must not be among the last five passwords
- Must not have more than two consecutive occurrences of a character
- Must contain at least one character from each of the following four character classes: uppercase letters, lowercase letters, digits, and special characters
- Must not contain the words "nutanix", "ntnx", "password", or any simple dictionary words.

7. Click **Update**.

Disabling Login Access to a Local User Account in Prism Central

You can disable login access to the local user account in Prism Central.

About this task

To disable login access to the local user account, follow these steps:

Procedure

1. Log in to Prism Central as an administrator.
2. From the [Application Switcher Function](#), select the **Admin Center** application and from the navigation bar, select **IAM**.
3. Select the **Identities** tab.
4. Select the checkbox associated with the local user account.
5. Select **Actions > Disable User**.
6. Click **Save**.

Enabling Login Access to a Local User Account in Prism Central

You can enable the login access for a disabled local user account. This action automatically restores all previously assigned permissions.

About this task

To enable login access to the local user account, follow these steps:

Procedure

1. Log in to Prism Central as an administrator.
2. From the [Application Switcher Function](#), select the **Admin Center** application and from the navigation bar, select **IAM**.
3. Select the **Identities** tab.
4. Select the checkbox associated with the local user account.
5. Select **Actions** > **Enable User**.
6. Click **Save**.

Resetting a Local User Account Password Using nCLI

Reset a local user account password using the Nutanix command-line interface (nCLI).

About this task

To reset the local user account password using nCLI, follow these steps:

Procedure

1. Do one of the following:
 - » SSH into any of the Controller VMs (CVMs) as a **nutanix** user:

```
$ ssh nutanix@cvm_ip_address
```
 - » SSH into Prism Central VM (PCVM) as a **nutanix** user:

```
$ ssh nutanix@pcvm_ip_address
```

2. Reset the local user account password:

```
» nutanix@cvm$ ncli user reset-password user-name='user_name'
password='new_password'
```

```
» nutanix@pcvm$ ncli user reset-password user-name='user_name'
password='new_password'
```

- Replace `'user_name'` with the local user account name.
- Replace `'new_password'` with the new password.

Ensure that your password meets the following requirements:

- At least eight characters long
- At least one lowercase letter
- At least one uppercase letter
- At least one digit
- At least one special character (allowed special characters are: "\$%&'()*+,-./:;<=>@[^_`{|}~!\)
- At least four characters different from the old password
- Must not be among the last five passwords
- Must not have more than two consecutive occurrences of a character
- Must contain at least one character from each of the following four character classes: uppercase letters, lowercase letters, digits, and special characters
- Must not contain the words "nutanix", "ntnx", "password", or any simple dictionary words.

Service Accounts

Service accounts provide controlled programmatic access to resources and operations in your Nutanix environment.

A service account is a special type of account used by Nutanix applications or services to perform automated tasks within the system. It is identified by a unique name, and is used to authenticate the application or service that owns it. The authentication is done using a shared secret key, which is generated and managed using IAM. Once configured, the service account can perform actions that it has been explicitly authorized to do. Administrators can assign role-based access control (RBAC) to a service account, allowing controlled access to specific resources or operations.

Service accounts are of the following types.

- User-managed service accounts: Created and maintained by administrators who have permission to manage them. These are typically used when custom automation or integration is needed.
- Service-managed accounts: Created automatically by backend services during setup.

To configure accounts for individual user access to Prism Central, use a local user account instead. For more information, see [Local User Account in Prism Central](#) on page 24.

Configuring a Service Account in Prism Central

You can configure a service account in Prism Central.

About this task

To configure a service account in Prism Central, follow these steps:

Procedure

1. Log in to Prism Central as an administrator.
2. From the [Application Switcher Function](#), select the **Admin Center** application and from the navigation bar, select **IAM**.
3. Select the **Identities** tab.
4. Select **Service Accounts**.
5. Click **+ Add New Service Account** and enter the following information:
 - **Service Account Name:** Enter a name for your service account.
 - **Service Account Email (optional):** Optionally, enter an email address for your service account.
 - **Description:** Optionally, enter a description for your service account.
6. Choose one of the following actions.
 - » **Save:** To save the service account without creating keys. This completes the service account creation. To configure keys later, see [Managing Service Account Keys](#) on page 30.
 - » **Save and Create Keys:** To save the service account and create keys. Proceed to the next step.
7. Create keys for your service account.
 - a. Click **+ Add New Key**.
 - b. Select a **Key Type** from the following options.
 - **Objects Key:** S3 compatible access key and secret key pair for Objects service.
 - **API Key:** Access key and secret key pair for use with any Nutanix service.
 - c. In the **Key Name** field, enter a name for the key.
 - d. Optionally, in the **Assigned To** field, enter the email address of the user to whom you want to assign the key.
 - e. Set key expiry.
 - » In the **Key Expiry Date** and **Key Expiry Time** fields, click the calendar icon and clock icon to select the key expiry date and time respectively.
 - » Enable the **Key never expires** checkbox to set the key without an expiration date.
 - f. Click **Generate Key**.

The key is generated successfully.

What to do next

Click the copy icon to copy the generated access key and secret access key. Alternatively, click **Download Key Details** to download the key details in the CSV file format.

Important: You cannot retrieve the key details after closing the dialog box. Ensure that you copy or download the key details before exiting.

Managing Service Account Keys

Add, revoke, or delete service account keys.

About this task

To manage keys associated with a service account, do the following:

Procedure

1. Log in to Prism Central as an administrator.
2. From the [Application Switcher Function](#), select the **Admin Center** application and from the navigation bar, select **IAM**.
3. Select the **Identities** tab.
4. Select **Service Accounts**.
5. From the list of configured service accounts, select the service account for which you want to add or update the key details.
6. Click **Manage Keys**.
7. Choose any of the following key management options depending on your requirement.
 - » To add a new key, click **+ Add New Key** and enter the key details. For more information, see step 5 on page 30 in [Configuring a Service Account in Prism Central](#) on page 29.

Important: Ensure that you copy or download the key details before exiting.

- » To revoke or delete the associated access key for a service account, click **Actions** and select **Revoke Key** or **Delete Key**.

Roles Management

Prism Central supports role-based access control (RBAC), which you can configure to provide customized access permissions for users based on their assigned roles.

Prism Central provides the following role types:

- **Built-in Role:** Includes a set of predefined roles designed for common IAM use cases. Built-in roles simplify role assignment by eliminating the need for administrators to manually configure granular permissions. To view the complete list of built-in roles, see the [Built-in Roles List](#) on page 32.

For example, the **VM Admin Role** provides comprehensive permissions for managing virtual machines, including creation, modification, and deletion.

Note: You cannot update or delete the built-in roles.

- **Custom Role:** In addition to the built-in roles, you can also create custom roles in Prism Central. The custom roles provide granular control necessary for fine-tuning permissions and to meet specific organizational needs, thus maintaining security and compliance. You can define precise access levels for the users based on specific job functions or tasks.

Note the following hypervisor-specific limitations for granular RBAC:

- **AHV Environment:** Granular RBAC is fully supported for all Prism Central entities.
- **ESXi Environment:** Granular RBAC is not supported for VM entities and applies only to a limited set of other Prism Central entities.

Note: Starting with the pc.2023.4 release, Prism Central provides additional built-in roles. If you created any custom roles with the same name as the new built-in roles, the system overrides the permissions of these custom roles with the

default role permissions after you upgrade. Before upgrading to pc.2023.4 or later, Nutanix recommends renaming or deleting these custom roles to prevent any permission issues after the upgrade.

Built-in Roles List

Prism Central offers several built-in roles, each designed for common administrative tasks.

The following table lists the built-in roles available in Prism Central:

Note: The **Self-Service Admin** role is deprecated and replaced by the **Project Manager** role. Use the **Project Manager** role for all new role assignments and update any existing authorization policies that reference the **Self-Service Admin** role.

Table 1: Prism Central Built-in Roles

Role	Description
Action Service User	Basic Playbook access for all users
Backup Admin	Full access for backup operations
Category Admin	Full access for category objects
Category Viewer	View access for category objects
Cluster Admin	Full access for cluster operations.
Cluster Viewer	View access for cluster operations.
Consumer	Launch blueprints and controls their life cycle and actions
CSI System	Full access for Kubernetes cluster infrastructure resources for CSI
Developer	Author blueprints, tests deployments, and publishes applications for other project members.
Disaster Recovery Admin	Full access for disaster recovery operations.
Disaster Recovery Viewer	View access for disaster recovery operations.
Domain Manager Admin	Full access for Domain Manager (Prism Central Management) v4 APIs
Domain Manager Viewer	View access for Domain Manager (Prism Central Management) v4 APIs
File Server Security Admin	Full access for Nutanix File server security related permissions.
File Server Share Admin	Full access for Nutanix File server share operations.
Files Admin	Full access for Files Storage operations.
Files Viewer	View access for Files Storage operations.
Flow Admin	Full access for Nutanix Flow operations including categories provisioning
Flow Viewer	View access for Nutanix Flow operations.
Flow Policy Author	Full access for flow operations except categories provisioning

Role	Description
Foundation Central Admin	Full access to perform all Nutanix Foundation Central operations
Foundation Central Viewer	View access to all API keys and nodes in Nutanix Foundation Central
Kubernetes Data Services System	Full access for Kubernetes cluster infrastructure resources for Kubernetes Data Services
Kubernetes Infrastructure Provision	Full access for Kubernetes cluster infrastructure VM resources
License Admin	Full access for Licensing operations
License Viewer	View access for Licensing operations
Local Account Manager Admin	Admin access for local account password related operations.
Local Account Manager Viewer	View access for local account password related operations.
Monitoring Admin	Full access to perform all monitoring operations
Monitoring Viewer	View access to all monitoring APIs
Network Infra Admin	Manage the infrastructure and underlay networking.
Network Shared Resources Viewer	View access for shared resources in underlay and overlay networking.
Objects Admin	Full access for Nutanix Objects store operations
Objects Editor	Edit access for Nutanix Objects store operations
Objects Viewer	View access for Nutanix Objects store operations
Operations Management Admin	Full access to perform all operations management
Operations Management Viewer	View access to all API in operations management
Operator	Manage the existing application deployments and exercise blueprint actions.
Prism Admin	Manage the infrastructure and platform, but cannot entitle other users to be admins.
Prism Viewer	View access to all infrastructure and platform features, but cannot make any changes
Project Admin	Manage end users within the project and has full access to their entities
Project Manager	Manage virtual infrastructure, oversees self service, and can delegate end user management.
Secure Policy Admin	Full access for secure policies operations
Secure Policy Disaster Recovery Admin	Manage associations for secure policies
Secure Policy Editor	Update access for secure policies
Secure Policy Viewer	View access for secure policies
Security Admin	Update access for security features like KMS and encryption

Role	Description
Security Dashboard Admin	Update access for security dashboard
Security Dashboard Viewer	View access for security dashboard
Security Viewer	View access for security features like KMS and encryption
Storage Admin	View and perform actions on storage entities.
Storage Viewer	View access for storage entities.
Super Admin	Manage Nutanix deployment, set-up, configure, and make use of every feature in the platform
	Note: Highest-level admin with full infrastructure and tenant access.
Virtual Machine Admin	Full access for virtual machine operations.
Virtual Machine Operator	Access day-to-day activities on virtual machines.
Virtual Machine Viewer	View access for virtual machines.
VPC Admin	Manage VPCs and related entities. Agnostic of the physical network infrastructure.

You can perform the following actions on built-in roles:

- Add an authorization policy. For more information, see [Adding an Authorization Policy to a Role](#) on page 38.
- Duplicate an existing built-in role. For more information, see [Duplicating a Built-in Role](#) on page 38.
- View the detailed list of permissions associated with a built-in role. For more information, see [Viewing Role Permissions](#) on page 40.

Note: You cannot update or delete the built-in roles.

Creating a Custom Role

In addition to built-in roles, you can also create a custom role in Prism Central.

About this task

To create the custom role in Prism Central, follow these steps:

Procedure

1. Log in to Prism Central as an administrator.
2. From the [Application Switcher Function](#), select the **Admin Center** application and from the navigation bar, select **IAM**.
3. Select the **Roles** tab.
4. Click **Create Role > New Role**.

5. In the **Role Name** field, enter a unique name for the role.

Note: You must enter a unique name for the custom role. Built-in role names are not allowed. For more information, see [Built-in Roles List](#) on page 32.

6. (Optional) In the **Description** field, enter a description for the role.

7. Do one of the following:

- To filter all the operations available for a specific resource type, select the **Entity Type** from the dropdown list and in the search box enter the name of the resource.

For example, when you select the **Entity Type** and enter *vm-vm anti-affinity policy* in the search box, the search result displays all the operations related to vm-vm anti-affinity policy resource.

- To filter the individual operations available for a specific resource type, select the **Operation** from the dropdown list and in the search box enter the name of the resource.

For example, when you select the **Operation** and enter *anti affinity* in the search box, the search result displays the individual operations that are performed on a vm-vm anti-affinity policy resource.

8. Click the plus icon for adding an individual operation or a group of operations to the role.

9. (Optional) To view the related operations, click the



related operations icon.

10. Select the recommended related operations and click **Add**.

Note: Nutanix recommends that you select all the related operations to ensure that the role is granted with sufficient permissions within the authorization policy.

11. Do one of the following:

- » To create the role, click **Save**.
- » To create the role and attach the authorization policy, click **Save & Create Authorization Policy**.

The system redirects to the **Create New Authorization Policy** page. For more information, see [Creating an Authorization Policy for Full Access](#) on page 53.

Creating a Custom Role from an Existing Role

You can use an existing built-in or custom role as a starting point when you create a custom role.

Before you begin

Before you create the custom role from an existing role, consider the following points:

- Any system operations in the existing role are not added to the new role that you create.
- Some operations are pre-selected if you are creating a role from an existing role.

About this task

To create the custom role from an existing role, follow these steps:

Procedure

1. Log in to Prism Central as an administrator.
2. From the [Application Switcher Function](#), select the **Admin Center** application and from the navigation bar, select **IAM**.
3. Select the **Roles** tab.
4. Click **Create Role > From Existing Role**.

Note: The new role does not include system operations from the existing role.

5. In the **Search roles** box, type the built-in or custom role name and select the role from the search result. The system displays the list of operations associated with the selected role.
6. Click **Next**.
7. On the **System Operations** window, click **Continue**.
8. In the **Role Name** field, enter a unique name for the role.

Note: You must enter a unique name for the custom role. Built-in role names are not allowed. For more information, see [Built-in Roles List](#) on page 32.

9. (Optional) In the **Description** field, enter a description for the role.

10. Do one of the following:

- To filter all the operations available for a specific resource type, select the **Entity Type** from the dropdown list and in the search box enter the name of the resource.

For example, when you select the **Entity Type** and enter *vm-vm anti-affinity policy* in the search box, the search result displays all the operations related to vm-vm anti-affinity policy resource.

- To filter the individual operations available for a specific resource type, select the **Operation** from the dropdown list and in the search box enter the name of the resource.

For example, when you select the **Operation** and enter *anti affinity* in the search box, the search result displays the individual operations that are performed on a vm-vm anti-affinity policy resource.

11. Click the plus icon to add an individual operation or a group of operations to the role.

12. (Optional) To view the related operations, click the



related operations icon.

13. Select the recommended related operations and click **Add**.

Note: Nutanix recommends that you select all the related operations to ensure that the role is granted with sufficient permissions within the authorization policy.

14. Do one of the following:

- » To create the role, click **Save**.
- » To create the role and attach the authorization policy, click **Save & Create Authorization Policy**.

The system redirects to the **Create New Authorization Policy** page. For more information, see [Creating an Authorization Policy for Full Access](#) on page 53.

Adding an Authorization Policy to a Role

The **Roles** tab provides an option to associate the authorization policy to both built-in and custom roles.

About this task

To add an authorization policy to a role, follow these steps:

Procedure

1. Log in to Prism Central as an administrator.
2. From the [Application Switcher Function](#), select the **Admin Center** application and from the navigation bar, select **IAM**.
3. Select the **Roles** tab.
4. Select the checkbox associated with the role.
5. Select **Actions > Add Authorization Policy**.
6. Configure the authorization policy fields as needed.

For more information on how to configure an authorization policy, see [Creating an Authorization Policy for Full Access](#) on page 53.

Duplicating a Built-in Role

You can duplicate a built-in role.

Before you begin

When you duplicate a built-in role, Prism Central displays a message indicating that it cannot duplicate internal operations. Prism Central excludes the internal operations from duplication to prevent misconfiguration or unintended behavior because these internal operations are tightly coupled with system-level logic and are subject to change. As a result, the duplicated role might include fewer operations and not provide access to specific features. Nutanix recommends that you verify the duplicated role's permissions before assigning it to users.

About this task

To duplicate a built-in role, follow these steps:

Procedure

1. Log in to Prism Central as an administrator.
2. From the [Application Switcher Function](#), select the **Admin Center** application and from the navigation bar, select **IAM**.
3. Select the **Roles** tab.
4. Select the checkbox associated with the built-in role.

5. Select **Actions > Duplicate**.
6. Click **Continue**.
7. Edit the fields in the **Duplicate Role** window.
8. Do one of the following:
 - » To save the role, click **Save**.
 - » To save the role and attach the authorization policy, click **Save & Create Authorization Policy**.

The system redirects to the **Create New Authorization Policy** page. For more information, see [Creating an Authorization Policy for Full Access](#) on page 53.

Duplicating a Custom Role

You can duplicate a custom role.

About this task

To duplicate a custom role, follow these steps:

Procedure

1. Log in to Prism Central as an administrator.
2. From the [Application Switcher Function](#), select the **Admin Center** application and from the navigation bar, select **IAM**.
3. Select the **Roles** tab.
4. Select the checkbox associated with the custom role.
5. Select **Actions > Duplicate**.
6. Edit the fields in the **Duplicate Role** window.
7. Do one of the following:
 - » To save the role, click **Save**.
 - » To save the role and attach the authorization policy, click **Save & Create Authorization Policy**.

The system redirects to the **Create New Authorization Policy** page. For more information, see [Creating an Authorization Policy for Full Access](#) on page 53.

Deleting a Role

You can delete a custom role in Prism Central.

About this task

To delete the custom role, follow these steps:

Procedure

1. Log in to Prism Central as an administrator.
2. From the [Application Switcher Function](#), select the **Admin Center** application and from the navigation bar, select **IAM**.

3. Select the **Roles** tab.
4. Select the checkbox associated with the role.
5. Select **Actions > Delete**.
6. On the **Delete Role** window, click **Delete**.

Updating a Custom Role

You can update a custom role in Prism Central.

About this task

To update the custom role, follow these steps:

Procedure

1. Log in to Prism Central as an administrator.
2. From the [Application Switcher Function](#), select the **Admin Center** application and from the navigation bar, select **IAM**.
3. Select the **Roles** tab.
4. Select the checkbox associated with the role.
5. Select **Actions > Update**.
6. Update the fields in the **Update Role** window as needed and click **Update**.

Viewing Role Permissions

You can view the permissions associated with both built-in and custom roles.

About this task

To view the permissions associated with a role, follow these steps:

Procedure

1. Log in to Prism Central as an administrator.
2. From the [Application Switcher Function](#), select the **Admin Center** application and from the navigation bar, select **IAM**.
3. Select the **Roles** tab.
4. Select the checkbox associated with the role.
The system displays all the permissions associated with the role.

Cluster or Category Independent Entities

The following list identifies the entities within Prism Central that do not support cluster or category-based scoping:

- License Configuration
- Cost Governance Registration
- Platform Dependent Field

- Prism Central
- XFit Policy
- Cluster Metrics
- Authorization Policy
- Directory Server Config
- Security Monitoring
- Network Security Rule
- Sync Policy
- Category Mapping
- Network Function Chain
- Identity Categorization Config
- Flow Upgrade
- OIDC Client
- Stats
- Simulation
- Secure Snapshot Approval Policy
- Stig Report
- Summary
- Stig Control
- Vulnerability
- Hardening Visibility Setting
- NC2 AWS VPC
- Network Controller Configuration
- NC2 AWS Subnet List Capability
- NC2 AWS Subnet
- Certificate Authentication Provider
- Role
- Entity
- Config Changeset
- Operation
- Client
- Service Account
- Alerts Policy
- Audit

- System Defined Alert Policy
- Event
- User Defined Alert Policy
- Alert Email Config
- Alert

Important: To configure authorization policy for these entities, you must explicitly allow access in the authorization policy configuration. For more information, see [Entities limitation in Authorization Policy](#).

Changes to Registered Operations and Custom Role Migration in Prism Central pc.7.3

Prism Central version pc.7.3 includes changes to several IAM operations.

The IAM operations are renamed, split into multiple operations, or consolidated to improve RBAC precision. For more information, see [Operations Changes in Prism Central pc.7.3](#) on page 43.

These changes to operations might affect any custom roles that reference older versions of these operations. To ensure custom roles remain valid after the upgrade, IAM automatically updates roles where a direct mapping is possible. However, you must manually migrate any custom roles that cannot be updated automatically. For more information, see [Migrating Custom Roles after Prism Central version pc.7.3 Upgrade](#) on page 42.

Migrating Custom Roles after Prism Central version pc.7.3 Upgrade

After upgrading to Prism Central version pc.7.3, some custom roles must be manually migrated to align with the changed operations in pc.7.3.

Before you begin

Ensure that you have one of the following built-in roles assigned to perform custom role migration.

- Super Admin
- Prism Admin
- Project Manager

About this task

To migrate custom roles containing outdated operations, do the following:

Procedure

1. Log in to Prism Central as an administrator.
2. From the [Application Switcher Function](#), select the **Admin Center** application and from the navigation bar, select **IAM**.
3. Select the **Roles** tab.
If there are roles that reference outdated operations, a banner containing the following message appears at the top of the page: **X custom roles in your system contain outdated operations and might not function as expected. To ensure full functionality, migrate your roles now.**
4. In the banner, click the **migrate your roles now** link.

What to do next

After migration, validate the role permissions to ensure they meet your access requirements.

Operations Changes in Prism Central pc.7.3

Updated IAM operations in Prism Central version pc.7.3.

The following table lists the updated IAM operations in Prism Central pc.7.3, along with their corresponding renamed, consolidated, or expanded forms.

Table 2: Updated IAM Operations in Prism Central pc.7.3

Operation Category	Existing Operation	Operation Renamed/ Consolidated	Operations Expanded
IAM	View User Key	View Service Account Api Key View User Buckets Access Key	
IAM	Delete User Key	Delete User Buckets Access Key Delete Service Account Api Key	
IAM	View Service Account Access Key	View Nutanix Access Key	
IAM	Delete User Buckets Access Key	Delete Buckets Access Key	
IAM	Create User Buckets Access Key	Add Buckets Access Key	
IAM	Create User Group	Add User Group	
IAM	Delete User Group	Remove User Group	
IAM	Update Operation	Update Permission	
IAM	Revoke Service Account Api Key	Revoke Nutanix Api Key	
IAM	Update Authorization Policy	Update ACP	
IAM	Delete Authorization Policy	Delete ACP	
IAM	Revoke User Key	Revoke User Buckets Access Key Revoke Service Account Api Key	
IAM	Change User State	Enable User	
IAM	Delete Saml Identity Provider	Delete SAML IdP Delete Identity Provider	
IAM	View Saml Identity Provider	View Identity Provider View SAML IdP	
IAM	View Operation	View Permission	
IAM	Create Saml Identity Provider	Create Identity Provider Create SAML IdP	
IAM	Create Service Account	Add Service Account	
IAM	Create Directory Service	Configure Directory Service	

Operation Category	Existing Operation	Operation Renamed/ Consolidated	Operations Expanded
IAM	View Client	View Service Client	
IAM	Create Service Account Api Key	Add Nutanix Api Key	
IAM	Create User	Add User	
IAM	Create Service Account Access Key	Add Nutanix Access Key	
IAM	View Service Account Api Key	View Nutanix Api Key	
IAM	Delete Directory Service	Delete Directory Service Configuration	
IAM	Create Authorization Policy	Create ACP	
IAM	Delete Service Account Access Key	Delete Nutanix Access Key	
IAM	Delete Service Account Api Key	Delete Nutanix Api Key	
IAM	Update Saml Identity Provider	Update Identity Provider Update SAML IdP	
IAM	Delete Operation	Delete Permission	
IAM	Reset User Password	Reset Password	
IAM	Create Operation	Create Permission	
IAM	View User Buckets Access Key	View Buckets Access Key	
IAM	Delete User	Remove User	
IAM	View Authorization Policy	View ACP	
IAM	Create User Key	Create Service Account Api Key Create User Buckets Access Key	
Files	View File Server Protected Pairs	View File Server Data Protected Pairs	
Cluster Management	View Cluster Credentials	View Config Credentials	
Cluster Management	View Storage Container Stats	View Container Stats View Container Stats	
Cluster Management	Create Storage Container	Create Container Create Container	
Cluster Management	Delete Storage Container	Delete Container Delete Container	
Cluster Management	View Storage Container	View Container View Container	

Operation Category	Existing Operation	Operation Renamed/ Consolidated	Operations Expanded
Cluster Management	Unmount Storage Container Datastore	Unmount Container Datastore Unmount Container Datastore	
Cluster Management	Mount Storage Container Datastore	Mount Container Datastore Mount Container Datastore	
Cluster Management	Update Storage Container	Update Container Update Container	
Cluster Management	View Storage Container Datastore	View Container Datastore View Container Datastore	
Volumes	View Volume Group iSCSI Attachments Internal		View Volume Group
Volumes	View Volume Group Category Associations		View Volume Group
Volumes	Update Volume Group Disk		Update Volume Group Virtual Disks Update Volume Group
Volumes	Update Volume Group Disk Internal		Update Volume Group Virtual Disks Update Volume Group
Volumes	Update Volume Group Details Internal	Update Volume Group Details	Update Volume Group
Volumes	Detach Volume Group From External iSCSI Client Internal		Update Connections with External iSCSI Clients Update Volume Group
Volumes	Attach Volume Group To External iSCSI Client		Update Connections with External iSCSI Clients Update Volume Group
Volumes	View Volume Group Details		View Volume Group
Volumes	Update External iSCSI Client Internal	Update External iSCSI Client	
Volumes	Detach Volume Group From External iSCSI Client		Update Connections with External iSCSI Clients Update Volume Group
Volumes	Create Volume Group Disk		Update Volume Group Virtual Disks Update Volume Group
Volumes	View Volume Group Metadata		View Volume Group
Volumes	View Volume Group Disk Stats		View Volume Group

Operation Category	Existing Operation	Operation Renamed/ Consolidated	Operations Expanded
Volumes	View Volume Group Stats		View Volume Group
Volumes	Associate Volume Group Categories		Update Volume Group Categories Update Volume Group
Volumes	View Volume Group Disks		View Volume Group
Volumes	View Volume Group VM Attachments		View Volume Group
Volumes	Attach Volume Group To AHV VM		Update Volume Group Update Connections with Direct-attach AHV VMs
Volumes	Disassociate Volume Group Categories		Update Volume Group Categories Update Volume Group
Volumes	Detach Volume Group From AHV VM		Update Connections with Direct-attach AHV VMs Update Volume Group
Volumes	Delete Volume Group Disk		Update Volume Group Virtual Disks Update Volume Group
Volumes	View Volume Group Metadata Internal		View Volume Group
Volumes	View Volume Group iSCSI Attachments		View Volume Group
Volumes	Detach Volume Group From AHV VM Internal		Update Connections with Direct-attach AHV VMs Update Volume Group
Networking	Delete Subnet	Delete Overlay Subnet Delete Overlay External Subnet Delete External Subnet Delete Overlay Subnet Delete Overlay External Subnet Delete External Subnet	
Networking	Unreserve Subnet Ip	Unreserve Vlan Subnet Ip Unreserve Overlay Subnet Ip Unreserve Vlan Subnet Ip Unreserve Overlay Subnet Ip	
Networking	View VPC	View Virtual Network View Transit VPC View Transit VPC	

Operation Category	Existing Operation	Operation Renamed/ Consolidated	Operations Expanded
Networking	View Subnet	View Overlay Subnet View Overlay External Subnet View External Subnet View Overlay Subnet View Overlay External Subnet View External Subnet	
Networking	Migrate vNIC	Migrate Overlay vNIC Migrate Overlay vNIC	
Networking	Update Network Gateway	Update Vpn Gateway	
Networking	View Network Gateway	View Vpn Gateway	
Networking	Update VPC	Update Transit VPC Update Virtual Network Update Transit VPC	
Networking	Delete VPC	Delete Virtual Network Delete Transit VPC Delete Transit VPC	
Networking	Update Subnet	Update Overlay Subnet Update Overlay External Subnet Update External Subnet Update Overlay Subnet Update Overlay External Subnet Update External Subnet	
Networking	Create Network Gateway	Create Vpn Gateway	
Networking	Create VPC	Create Virtual Network Create Transit VPC Create Transit VPC	
Networking	Reserve Subnet Ip	Reserve Overlay Subnet Ip Reserve Vlan Subnet Ip Reserve Overlay Subnet Ip Reserve Vlan Subnet Ip	
Networking	Delete Network Gateway	Delete Vpn Gateway	
Networking	Create Subnet	Create Overlay External Subnet Create External Subnet Create Overlay Subnet Create Overlay External Subnet Create External Subnet Create Overlay Subnet	
Microsegmentation	View Directory Server Config	View Directory Server	
Microsegmentation	Update Directory Server Config	Update Directory Server	

Operation Category	Existing Operation	Operation Renamed/ Consolidated	Operations Expanded
Microsegmentation	Policy Preview Flow Upgrade	Preview Migration Flow Migrator	
Microsegmentation	Upgrade Config Flow Upgrade	Migrate Config Flow Migrator	
Microsegmentation	Summarise Flow Upgrade	Summarise Migration Flow Migrator	
Microsegmentation	Delete Directory Server Config	Delete Directory Server	
Microsegmentation	Create Directory Server Config	Create Directory Server	
Prism	View Category	View Name Category View Value Category	
Prism	Delete Category	Delete Name Category Delete Value Category	
Prism	View Domain Manager	View Prism Central	
Virtual Machine Management	Power On Virtual Machine		Update Virtual Machine Power State Update Virtual Machine Allow Virtual Machine Power On
Virtual Machine Management	Create Virtual Machine NIC		Update Virtual Machine NIC List Update Virtual Machine
Virtual Machine Management	Delete Virtual Machine CD ROM		Update Virtual Machine Update Virtual Machine Disk List
Virtual Machine Management	View Virtual Machine Disk		View Virtual Machine
Virtual Machine Management	ACPI Shutdown Virtual Machine		Allow Virtual Machine Power Off Update Virtual Machine Power State Update Virtual Machine
Virtual Machine Management	Power Off Virtual Machine		Update Virtual Machine Allow Virtual Machine Power Off Update Virtual Machine Power State
Virtual Machine Management	Migrate Virtual Machine NIC		Update Virtual Machine NIC List Update Virtual Machine
Virtual Machine Management	Create Virtual Machine Disk		Update Virtual Machine Update Virtual Machine Disk List

Operation Category	Existing Operation	Operation Renamed/ Consolidated	Operations Expanded
Virtual Machine Management	Guest Reboot Virtual Machine		Allow Virtual Machine Reboot Update Virtual Machine Power State Update Virtual Machine
Virtual Machine Management	Associate Virtual Machine Categories		Update Virtual Machine Categories Update Virtual Machine
Virtual Machine Management	Update Virtual Machine NIC		Update Virtual Machine NIC List Update Virtual Machine
Virtual Machine Management	Create Virtual Machine GPU		Update Virtual Machine GPU List Update Virtual Machine
Virtual Machine Management	Create Virtual Machine Serial Port		Update Virtual Machine
Virtual Machine Management	Power Off ESXi Virtual Machine		Update ESX Virtual Machine
Virtual Machine Management	Insert Virtual Machine CD ROM		Mount Virtual Machine CDROM Update Virtual Machine Disk List Update Virtual Machine
Virtual Machine Management	View Existing Virtual Machine		View Virtual Machine
Virtual Machine Management	View Virtual Machine Disk Stats		View Virtual Machine
Virtual Machine Management	Migrate Virtual Machine To Host		Update Virtual Machine
Virtual Machine Management	Migrate Virtual Machine Disk		Initiate Virtual Machine Disk Migration Update Virtual Machine Disk List Update Virtual Machine
Virtual Machine Management	Create Virtual Machine CD ROM		Update Virtual Machine Disk List Update Virtual Machine
Virtual Machine Management	View Virtual Machine NIC		View Virtual Machine
Virtual Machine Management	Power Cycle Virtual Machine		Allow Virtual Machine Reboot Update Virtual Machine Power State Update Virtual Machine
Virtual Machine Management	Eject Virtual Machine CD ROM		Update Virtual Machine Unmount Virtual Machine CDROM Update Virtual Machine Disk List

Operation Category	Existing Operation	Operation Renamed/ Consolidated	Operations Expanded
Virtual Machine Management	Update Virtual Machine Disk		Update Virtual Machine Disk List Update Virtual Machine
Virtual Machine Management	Disassociate Virtual Machine Categories		Update Virtual Machine Categories Update Virtual Machine
Virtual Machine Management	View VM Host Affinity Policy VM Compliances		View VM Host Affinity Policy
Virtual Machine Management	Insert Virtual Machine NGT ISO		Update Virtual Machine NGT Config Update Virtual Machine
Virtual Machine Management	View Virtual Machine CD ROM		View Virtual Machine
Virtual Machine Management	Reset ESXi Virtual Machine		Update ESX Virtual Machine
Virtual Machine Management	Create New Virtual Machine		Create Virtual Machine
Virtual Machine Management	Guest Shutdown Virtual Machine		Allow Virtual Machine Power Off Update Virtual Machine Power State Update Virtual Machine
Virtual Machine Management	Delete Existing Virtual Machine		Delete Virtual Machine
Virtual Machine Management	View Virtual Machine NIC Stats		View Virtual Machine
Virtual Machine Management	Reset Virtual Machine		Allow Virtual Machine Reset Allow Virtual Machine Reboot Update Virtual Machine Power State Update Virtual Machine
Virtual Machine Management	Guest Shutdown ESXi Virtual Machine		Update ESX Virtual Machine
Virtual Machine Management	Clone Existing Virtual Machine		Clone Virtual Machine
Virtual Machine Management	View Virtual Machine Stats		View Virtual Machine
Virtual Machine Management	Delete Virtual Machine NIC		Update Virtual Machine NIC List Update Virtual Machine
Virtual Machine Management	Guest Reboot ESXi Virtual Machine		Update ESX Virtual Machine
Virtual Machine Management	Delete Virtual Machine GPU		Update Virtual Machine Update Virtual Machine GPU List

Operation Category	Existing Operation	Operation Renamed/ Consolidated	Operations Expanded
Virtual Machine Management	Assign Virtual Machine Owner		Update Virtual Machine Update Virtual Machine Owner
Virtual Machine Management	Update Virtual Machine NGT		Update Virtual Machine NGT Config Update Virtual Machine
Virtual Machine Management	Delete Virtual Machine Disk		Update Virtual Machine Update Virtual Machine Disk List
Virtual Machine Management	ACPI Reboot Virtual Machine		Allow Virtual Machine Reboot Update Virtual Machine Power State Update Virtual Machine
Virtual Machine Management	Suspend ESXi Virtual Machine		Update ESX Virtual Machine
Virtual Machine Management	View Virtual Machine Serial Port		View Virtual Machine
Virtual Machine Management	Delete Virtual Machine Serial Port		Update Virtual Machine
Virtual Machine Management	Power On ESXi Virtual Machine		Update ESX Virtual Machine
Virtual Machine Management	Update Virtual Machine Serial Port		Update Virtual Machine
Virtual Machine Management	Assign Virtual Machine NIC IP		Update Virtual Machine NIC List Update Virtual Machine
Virtual Machine Management	View Virtual Machine GPU		View Virtual Machine
Virtual Machine Management	Install Virtual Machine NGT		Update Virtual Machine NGT Config Update Virtual Machine
Virtual Machine Management	Upgrade Virtual Machine NGT		Update Virtual Machine NGT Config Update Virtual Machine
Virtual Machine Management	Customise Virtual Machine Guest		Update Virtual Machine
Virtual Machine Management	Reenforce VM Host Affinity Policy		Update VM Host Affinity Policy
Virtual Machine Management	Update Virtual Machine Basic Config		Update Virtual Machine
Virtual Machine Management	View Virtual Machine NGT		View Virtual Machine

Operation Category	Existing Operation	Operation Renamed/ Consolidated	Operations Expanded
Virtual Machine Management	Uninstall Virtual Machine NGT		Update Virtual Machine NGT Config Update Virtual Machine
Virtual Machine Management	View VM Host Affinity Policy VM Compliance States		View VM Host Affinity Policy View VM Host Affinity Policy VM Compliances
Virtual Machine Management	Release Virtual Machine NIC IP		Update Virtual Machine NIC List Update Virtual Machine
Licensing	View Licenses	View License Metadata	
Licensing	View License Entitlements	View License Entitlement	
Operations Management	Create Report	Create Report Instance	
Operations Management	Upload Report Artifact		Create Global Report Setting
Operations Management	Delete Report	Delete Report Instance	
Operations Management	View Report Artifact File		View Global Report Setting
Operations Management	Update Global Report Setting	Update Common Report Config	
Operations Management	View Report	View Report Instance	
Operations Management	Delete Global Report Setting	Delete Common Report Config	
Operations Management	Notify Recipients Report	Notify Report Instance	
Operations Management	Download Report		View Report
Operations Management	Create Global Report Setting	Create Common Report Config	
Operations Management	View Global Report Setting	View Common Report Config	
Monitoring	View System Defined Alert Policy Cluster Configs		View Alerts Policy
Monitoring	Delete User Defined Alert Policy		Delete Alerts Policy
Monitoring	Create User Defined Alert Policy		Create Alerts Policy
Monitoring	View User Defined Alert Policy		View Alerts Policy
Monitoring	Update User Defined Alert Policy		Update Alerts Policy

Operation Category	Existing Operation	Operation Renamed/ Consolidated	Operations Expanded
Monitoring	Update System Defined Alert Policy Cluster Configs		Update Alerts Policy
Monitoring	View System Defined Alert Policy		View Alerts Policy
Objects	View Object Store Certificate	Download Certificate Object Store	
Objects	Create Object Store Certificate	Replace Certificates Object Store	
AIOps	View Scenario	View Whatif	
AIOps	View Stats Sources		View Stats
AIOps	View Stats Entities		View Stats
AIOps	Share Scenario	Share Whatif	
AIOps	Update Scenario	Update Whatif	
AIOps	Update Ignore Window	Update Blackout	
AIOps	Calculate Ignore Runway	Calculate Runway	
AIOps	View Stats Entity Types		View Stats
AIOps	View Stats Entity Descriptors		View Stats
AIOps	Delete Scenario	Delete Whatif	
AIOps	View Ignore Window	View Blackout	
AIOps	Create Ignore Window	Create Blackout	
AIOps	Delete Ignore Window	Delete Blackout	
AIOps	Create Scenario	Create Whatif	

Authorization Policies

Authorization policy is an identity and access management (IAM) mechanism to define access to Nutanix resources using Prism Central.

You create an authorization policy to assign a role (built-in or custom) to an identity (user or user group) for a global or customized scope of allowed actions. Authorization policy ensures that the identity accessing a resource for a specific operation has the appropriate permissions.

After you configure user authentication, by default no permissions are granted to the users or user groups. You must explicitly assign the permissions to users by creating an authorization policy.

Note: The System Admin, Prism Admin, and Prism Viewer roles include default authorization policies. You cannot modify or delete these policies.

Creating an Authorization Policy for Full Access

Create an authorization policy with full access to all entity types and instances for the added users in the associated role.

About this task

To create an authorization policy, follow these steps:

Procedure

1. Log in to Prism Central as an administrator.
2. From the [Application Switcher Function](#), select the **Admin Center** application and from the navigation bar, select **IAM**.
3. Select the **Authorization Policies** tab.
4. Click **+ Create Authorization Policy**.
5. (Optional) To edit the name, click the pencil icon next to the default name, edit the name, and click the checkmark icon.
6. On the **Select Role** search box, enter the built-in or custom role's name and select the role from the list of suggestions.
The system displays role details for the selected role.
7. Click **Next**.
8. Select **Full Access: all entity type & instances**.

Note:

After updating to pc.2024.1, when you create an Authorization Policy for the Prism Admin role, the LCM page sometimes stall with Waiting for LCM framework to start message. The workaround is to select the scope as **Full access: all entity types & instances#Automatically grant access to new entity types that are added to this role in the future**.

9. (Optional): To automatically gain access to any new entity types that are added to your selected role in the future, select the **Automatically grant access to new entity types that are added to this role in the future** checkbox.
10. Click **Next**.
11. From the **Users** dropdown list, select the user type.
12. On the **Search** box, enter the first few letters of the user or user group's name and select the correct user or user group from the list of suggestions.
13. Click **Save**.

Creating an Authorization Policy for Configurable Access

Create an authorization policy with configurable access to the selected entity types and instances for the added users in the associated role.

About this task

To create an authorization policy, follow these steps:

Procedure

1. Log in to Prism Central as an administrator.

2. From the [Application Switcher Function](#), select the **Admin Center** application and from the navigation bar, select **IAM**.
3. Select the **Authorization Policies** tab.
4. Click **+ Create Authorization Policy**.
5. (Optional) To edit the name, click the pencil icon next to the default name, edit the name, and click the checkmark icon.
6. On the **Select Role** search box, enter the built-in or custom role's name and select the role from the list of suggestions.
The system displays role details for the selected role.
7. Click **Next**.
8. Select **Configure access: select entity types & instances** and follow these steps:
 - a. From the **Entity Type** dropdown list, select the entity type.
The list of available entities depends on the role that you select.
 - b. From the **Filters** dropdown list, select the filter type.
The list of available filter options depends on the **Entity Type** that you select.

Starting with AOS 7.0 and pc.2024.3, the **Filters** dropdown list includes a new filter called **Advanced** for some of the entity types. The **Advanced Filter** creates a set of filters with the AND operator.

You can repeat the **Entity Type** and **Filter** selections for any combination of entity and filter. Defining multiple entity-filter-search combinations by repeating the entity type and filter selection creates a set of filters with an OR operator.
 - c. If you selected the **Advanced** filter, perform the following actions:
 - From the **Filters** dropdown list, filter the entity type.
 - On the **Search** box, select the entity.
 - Click **+ Condition**.
 - From the **Filters** dropdown list, filter the entity type.
 - On the **Search** box, select the entity.
 - Click **Add to Policy**.
 - d. On the **Search** box, enter the first few letters of the specific entities and select the correct entities from the list of suggestions.
9. (Optional) To allow the assigned users to access any new entity instances created by them, select the **Allow users access to entities created by them** checkbox.
For example, if a user has permission to create a new VM, the user must also have access to the VM.
10. Click **Next**.
11. From the **Users** dropdown list, select the user type.
12. On the **Search** box, enter the first few letters of the user or user group's name and select the correct user or user group from the list of suggestions.
13. Click **Save**.

Editing an Authorization Policy

You can edit an authorization policy.

About this task

To edit an authorization policy, follow these steps:

Procedure

1. Log in to Prism Central as an administrator.
2. From the [Application Switcher Function](#), select the **Admin Center** application and from the navigation bar, select **IAM**.
3. Select the **Authorization Policies** tab.
4. Select the checkbox associated with an authorization policy.
5. Select **Actions > Edit**.
6. In the **Edit Authorization Policy** window, edit the fields as needed and click **Save**.

Duplicating an Authorization Policy

You can create a copy of predefined or custom authorization policy.

About this task

To duplicate an authorization policy, follow these steps:

Procedure

1. Log in to Prism Central as an administrator.
2. From the [Application Switcher Function](#), select the **Admin Center** application and from the navigation bar, select **IAM**.
3. Select the **Authorization Policies** tab.
4. Select the checkbox associated with an authorization policy.
5. Select **Actions > Duplicate**.
6. In the **Duplicate Authorization Policy** window, edit the fields as needed and click **Save**.

Deleting an Authorization Policy

You can delete an authorization policy.

Before you begin

Before you delete an authorization policy, consider the following points:

- You can not delete any predefined system authorization policies.
- Any users associated with the authorization policy lose their assigned access after you delete the authorization policy.

About this task

To delete an authorization policy, follow these steps:

Procedure

1. Log in to Prism Central as an administrator.
2. From the [Application Switcher Function](#), select the **Admin Center** application and from the navigation bar, select **IAM**.
3. Select the **Authorization Policies** tab.
4. Select the checkbox associated with an authorization policy.
5. Select **Actions > Delete**.
6. Click **Delete**.

Deleting a User or User Group

Delete local user and identity provider user or users group using nuclei commands.

Before you begin

Before you delete a user or users group, consider the following points:

- Remove any authorization policies associated with a user or user group, whether directly assigned (Prism Central version pc.2024.1 or later) or through project or role mapping (Prism Central versions prior to pc.2024.1).
- Verify that the user you want to delete does not own any VMs or other entities. You must transfer or delete the entity ownership before deleting an account, see [Editing an Authorization Policy](#) on page 56.

About this task

To delete Prism Central user or users group, follow these steps:

Procedure

1. SSH into the Prism Central VM (PCVM) as a `nutanix` user:

```
$ ssh nutanix@pcvm_ip_address
```

2. List all users or user groups UUIDs:

» List all users:

```
nutanix@PCVM$ nuclei user.list
```

» List all user groups:

```
nutanix@PCVM$ nuclei user_group.list
```

If there are more than 20 users or user groups, use the count flag to specify the number to list. For example, to list 100 users or user groups:

```
nutanix@PCVM$ nuclei user.list count=100
```

```
nutanix@PCVM$ nuclei user_group.list count=100
```

3. Delete the user or user group:

- » Delete a user by UUID.

```
nutanix@PCVM$ nuclei user.delete user_UUID
```

- » Delete a user group by UUID.

```
nutanix@PCVM$ nuclei user_group.delete user_group_UUID
```

Replace *user_UUID* or *user_group_UUID* with the actual UUID of the user or users group.

Configuring Your Profile

Configure your profile information in Prism Central.

In the Prism Central web console, you can manage your profile settings by updating personal information and credentials. You can modify your first name, last name, email address, and preferred language. You can also change your password to strengthen security, keeping your account details accurate and ensuring reliable access within the environment.

Updating Your Profile

You can update your profile information in Prism Central.

About this task

To update your profile information in Prism Central, follow these steps:

Procedure

1. Log in to Prism Central as an administrator.
2. Click **User Menu > Update Profile**
3. On the **Update Profile** window, enter the following information:
 - **First Name:** Enter a first name.
 - **Last Name:** Enter a last name.
 - **Email Address:** Enter a valid user email address.
 - **Language:** From the dropdown list, select the language setting.
4. Click **Save**.

Changing Your Password

You can change your profile's password in Prism Central.

About this task

To change your profile's password in Prism Central, follow these steps:

Procedure

1. Log in to Prism Central as an administrator.
2. Click **User Menu > Change Password**

3. On the **Change Password** window, enter the following information:

- **Current Password:** Enter the current password.
- **New Password:** Enter the new password.
- **Confirm Password:** Re-enter the new password.

4. Click **Save**.

Centralized Password Management Using System Accounts

Prism Central System Accounts page provides a centralized way to view, update, and track system account passwords across Prism Central, Controller VMs, and AHV.

System Accounts Overview

You can use the System Accounts page to centrally manage all the system account passwords using Prism Central. The centralized management of passwords ensures enhanced account security by providing a direct view of the status of passwords (default or secure) and the ability to change the passwords of system accounts that are grouped under Prism Central, Controller VM, or AHV scope.

You can also use different viewing options for the system account passwords using the **Filters** option. Using filters, you can refine the view based on the following categories:

Status of the account password

- **Default:** The password is default.
- **Secure:** The password is custom and secure.

Account Type

- Root (AHV-only)
- Admin
- Nutanix

Tip: You can view the password change history using the [Audit Dashboard](#) or [Tasks](#) for traceability.

Password Management Requirements

Supported Software Versions:

- Prism Central version pc.2023.4 or later
- AOS version 6.7.1 or later

Password Management for AHV system accounts is supported for the following software versions:

- Prism Central version pc.2024.3 or later
- AOS version 7.0 or later
- AHV version 10.0 or later

Node-level password management for AHV root and admin (only if password is set) accounts is supported for the following software versions:

- Prism Central version pc.7.3 or later

- AOS version 7.3 or later
- AHV version 10.3 or later

Password Management Limitations

- The account passwords for a maximum of 10 clusters can be changed simultaneously.
- Bulk password change for multiple user accounts on different clusters is not supported.
- Password change for AHV system accounts is supported on clusters containing at least one hyperconverged infrastructure (HCI) node. However, AHV password change is not available on clusters composed entirely of compute-only (CO) nodes.
- In case of successful AHV system account password change, the status of the password on the **System Accounts** page is updated immediately.

Changing System Account Passwords

Use the System Accounts page in Prism Central to view and update passwords for system accounts across Prism Central, Controller VMs, and AHV hosts.

About this task

To change the password of one or multiple system accounts, follow these steps:

Procedure

1. Log on to Prism Central as an administrator.
2. Select the **Infrastructure** application from [Application Switcher Function](#), and go to **Network & Security > System Accounts** from the **Navigation Bar**. Alternatively, go to the [Accessing the Security Dashboard](#) on page 80 and click **View all System Account Passwords**.

The **System Accounts** page opens. This page provides the list of accounts and the password status for all the accounts configured on the registered clusters.

The system accounts are grouped by component type: Prism Central, Controller VM (CVM), and Hypervisor (AHV). AHV accounts are displayed at the host level, allowing you to view and manage passwords per host where supported.

3. To change the password of accounts on Prism Central and Controller VMs, select the account from the **Account** column and click **Change Password**.

For AHV accounts, expand the cluster row to view associated hosts, then select the specific hosts where you need to change the password.

4. In the **Change Password** window, enter the following and click **Change Password**.

- Existing account password
- New password
- Confirm new password

The passwords are updated for the selected accounts.

SSL Certificate Management in Prism Central

Manage SSL certificates in Prism Central to ensure secure access.

Prism Central supports SSL certificate-based authentication for console access. Prism Central includes a default self-signed SSL certificate to enable secure communication with a cluster. You can replace the default self-signed SSL

certificate with your own self-signed SSL certificate or a certificate authority (CA) signed SSL certificate from either an internal third-party trusted CA or a public trusted CA of your choice.

For production environments, Nutanix recommends that you replace the default self-signed certificate with a CA-signed SSL certificate.

Note: Prism Central supports only a cluster-wide SSL certificate. You cannot customize the SSL certificate for an individual Controller VM (CVM). Nutanix recommends that you check for the validity of the certificate periodically and replace the certificate if it is invalid.

Generating a Self-Signed SSL Certificate with Subject Alternative Name

Generate a self-signed SSL certificate with Subject Alternative Name (SAN) using OpenSSL to secure Prism Central.

Before you begin

- The certificate import process validates that the key and certificate pair uses the correct signature algorithm to comply with NIST SP800-131a and RFC 6460 (NSA Suite B) standards. Use the correct key types, sizes, curves, and signature algorithms. For more information, see [Supported Key Configurations for SSL Certificates](#) on page 71.
- Nutanix recommends including a DNS name for all Prism Central VMs (PCVMs) in the self-signed SSL certificate using the SAN extension to avoid SSL certificate errors when accessing a PCVM through its DNS name instead of the shared cluster IP address.

About this task

To generate a self-signed SSL certificate with SAN, follow these steps:

Procedure

1. SSH into the Prism Central VM (PCVM) as a `nutanix` user:

```
$ ssh nutanix@pcvm_ip_address
```

2. Generate a private key:

- RSA private key with a bit length of 2048:

```
nutanix@pcvm$ openssl genrsa -out my_key_name.key 2048
```

- RSA private key with a bit length of 4096:

```
nutanix@pcvm$ openssl genrsa -out my_key_name.key 4096
```

- ECDSA private key using the prime256v1 curve:

```
nutanix@pcvm$ openssl ecparam -name prime256v1 -genkey -out my_key_name.pem
```

Important: While you are generating the private key, ensure that the private key is not password protected.

3. Generate the Certificate Signing Request (CSR):

- For RSA 2048 and RSA 4096 private keys:

```
nutanix@pcvm$ openssl req -new -nodes -key my_key_name.key -signature_algorithm -  
out my_csr_name.csr
```

signature_algorithm: Specify sha256 or sha384 or sha512.

For example, to generate a CSR for RSA 2048 private key using SHA-256 signature algorithm:

```
nutanix@pcvm$ openssl req -new -nodes -key my_key_name.key -sha256 -out  
my_csr_name.csr
```

- For ECDSA 256 private key:

```
nutanix@pcvm$ openssl req -new -nodes -key my_key_name.pem -sha256 -out  
my_csr_name.csr
```

For example, to generate a CSR for ECDSA 256 private key using SHA-256 signature algorithm:

```
nutanix@pcvm$ openssl req -new -nodes -key my_key_name.pem -sha256 -out  
my_csr_name.csr
```

4. Enter the information in the command output to incorporate into your certificate request:

```
You are about to be asked to enter information that will be incorporated  
into your certificate request.  
What you are about to enter is what is called a Distinguished Name or a DN.  
There are quite a few fields but you can leave some blank  
For some fields there will be a default value,  
If you enter '.', the field will be left blank.  
-----  
Country Name (2 letter code) []:# Enter the 2 letter country code  
State or Province Name (full name) []:# Enter the full name of the state or province  
Locality Name (eg, city) []:# Enter the name of the city or locality  
Organization Name (eg, company) []: # Enter the legal name of your organization or  
company  
Organizational Unit Name (eg, section) []: # Enter the department or business unit  
Common Name (eg, fully qualified host name) []:# Enter the fully qualified domain  
name (FQDN) of the server  
Email Address []:# Enter a valid email address  
Please enter the following 'extra' attributes  
to be sent with your certificate request  
A challenge password []: # (Optional) Enter your password
```

5. Create a configuration file in your home directory with your preferred text editor named *san.cnf* that contains the following text:

```
[req]  
distinguished_name = req_distinguished_name  
req_extensions = v3_req  
  
[req_distinguished_name]  
[v3_req]  
basicConstraints = CA:FALSE  
keyUsage = nonRepudiation, digitalSignature, keyEncipherment  
subjectAltName = @alt_names  
  
[alt_names]  
DNS.0 = example1.domain.com # Primary domain or fully qualified domain name (FQDN) of  
the server  
DNS.1 = example2.domain.com # Secondary domain or alias for the server  
DNS.2 = example3.domain.com # Additional domain name that should be trusted
```

```
DNS.3 = *.domain.com # Wildcard entry to match any subdomain under domain.com
IP.0 = x.x.x.x # IP address of the Prism Central
IP.1 = y.y.y.y # Additional IP address of the Prism Central
```

[*alt_names*]

Specify your DNS and IP addresses. If you have a range of hosts, use wildcards (*) to match any subdomain of the domain name.

6. Generate a self-signed certificate:

```
nutanix@pcvm$ openssl x509 -req -days number_of_days -in my_csr_name.csr -signkey
my_key_name.key -out my_crt_name.crt -signature_algorithm -extensions v3_req -
extfile san.cnf
```

number_of_days: Specify the number of days until a newly generated certificate expires.

signature_algorithm: Specify sha256, sha384, or sha512. Ensure that you use the same signature algorithm that was used to generate the CSR.

Example:

```
nutanix@pcvm$ openssl x509 -req -days 1460 -in my_csr_name.csr -signkey
my_key_name.key -out my_crt_name.crt -sha256 -extensions v3_req -extfile san.cnf
```

7. Copy *my_key_name.key* and *my_crt_name.crt* from the PCVM to your local machine:

```
nutanix@pcvm$ scp my_key_name.key my_crt_name.crt username@local-machine:/
local_file_path/
```

What to do next

After generating the self-signed certificate with a private key, follow the procedure described in [Importing a Self-Signed SSL Certificate in Prism Central](#) on page 67 to replace the default certificate with your self-signed SSL certificate. The following table lists certificate components and its corresponding file type to choose when SSL certificate window prompts:

Table 3: SSL Certificate Import Files

Certificate Components	File type
Private Key	my_key_name.key
Public Certificate	my_crt_name.crt
CA Certificate/Chain	my_crt_name.crt

Generating a Certificate Signing Request with Subject Alternative Name for Submission to Certificate Authority

Generate a Certificate Signing Request (CSR) with Subject Alternative Name (SAN) using OpenSSL for Certificate Authority (CA) submission.

Before you begin

- The certificate import process validates that the key and certificate pair uses the correct signature algorithm to comply with NIST SP800-131a and RFC 6460 (NSA Suite B) standards. Use the correct key types, sizes, curves, and signature algorithms. For more information, see [Supported Key Configurations for SSL Certificates](#) on page 71.

- Nutanix recommends including a DNS name for all Prism Central VMs (PCVMs) in the CSR using the SAN extension to avoid SSL certificate errors when accessing a PCVM through its DNS name instead of the shared cluster IP address.

About this task

To generate a CSR with SAN, follow these steps:

Procedure

1. SSH into the Prism Central VM (PCVM) as a `nutanix` user:

```
$ ssh nutanix@pcvm_ip_address
```

2. Create a configuration file in your home directory with your preferred text editor named `ssl.cnf` that contains the following text:

```
[req]
distinguished_name = req_distinguished_name
req_extensions = v3_req
prompt = no

[req_distinguished_name]
countryName = # Country Name (2 letter code)
stateOrProvinceName = # State or Province Name (full name)
localityName = # Locality Name (eg, city)
organizationName = # Organization Name (eg, company)
organizationalUnitName = # Organizational Unit Name (eg, BU)
commonName = # Common Name (e.g. server FQDN or YOUR name)
emailAddress = # Email Address

[v3_req]
subjectAltName = @alt_names

[alt_names]
DNS.0 = example1.domain.com # Primary domain or fully qualified domain name (FQDN) of
the server
DNS.1 = example2.domain.com # Secondary domain or alias for the server
DNS.2 = example3.domain.com # Additional domain name that should be trusted
DNS.3 = *.domain.com # Wildcard entry to match any subdomain under domain.com
IP.0 = x.x.x.x # IP address of the Prism Central
IP.1 = y.y.y.y # Additional IP address of the Prism Central
```

[alt_names] - Specify your DNS and IP addresses. If you have a range of hosts, use wildcards (*) to match any subdomain of the domain name.

3. Generate a private key:

- RSA private key with a bit length of 2048:

```
nutanix@pcvm$ openssl genrsa -out my_key_name.key 2048
```

- RSA private key with a bit length of 4096:

```
nutanix@pcvm$ openssl genrsa -out my_key_name.key 4096
```

- ECDSA private key using the prime256v1 curve:

```
nutanix@pcvm$ openssl ecparam -name prime256v1 -genkey -out my_key_name.pem
```

Important: While you are generating the private key, ensure that the private key is not password protected.

4. Generate a CSR:

- For RSA 2048 and RSA 4096 private keys:

```
nutanix@pcvm$ openssl req -new -nodes -key my_key_name.key -signature_algorithm -  
out my_csr_name.csr -config ssl.cnf
```

signature_algorithm: Specify sha256 or sha384 or sha512.

For example, to generate a CSR for RSA 2048 private key using SHA-256 signature algorithm:

```
nutanix@pcvm$ openssl req -new -nodes -key my_key_name.key -sha256 -out  
my_csr_name.csr -config ssl.cnf
```

- For ECDSA 256 private key:

```
nutanix@pcvm$ openssl req -new -nodes -key my_key_name.pem -sha256 -out  
my_csr_name.csr -config ssl.cnf
```

For example, to generate a CSR for ECDSA 256 private key using SHA-256 signature algorithm:

```
nutanix@pcvm$ openssl req -new -nodes -key my_key_name.pem -sha256 -out  
my_csr_name.csr -config ssl.cnf
```

5. Copy `my_key_name.key` and `my_csr_name.csr` from the PCVM to your local machine:

```
nutanix@pcvm$ scp my_key_name.key my_csr_name.csr username@local-machine:/  
local_file_path/
```

6. Log out of the PCVM.

7. Send your CSR file to the CA of your choice.

After receiving your CSR, the CA sends the following files:

- CA signed public certificate
- CA's public certificate
- Root CA public certificate (if the CA is intermediate)

The issuing CA validates the public certificate. If the issuing CA is intermediate, the root CA validates the issuing CA certificate. The system validates the certificate chain to establish trust.

8. Download all the certificate files you receive from CA to the local file directory.

9. (Optional) If the CA chain certificate provided by the certificate authority is not in a single file, run the following command to concatenate the list of CA certificates into a chain file:

```
$ cat intermediateCAcert.crt rootCAcert.crt > ca_chain_certs.crt
```

Start the chain with the signer's certificate and end it with the root CA certificate.

Ensure that the chain file only has the root and intermediate certificates. Prism Central fails to import the chain file if it contains public or private certificates.

What to do next

Follow [Importing a CA-Signed SSL Certificate in Prism Central](#) on page 69 section to replace the default certificate with a CA-signed certificate. The following table lists certificate components and its corresponding file type to choose when SSL certificate window prompts:

Table 4: SSL Certificate Import Files

Certificate Components	File type
Private Key	my_key_name.key
Public Certificate	ca_signed_public_cert.cer
CA Certificate/Chain	ca_public_cert.crt or ca_chain_certs.crt

Verifying the Certificate Generation Request

Run the following commands to verify the certificate generation request.

- Verify that the CA certificate chain is valid:

```
nutanix@pcvm$ openssl verify -CAfile ca_chain_certs.crt myPublicCert.cer
```

Example output:

```
myPublicCert.cer: OK
```

- Verify the private key and signature algorithm details:

```
nutanix@pcvm$ openssl x509 -in my_cert_name.crt -text -noout | grep -i 'rsa\|ecdsa\|Public'
```

Example output:

```
Signature Algorithm: ecdsa-with-SHA256
      Subject Public Key Info:
        Public Key Algorithm: id-ecPublicKey
        Public-Key: (256 bit)
        Signature Algorithm: ecdsa-with-SHA256
```

- Verify that the CA certificate chain uses SHA 256 as the signature algorithm:

```
nutanix@pcvm$ openssl crl2pkcs7 -nocrl -certfile ca_chain_certs.crt | openssl pkcs7 -print_certs -noout -text | grep -Ew '(Subject|Issuer|Signature Algorithm):' | grep -C1 Issuer
```

Troubleshooting the Certificate Generation Request

The following troubleshooting tips can help you resolve common issues that can occur when generating certificates.

Chain certificate format

If your chain certificate file has public or private certificates, it will fail to import in Prism Central. Ensure that the chain certificate file only has the root and intermediate certificates.

For example, if a public certificate is present in a chain file, you can remove it by opening your chain file in your preferred text editor. Ensure that there are no extra white spaces at the bottom of the file.

DER-encoded certificate issue

If the certificate is DER encoded, it fails to import in Prism Central. You can resolve the issue by converting it to PEM-encoded ASCII format.

- Ensure that the certificate is DER encoded:

```
nutanix@pcvm$ openssl x509 -in cert.crt -inform der -text -noout
```

- If the certificate is DER encoded, run the following command to convert the certificate from DER to PEM-encoded ASCII format:

```
nutanix@pcvm$ openssl x509 -in certDER.crt -inform der -outform pem -out cert.crt
```

Certificate format

Ensure that all the certificates do not have any extra data (or custom attributes) before the beginning (-----BEGIN CERTIFICATE-----) or after the end (-----END CERTIFICATE-----) of the block.

Importing a Self-Signed SSL Certificate in Prism Central

Import a self-signed SSL certificate into Prism Central.

Before you begin

Ensure that you generate a self-signed SSL certificate. For information, see [Generating a Self-Signed SSL Certificate with Subject Alternative Name](#) on page 61.

About this task

To import a self-signed SSL certificate into Prism Central, follow these steps:

Procedure

1. Log in to Prism Central as an administrator.
2. From the [Application Switcher Function](#), select the **Admin Center** application and from the navigation bar, select **Settings**.
3. Under Security, select **SSL Certificate**.
4. Click **Replace Certificate**.
5. Select **Import Key and Certificate** and click **Next**.

6. On the SSL certificate window, provide the following information:

- **Private Key Type:** From the dropdown list, select the appropriate private key type for the self-signed certificate.

Note: EC DSA 521 and EC DSA 384 bit private keys are not supported.

- **Private Key:** Click **Choose file** and select the private key.

Note: The private key that you import must be unencrypted. Password-protected private keys are not supported.

- **Public Certificate:** Click **Choose file** and select the self-signed certificate corresponding to the private key.
- **CA Certificate/Chain:** Click **Choose file** select the self-signed certificate corresponding to the private key.

SSL Certificate ?

Note: Any change made to a certificate will restart the prism session and the user will be logged out automatically.

Guidelines for RSA: Please use a SHA-256, SHA-384, or SHA-512 signature algorithm with RSA certificates for security and performance.

Private Key Type

RSA 2048 bit

Private Key ?

Choose file my_key_name.key

Public Certificate

Choose file my_cert_name.crt

CA Certificate/Chain

Choose file my_cert_name.crt

Cancel Import Files

Figure 3: Importing self-signed certificate

The following table lists certificate components and its corresponding file type to choose when SSL certificate window prompts:

Certificate Components	File type
Private Key	my_key_name.key
Public Certificate	my_cert_name.crt
CA Certificate/Chain	my_cert_name.crt

7. Click **Import Files**.

Note: Prism Central stores only one custom SSL certificate. When you upload a new certificate, Prism Central replaces the existing certificate.

After you import the new certificate, the Prism Central web console restarts. If the certificate and credentials are valid, Prism Central uses the new certificate immediately. All open browser sessions becomes invalid until you reload the page and accept the new certificate. If the certificate is invalid due to corrupted file or wrong certificate type, Prism Central discards the new certificate and reverts to the default certificate provided by Nutanix.

Importing a CA-Signed SSL Certificate in Prism Central

Import a certificate authority (CA) signed SSL certificate into Prism Central.

Before you begin

Ensure that you generate a Certificate Signing Request (CSR) for submission to a CA. For more information, see [Generating a Certificate Signing Request with Subject Alternative Name for Submission to Certificate Authority](#) on page 63.

About this task

To import a CA-signed SSL certificate into Prism Central, follow these steps:

Procedure

1. Log in to Prism Central as an administrator.
2. From the [Application Switcher Function](#), select the **Admin Center** application and from the navigation bar, select **Settings**.
3. Under Security, select **SSL Certificate**.
4. Click **Replace Certificate**.
5. Select **Import Key and Certificate** and click **Next**.

6. On the SSL certificate window, provide the following information:

- **Private Key Type:** From the dropdown list, select the appropriate private key type for the CA signed certificate.

Note: Prism Central does not support EC DSA 521 and EC DSA 384 bit private keys.

- **Private Key:** Click **Choose file** and select the private key.

Note: The private key that you import must be unencrypted. Password-protected private keys are not supported.

- **Public Certificate:** Click **Choose file** and select the CA signed public portion of the certificate corresponding to the private key.

- **CA Certificate/Chain:** Click **Choose file** and select the certificate or chain of the signing authority for the public certificate.

Note: For more information on how to create a chain file from the list of CA certificates, see [Generating a Certificate Signing Request with Subject Alternative Name for Submission to Certificate Authority](#) on page 63.

SSL Certificate ?

Note: Any change made to a certificate will restart the prism session and the user will be logged out automatically.

Guidelines for RSA: Please use a SHA-256, SHA-384, or SHA-512 signature algorithm with RSA certificates for security and performance.

Private Key Type

RSA 2048 bit

Private Key ⓘ

Choose file my_key_name.key

Public Certificate

Choose file ca_signed_public_cert.cer

CA Certificate/Chain

Choose file ca_public_cert.crt

Cancel Import Files

Figure 4: Importing CA-signed certificate

The following table lists certificate components and its corresponding file type to choose when SSL certificate window prompts:

Table 5: CA Signed Certificate File Type

Certificate Components	File type
Private Key	my_key_name.key
Public Certificate	ca_signed_public_cert.cer
CA Certificate/Chain	ca_public_cert.crt or ca_chain_certs.crt

7. Click **Import Files**.

Note: Prism Central stores only one custom SSL certificate. When you upload a new certificate, Prism Central replaces the existing certificate.

After you import the new certificate, the Prism Central web console restarts. If the certificate and credentials are valid, Prism Central uses the new certificate immediately. All open browser sessions becomes invalid until you reload the page and accept the new certificate. If the certificate is invalid due to corrupted file or wrong certificate type, Prism Central discards the new certificate and reverts to the default certificate provided by Nutanix.

Regenerating a Self-Signed Certificate in Prism Central

Regenerate a self signed SSL certificate in Prism Central.

About this task

To regenerate a self signed certificate in Prism Central, follow these steps:

Procedure

1. Log in to Prism Central as an administrator.
2. From the [Application Switcher Function](#), select the **Admin Center** application and from the navigation bar, select **Settings**.
3. Under Security, select **SSL Certificate**.
4. Click **Replace Certificate**.
5. Select **Regenerate Self Signed Certificate** and click **Apply**.
6. Click **OK**.
Prism Central generates and applies a new RSA 2048 bit self-signed certificate.

Supported Key Configurations for SSL Certificates

Nutanix supports only the following key types, sizes or curves, and signature algorithms for SSL certificates:

Table 6: SSL Certificate Key Type Options

Key Type	Size or Curve	Signature Algorithm
RSA	4096	SHA-256, SHA-384 or SHA512
RSA	2048	SHA-256, SHA-384 or SHA512

Key Type	Size or Curve	Signature Algorithm
EC DSA 256	prime256v1	ecdsa-with-sha256

In Prism Central versions prior to pc.2024.3, uploading an RSA 4096-bit certificate might cause some issues. For more information, see [KB 12775](#).

Availability Zone (AZ) pairing between Prism Central clusters supports RSA 2048-bit certificates only. RSA 4096-bit certificates are not supported.

Nutanix does not support SHA-1 certificates, including root CAs.

Client and CAC authentication support only RSA 2048-bit certificates.

Cluster Lockdown in Prism Central

Cluster lockdown in Prism Central disables password-based SSH login and requires SSH keys for remote access.

Cluster lockdown in Prism Central enhances security by disabling the password-based SSH login. Instead, authorized user SSH keys provide access to Prism Central, which improves the overall security posture.

Nutanix supports the following key-based SSH encryption algorithms:

- AES128-CTR
- AES192-CTR
- AES256-CTR

Nutanix supports the following key types:

- RSA
- ECDSA

Cluster Lockdown Considerations

Before you configure cluster lockdown in Prism Central, consider the following points:

- Generate the public key using the `ssh-keygen` command on Mac or Linux, or the PuTTY application on Windows. For more information, see [KB-1895](#).
- When you enable cluster lockdown, Prism Central does not store the passwords for both the Prism Central VM (PCVM) and host. You cannot change these passwords to access the cluster resources.
- When you enable cluster lockdown, password-based SSH login no longer works for the PCVM and host. Users must sign-in using SSH based keys.
- Adding a user SSH key enables SSH access for both the `nutanix` and `admin` accounts on the PCVM and host.
- You can configure multiple user SSH keys.
- Disabling remote login and deleting all user SSH keys locks down cluster SSH access.
- For additional security, you can configure SSH security level for the PCVM. For more information, see [PCVM Security Hardening](#) on page 156.

Configuring Cluster Lockdown in Prism Central

Configure cluster lockdown to enable SSH key-based access to your cluster.

Before you begin

Ensure that you generate the public key using `ssh-keygen` command on Mac or Linux, or the PuTTY application on Windows. For more information, see [KB-1895](#).

About this task

To configure the cluster lockdown in Prism Central, follow these steps:

Procedure

1. Log in to Prism Central as an administrator.
2. From the [Application Switcher Function](#), select the **Admin Center** application and from the navigation bar, select **Settings**.
3. Under Security, select **Cluster Lockdown**.
4. Deselect the **Enable Remote Login with Password** checkbox.
5. Click **+ New user SSH Key** and enter the following information:
 - **Name:** Enter a key name.
 - **Key:** Paste the public key value.
6. Click **Save**.

Deleting a User SSH Key in Prism Central

You can disable the SSH key-based access to your cluster.

About this task

Caution: Disabling remote login and deleting all user SSH keys locks down the cluster SSH access. Ensure to select the **Enable Remote Login with Password** checkbox before deleting all the user SSH keys.

To remove the user SSH key, follow these steps:

Procedure

1. Log in to Prism Central as an administrator.
2. From the [Application Switcher Function](#), select the **Admin Center** application and from the navigation bar, select **Settings**.
3. Under Security, select **Cluster Lockdown**.
4. Select the **Enable Remote Login with Password** checkbox.
5. Click the **X** remove icon.
6. Select **OK**.

Security Policies using Nutanix Flow

Nutanix Flow secures traffic with microsegmentation policies.

Nutanix Flow provides a policy-driven security framework to control and inspect application traffic within the data center. You can define microsegmentation policies that restrict east-west traffic between VMs, enforce application isolation, and apply service-specific rules. These policies improve visibility, reduce attack surfaces, and ensure

compliance with organizational security requirements. For more information, see the [Flow Microsegmentation Guide](#).

Data-in-Transit Encryption

Data-in-transit encryption encrypts service-level traffic between cluster nodes to protect data from unauthorized access.

Data-in-Transit encryption, along with data-at-rest encryption, protects the entire life cycle of data and is an essential countermeasure against unauthorized access of critical data.

For information on licensing Data-in-transit encryption and other Nutanix products, see [Nutanix Cloud Platform Software Options](#).

Before you configure data-in-transit encryption, consider the following points:

- Data-in-transit encryption might have an impact on I/O latency and CPU performance.
- Intra-cluster traffic encryption is supported only for the Stargate service.
- RDMA traffic encryption is not supported.
- When a CVM goes down, the traffic from guest VM to remote CVM is not encrypted.
- Traffic between guest VMs connected to volume groups is not encrypted when the target disk is on a remote CVM.

Enabling Data-in-Transit Encryption

Encrypt service-level traffic between the cluster nodes.

About this task

To enable data-in-transit encryption, follow these steps:

Procedure

1. Log in to Prism Central as an administrator.
2. From the [Application Switcher Function](#), select the **Infrastructure** application and from the navigation bar, select **Hardware > Clusters**.
3. Select the checkbox associated with the cluster.
4. Click **Actions > Enable Data-in-Transit Encryption**.
5. Click **Enable**.

Disabling Data-in-Transit Encryption

You can disable data-in-transit encryption.

About this task

To disable data-in-transit encryption, follow these steps:

Procedure

1. Log in to Prism Central as an administrator.
2. From the [Application Switcher Function](#), select the **Infrastructure** application and from the navigation bar, select **Hardware > Clusters**.

3. Select the checkbox associated with the cluster.
4. Click **Actions** > **Disable Data-in-Transit Encryption**.
5. Click **Disable**.

Securing AHV VMs with a Virtual Trusted Platform Module

You can configure VMs with a vTPM (virtual Trusted Platform Module) in Prism Central.

A vTPM is a feature available on AHV that allows VMs to use the software-based TPM 2.0 specification that works as a virtual device without requiring physical hardware TPMs. The vTPM provides the same functionality as a physical TPM but within the virtualized environment that allows VMs to leverage TPM capabilities without needing dedicated hardware.

You can also enable the vTPM using aCLI. For more information, see [Securing AHV VMs with Virtual Trusted Platform Module \(aCLI\)](#) in the *AHV Administration Guide*.

Virtual Trusted Platform Module Use Cases

A vTPM provides virtualization-based security support for the following primary use cases:

- **Microsoft Credential Guard:** vTPM is utilized to protect Windows credentials. For more information on Microsoft Windows Defender Credential Guard, see Microsoft documentation.
- **Windows 11 Installation:** Windows 11 requires TPM 2.0 and Secure Boot enabled for optimal security. For more information on Windows 11 specifications, features, and system requirements, see Microsoft documentation.
- **Secure Boot for Windows:** vTPM is compatible with Secure Boot, enhancing the system security. For information on how to create and update a guest VM with Secure Boot enabled, see [Creating a VM through Prism Central \(AHV\)](#) and [Managing a VM through Prism Central \(AHV\)](#) sections in *Prism Central Infrastructure Guide*.
- **Windows BitLocker:** vTPM supports Windows BitLocker with certain restrictions. For more information, see [Virtual Trusted Platform Module Limitations](#) on page 75.

Virtual Trusted Platform Module Requirements

Understand the requirements for a virtual Trusted Platform Module (vTPM) in Prism Central.

Supported software versions:

- pc.2022.9 or later
- AHV version 20220304.242 or later
- AOS version 6.5.1 or later

VM requirements:

- You must enable UEFI firmware. For more information, see [UEFI Support for VM](#).
- You must enable Secure Boot if you are using Windows BitLocker. For more information, see [Creating/Updating a VM with Secure Boot Enabled](#).

Virtual Trusted Platform Module Limitations

Understand the limitations of a virtual Trusted Platform Module (vTPM) in Prism Central.

- The Secure Boot limitations apply to vTPM-enabled VMs. For more information, see [Secure Boot limitations](#).

- The Nutanix Disaster Recovery limitations apply when protecting vTPM-enabled VMs. For more information, see [Disaster Recovery limitations](#).
- If you take a third-party backup of the vTPM-enabled guest VMs, the vTPM data including crucial encryption keys is not protected because third-party backup software does not have access to the data. Therefore, third-party backups of vTPM-enabled guest VMs cannot protect the vTPM device data.
- If the data stored in the vTPM device is critical, such as encryption keys for Windows BitLocker, restoring a VM from a third-party backup might render it unusable and result in unrecoverable data loss.
- If the data in the vTPM device is non critical, such as for Windows Secure Boot, the backup software or user must attach a new vTPM device during the restore process.

Configuring a VM with a Virtual Trusted Platform Module

Enable a virtual Trusted Platform Module (vTPM) feature to secure virtual machines running on AHV.

Before you begin

For more information on VM creation settings, see [Creating a VM](#) in *Prism Central Infrastructure Guide*.

About this task

To enable the vTPM when creating a VM, follow these steps:

Procedure

1. Log in to Prism Central as an administrator.
2. From the [Application Switcher Function](#), select the **Infrastructure** application and from the navigation bar, select **Compute > VMs**.
3. Select the **List** tab.
4. Click **Create VM**.
5. Navigate to the **Resources** tab and perform the following actions:
 - **Boot Configuration:** Select **UEFI BIOS Mode** option.
 - **Shield VM Security Settings:** Select the **Attach vTPM** checkbox.

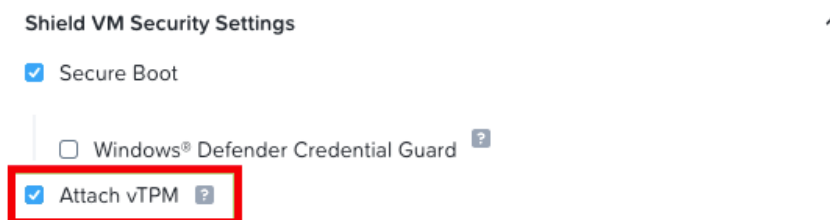


Figure 5: Attach vTPM

6. At the subsequent VM setting tabs, click **Next**.
7. Click **Save**.

What to do next

Power on the VM to verify if the vTPM configuration is applied on the VM.

Enabling a Virtual Trusted Platform Module on an Existing VM

Update the settings of an existing VM to enable a virtual Trusted Platform Module (vTPM).

Before you begin

Before you enable the vTPM on an existing VM, consider the following points:

- The UEFI BIOS mode is enabled for boot configuration in the existing VM.
- Power off the VM before you enable the vTPM.

About this task

To enable the vTPM on the existing VM, follow these steps:

Procedure

1. Log in to Prism Central as an administrator.
2. From the [Application Switcher Function](#), select the **Infrastructure** application and from the navigation bar, select **Compute > VMs**.
3. Select the **List** tab.
4. Select the checkbox associated with the VM.
5. Click **Actions > Update**
6. Navigate to the **Resources** tab and select **Attach vTPM** checkbox.

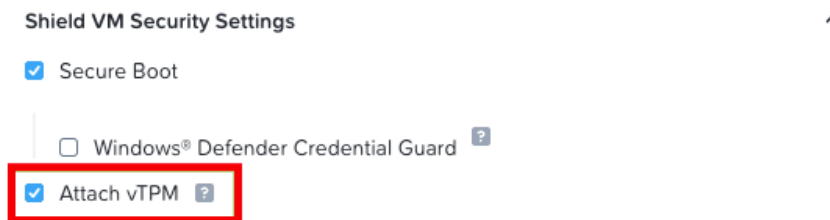


Figure 6: Attach vTPM

7. At the subsequent VM setting tabs, click **Next**.
8. Click **Save**.

What to do next

Power on the VM to verify if the vTPM configuration is applied on the VM.

Disabling a Virtual Trusted Platform Module

Disable a virtual Trusted Platform Module (vTPM) in a VM.

Before you begin

Before you disable the vTPM, consider the following points:

- Disabling a vTPM might severely affect VM functionality or result in data loss. For example, if your Windows BitLocker key is stored in a vTPM, ensure that you have the recovery key before disabling a vTPM.

- Power off the VM before disabling a vTPM.

About this task

To disable the vTPM, follow these steps:

Procedure

1. Log in to Prism Central as an administrator.
2. From the [Application Switcher Function](#), select the **Infrastructure** application and from the navigation bar, select **Compute > VMs**.
3. Select the **List** tab.
4. Select the checkbox associated with the VM.
5. Click **Actions > Update**
6. Navigate to the **Resources** tab and deselect **Attach vTPM** checkbox.
7. At the subsequent VM setting tabs, click **Next**.
8. Click **Save**.

Security Dashboard

Monitor and manage the security posture of your registered clusters with the customizable Security Dashboard, tracking key metrics and compliance.

The Security Dashboard provides dynamic summary of the security posture across all the registered clusters. The Security Dashboard allows you to view the most critical security parameters like cluster-based issue summary, STIG policy compliance, security hardening, identified vulnerabilities, and the local account password status. The security dashboard is divided into multiple widgets to represent different security focus areas.

Note: The Security Dashboard does not report security issues for user VMs.

The following figure provides a sample view of the Security Dashboard with the default widgets:

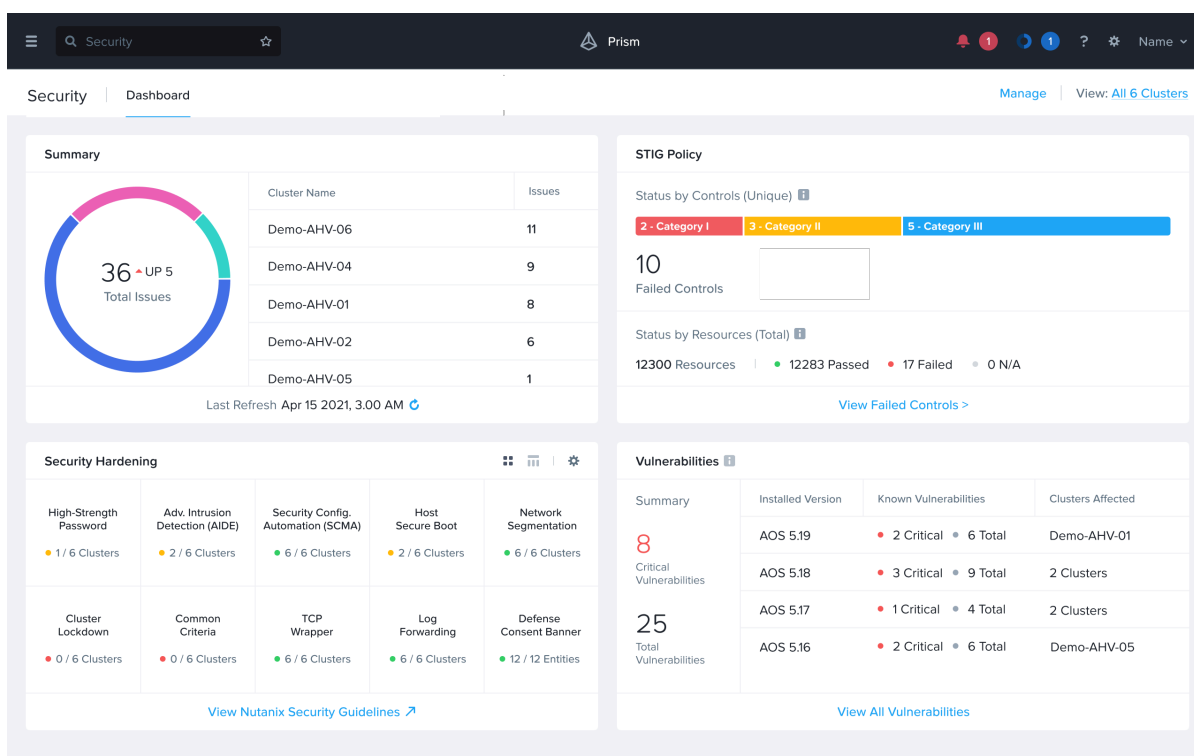


Figure 7: Security Dashboard Overview

Security Dashboard Requirements

Understand the requirements of the Security Dashboard feature:

- Your cluster must run AOS 6.6 or later and pc.2022.9 or later.

Note: Ensure your cluster uses a compatible AHV or ESXi hypervisor. For software compatibility details, see the [Compatibility and Interoperability Matrix](#) on the Nutanix Support portal.

- Starting from Prism Central version pc.2024.3.1, certain features of the Security Dashboard, such as the **Summary** widget, are available only on clusters running AOS 7.0 or later.
- Only users with the Prism Admin role can access the Security Dashboard feature. Log in with an account that has Prism Admin role assignment to view and manage the Security Dashboard feature.
- You can enable the Security Dashboard on a dark site. For more information, see [Enabling Security Dashboard on Dark Sites](#) on page 85.
- Security Dashboard feature is not available on the Prism Central instances deployed on Nutanix Cloud Clusters (NC2).
- Certain Security Dashboard widgets are available only on specific AOS versions:
 - Security Hardening and Vulnerabilities** widget requires AOS 6.6 or later.
 - System Account Passwords** widget requires AOS 6.7.1 or later.

Security Dashboard Widget in Prism Central Dashboard

View security issue summaries across clusters from the Prism Central main dashboard.

The Security Dashboard widget on the Prism Central main dashboard displays the total number of security issues across your clusters. The issue count is also available based on categories including security hardening, STIG issues, and vulnerabilities.

Click **View All Issues** to access the [Security Dashboard](#) for detailed information on the security posture of all registered clusters.

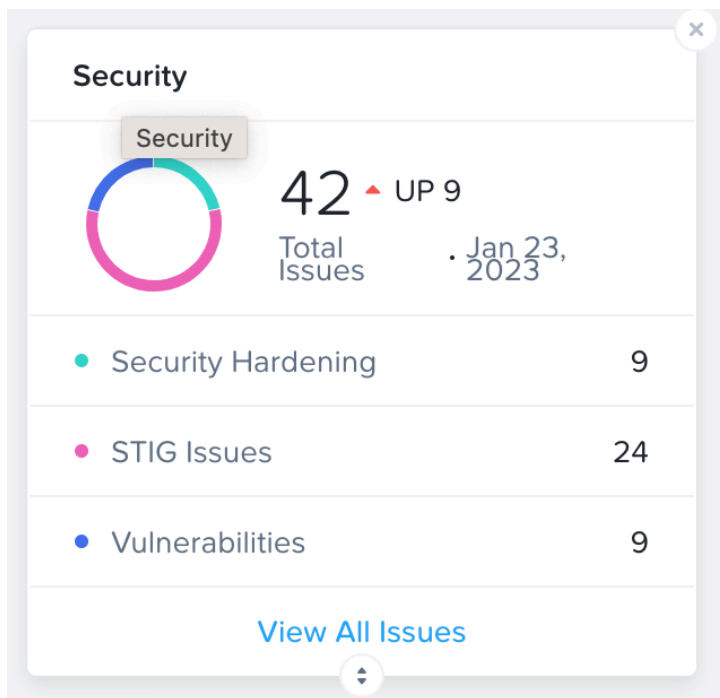


Figure 8: Security Dashboard Widget

Security Dashboard Wizard

Take a guided tour of the Security Dashboard in Prism Central.

The Security Dashboard wizard allows you to take a guided tour and navigate through the various tasks and workflows in the **Security Dashboard** in Prism Central.

The Security Wizard automatically presents as a dialog-box when you access the dashboard feature for the first time. Click **Start Tour** to begin the dashboard walkthrough.

Optionally, you can click **Skip for Now** to access the dashboard directly. To access this tour later, click the help menu icon (?), expand **New in Prism Central**, and select **Security Dashboard**.

Accessing the Security Dashboard

Access the Security Dashboard from Prism Central.

About this task

To access the Security Dashboard from Prism Central, follow these steps:

Procedure

1. Log in to Prism Central as an administrator.

2. Do one of the following:

- » From the [Application Switcher Function](#), select the **Infrastructure** application and from the navigation bar, select **Network & Security > Security Dashboard**.
- » From the Prism Central main dashboard, under **Security** widget, click **View All Issues**.

Security Dashboard Details Page

Monitor your cluster security posture in a single customizable view.

Security Dashboard is a customizable and dynamic console that provides a unified view of your Nutanix infrastructure's security.

By default, the Security Dashboard displays the following information widgets:

- [Summary](#) on page 81
- [STIG Policy](#) on page 81
- [Security Hardening](#) on page 82
- [Vulnerabilities](#) on page 83
- [System Account Passwords](#) on page 83

Use the **View** menu on the dashboard to switch between different cluster views. The widget data updates dynamically based on the following options:

- All clusters
- Individual cluster
- Selection of clusters

Summary

The **Summary** widget allows you to view your open security issues or focus on the clusters that have the most number of security issues. You can click the Summary pie graph to view the following information.

- Total number of issues in the clusters
- Number of issues in the clusters categorized based on the following issue categories:
 - Security Hardening
 - STIG Issues
 - Vulnerabilities
 - Local Account Passwords

Tip: The Security Dashboard refreshes once daily. Updating the STIG check based vulnerabilities information requires a manual refresh. You can initiate a manual refresh by clicking the refresh icon at the bottom of the widget. Manual refresh process takes some approximately 20 minutes to 2 hours to complete and depends on the number of clusters in your environment.

STIG Policy

STIG helps detect deviation from the security baseline configuration of the operating system and hypervisor to remain in compliance. Nutanix has implemented the Controller VM and Prism Central VM to support STIG compliance with the RHEL 8 STIG as published by DISA. For more information, see [Security Dashboard STIG Guidance Reference](#).

The **STIG Policy** widget provides an accurate snapshot of policy violations or deviations (resulting in failure) from the baseline STIG policy. This widget displays the STIG policy compliance status according to controls and resources.

Note: The STIG checks are run only on the node running primary Prism Element.

Status by Controls (Unique)	Number of unique STIG controls that are not met.
Status by Resources (Total)	Total number of individual resources that have failed a set of STIG controls.

Click **View Failed Controls** to view the list of STIG controls that are not met.

STIG Policy - Failed Controls

The STIG Policy -Failed Controls dialog box allows you to view STIG controls that do not meet the required settings.

Security Hardening

Use the **Security Hardening** widget to view the status of security hardening controls that are applied to your clusters and to configure multiple security hardening controls directly.

You can configure the following security hardening configurations from the **Security Hardening** widget.

High-Strength Password

Enables the Nutanix-recommended high-strength password policy. After enabling this setting, you must update the Controller VM and AHV host passwords.

Advanced Intrusion Detection Environment (AIDE)

Enables AIDE to monitor operating system files on cluster nodes. AIDE runs weekly to detect unauthorized file modifications.

Important: Nutanix recommends enabling AIDE for all clusters.

Security Configuration Management Automation (SCMA)

Enables SCMA frequency for AHV hosts and Controller VM. SCMA checks multiple security entities for both Nutanix storage and AHV. For more information, see [Security Configuration Management Automation Implementation](#) on page 7.

By default, SCMA scanning is set to **Daily** for both AHV and Controller VM. You can configure the scan frequency to **Hourly**, **Daily**, **Weekly**, or **Monthly** individually for Controller VM and AHV hosts.

Note: Higher frequency SCMA scans can decrease performance on the respective cluster.

Cluster Lockdown

Enables **Cluster Lockdown** mode to restrict administrative access and enhance security. For more information, see [Configuring Cluster Lockdown using Security Dashboard](#) on page 83.

Defense Consent Banner

Enables **Defense Consent Banner**. Once enabled, the system displays the consent banner of the US Department of Defense when connecting with SSH. You must enable the Defense Consent Banner individually for Controller VM and AHV hosts. This banner is customizable, but you must reconfigure it after every AOS upgrade.

Host Secure Boot

Enables **Host Secure Boot** to prevent unauthorized firmware or software from running on AHV or ESXi hosts during the boot process. Host secure boot is a unified extensible firmware interface (UEFI) firmware

feature that ensures only trusted system components are allowed to run at startup. For more information on enabling and verifying host secure boot, see [Host Secure Boot \(UEFI\)](#).

You can also configure security hardening setting using the nCLI interface. For more information, see [Hardening Instructions Using nCLI](#) on page 151.

Configuring Cluster Lockdown using Security Dashboard

You can configure cluster lockdown directly using the Security Dashboard.

About this task

To configure cluster lockdown using the **Security Dashboard**, follow these steps:

Procedure

1. Log in to Prism Central as an administrator.
2. From the [Application Switcher Function](#), select the **Admin Center** application.
3. From the navigation bar, go to **Network & Security > Security Dashboard**.
4. In the **Security Hardening** widget, click **Cluster Lockdown**.
5. From the **Your Clusters** list, click **Lock Down** for the required cluster.
6. Disable remote log on using the **Enable Remote Login with Password** checkbox.
7. Delete the SSH key.
To delete the SSH key, click the cross button from the **Your Keys** list.
Your preferences are automatically saved upon configuration.

Vulnerabilities

The **Vulnerabilities** widget displays a list of vulnerabilities (or CVEs) associated with your clusters based on the AOS versions.

The **Vulnerabilities** widget displays vulnerability data for the Prism Central instance and clusters running AOS and AHV only.

Click **View All Vulnerabilities** to view the list of all vulnerabilities and the recommended upgrade path for mitigating the identified vulnerabilities.

Note: Not all the listed CVEs might be resolved when upgraded to a new release.

Starting with Prism Central 2024.1 release, you can update the **Vulnerabilities** information available on the **Security Dashboard**. Upgrade the LCM framework to LCM 3.0 release, and then upgrade the **Security Dashboard CVE Data** using either the connected site or dark site method (web server-based method only).

For more information, see [Life Cycle Manager Guide](#).

System Account Passwords

View system accounts and their password state.

The **System Account Passwords** widget displays a list of local accounts and the associated password state (default or secure).

You can view the local accounts passwords **By Component** or **By Cluster**.

Click **View all System Account Passwords** to change the account passwords using the [Centralized Password Management Using System Accounts](#) on page 59 page.

Manually Refresh the Security Dashboard

Manually refresh the Security Dashboard only when you are aware that updates or changes are made to the system or data since the last automatic refresh.

About this task

To manually refresh the Security Dashboard, follow these steps:

Procedure

1. Log in to Prism Central as an administrator.
2. From the [Application Switcher Function](#), select the **Infrastructure** application and from the navigation bar, select **Network & Security > Security Dashboard**.
3. Click **Manage Dashboard**.
4. Click **Refresh Dashboard**.

Note: The system might take some time to refresh the dashboard.

Ensure that your Security Dashboard version is updated in Life Cycle Manager (LCM) to ensure security details are correct for all your clusters.

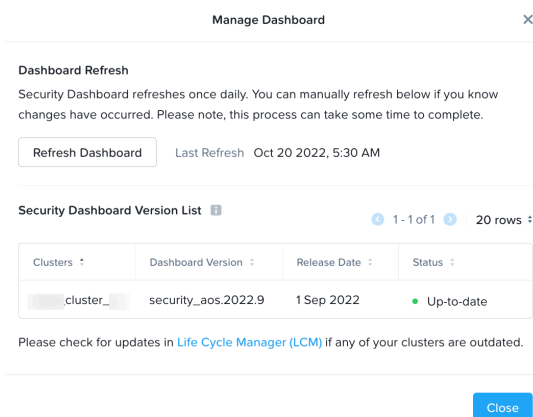


Figure 9: Manage Security Dashboard Window

Manually Upgrade the Security Dashboard Using LCM

The Life Cycle Manager (LCM) allows you to manually upgrade the security dashboard using PC core services module

Before you begin

- Ensure that port 443 is open in your firewall and allow access to <https://objects.githubusercontent.com/>. For more information, see [Prism Central Port and Protocols](#).
- Manual upgrade of security dashboard is supported in the following software versions:
 - pc.2022.9 and later
 - Life Cycle Manager version 2.6 and later

About this task

To upgrade the security dashboard manually using LCM, follow these steps:

Procedure

1. Log in to Prism Central as an administrator.
2. From the [Application Switcher Function](#), select the **Admin Center** application and from the navigation bar, select **LCM**.
3. Perform the inventory.
For more information, see [LCM Inventory](#).
4. From the list of available upgrades, select **PC Core Services**.

Note: Not every Prism Central core services module upgrade includes security dashboard updates. Some upgrades might only include internal service upgrades.

Starting from pc.2024.1, you can update the [Vulnerabilities](#) on page 83 widget information available on the Security Dashboard by upgrading the LCM framework to LCM 3.0 release and the Security Dashboard CVE data using either the connected site or web server-based dark site. For more information, see [Life Cycle Manager Guide](#).

5. Click **Upgrade**.

Enabling Security Dashboard on Dark Sites

Enable the Security Dashboard on dark sites.

Before you begin

Ensure that you have set up the web server-based dark site using LCM. For more information, see the [LCM Settings for Dark Sites](#) in *Life Cycle Manager Guide*.

About this task

To enable the Security Dashboard on dark sites, follow these steps:

Procedure

1. Log in to [Nutanix Portal](#).
2. From the [Entities Menu](#), select **Downloads**.
3. Select **Prism Central** product.
4. Next to **Prism Central Onprem Apps Bundle**, click **Download**.
5. Transfer the downloaded file to the LCM dark site server and place it in the release directory.

Note: Do not extract the downloaded bundle. The pc_platform_bootstrap service requires the bundle to remain in compressed format.

6. SSH into the Prism Central VM (PCVM) as a `nutanix` user:

```
$ ssh nutanix@pcvm_ip_address
```

7. Stop the `pc_platform_bootstrap` service on the dark site cluster:

```
nutanix@pcvm$ allssh genesis stop pc_platform_bootstrap
```

8. Restart the `pc_platform_bootstrap` service on the dark site cluster:

```
nutanix@pcvm$ allssh genesis restart
```

The restart process takes around five minutes to complete.

9. Verify the deployment status of the Security Dashboard:

```
nutanix@pcvm$ sudo kubectl get pods -A | grep security
```

Note: Ensure the output shows the Security Dashboard pod in a running state.

Portal Proxy Connection in Prism Central

The portal proxy connection enables secure communication between Prism Central and Nutanix Support Portal using API keys.

The portal proxy connection provides the following capabilities:

- Direct support case creation from the Prism Central web console
- Automatic updates to the cluster's license status, which eliminates the manual downloading and uploading of the licensing files

Portal Proxy Connection Considerations

Before you configure the portal proxy connection, consider the following points:

- You can configure only one API key in the portal proxy connection.
- When you disable the portal proxy connection, the associated API key becomes invalid. You cannot reuse the same API key for a new portal proxy connection.

Configuring the Portal Proxy Connection in Prism Central

Configure the portal proxy connection in Prism Central.

Before you begin

Ensure that you generated an API Key and downloaded the optional SSL key from the My Nutanix portal. For more information, see [Creating an API Key](#).

About this task

To configure the portal proxy connection in Prism Central, follow these steps:

Procedure

1. Log in to Prism Central as an administrator.
2. From the [Application Switcher Function](#), select the **Admin Center** application and from the navigation bar, select **Settings**.
3. Under Setup, select the **Portal Proxy**.
4. On the **Portal Connection** page, enter the following information:
 - a. **API Key:** Enter the API key that you generated from the My Nutanix portal.
 - b. (Optional) **Public Key:** To upload the optional SSL key that you downloaded from the My Nutanix portal, click **Choose File**.

5. Click **Save**.

Updating the Portal Proxy Connection in Prism Central

Update the portal proxy connection in Prism Central.

About this task

To update the portal proxy connection in Prism Central, follow these steps:

Procedure

1. Log in to Prism Central as an administrator.
2. From the [Application Switcher Function](#), select the **Admin Center** application and from the navigation bar, select **Settings**.
3. Under Setup, select the **Portal Proxy**.
4. Click **Update**.
5. Update the fields in the **Portal Connection** window as needed and click **Save**.

For more information on the fields available in the **Portal Connection** window, see [Configuring the Portal Proxy Connection in Prism Central](#) on page 86.

Disabling the Portal Proxy Connection in Prism Central

Disable the portal proxy connection in Prism Central.

Before you begin

You cannot reuse the disabled API key for a new portal proxy connection.

About this task

To disable the portal proxy connection in Prism Central, follow these steps:

Procedure

1. Log in to Prism Central as an administrator.
2. From the [Application Switcher Function](#), select the **Admin Center** application and from the navigation bar, select **Settings**.
3. Under Setup, select the **Portal Proxy**.
4. Click **Disable**.
5. Click **Yes**.

SECURITY MANAGEMENT USING PRISM ELEMENT

Nutanix provides several mechanisms to maintain security in a cluster using Prism Element.

Prism Element provides cluster security by supporting multiple authentication methods, including Active Directory, OpenLDAP, and Common Access Card (CAC). You can configure role mappings, manage local or emergency accounts, and enforce session timeouts. Prism Element also supports SSL certificate management for secure communication, cluster lockdown with SSH key-based access, and data-at-rest encryption using either self-encrypting drives or software encryption with key management. These controls protect access, data, and cluster operations.

Authentication in Prism Element

Authentication in Prism Element supports Active Directory, OpenLDAP, and Common Access Card (CAC).

Prism Element supports multiple authentication methods to secure access to the cluster. You can configure Active Directory for centralized credential management, or use OpenLDAP for directory-based authentication. For environments requiring stronger assurance, Prism Element supports Common Access Card (CAC) authentication with client chain certificates, enabling certificate-based login instead of passwords.

Configuring Active Directory Authentication in Prism Element

Configure Active Directory (AD) as the identity provider in Prism Element to enable AD credential-based login for users.

Before you begin

Before you configure AD in Prism Element, consider the following points:

- The service account for AD must have full read permissions on the directory service.
- Users with the User must change password at next logon attribute enabled cannot authenticate to Prism Element. Ensure that users change their password on a domain workstation before logging into Prism Element.
- If SSL is enabled on the AD server, ensure that Prism Element's firewall allows access to the appropriate port.
- Use of the Protected Users group is not supported for Prism Element authentication. For more information, see *Guidance about how to configure protected accounts* on Microsoft documentation website.
- Prism Element supports AD with LDAP v2 on Windows Server 2012 R2, Windows Server 2016, and Windows Server 2019.

Important: Prism Element does not support insecure SSLv2 and SSLv3 ciphers. To prevent the SSL fallback situation and being denied access to Prism Element, configure the following settings in your browser:

- Disable SSLv2 and SSLv3
- Enable TLS

About this task

To configure AD in Prism Element, follow these steps:

Procedure

1. Log in to Prism Element web console as an administrator.

2. From the main menu, click the gear icon.
3. On the left navigation bar, select **Authentication**.
4. Select the **Authentication Types** tab and select **Directory Service** checkbox.
5. Select the **Directory List** tab.

6. Click **+ New Directory** and enter the following information:

- **Directory Type:** From the dropdown list, select **Active Directory**.
- **Name:** Enter a name to identify the directory service on Prism Element.
- **Domain:** Enter the domain or sub-domain of the directory service.

For example nutanix.com

- **Directory URL:** Enter the URL address of the directory in `ldap://host:ldap_port_num` format.

host: The host value is either the IP address or fully qualified domain name.

ldap_port_num: The default LDAP port number is 389. Nutanix also supports ports 636 for LDAPS, 3268 and 3269 for LDAP/S global catalog servers.

Examples:

- Default LDAP (Port 389), when the configuration is single domain, single forest, and not using SSL:

`ldap://ldap.example.com` or `ldap://10.1.4.111:389`

- LDAPS (Port 636), when the configuration is single domain, single forest, and using SSL:

`ldaps://ldap.example.com` or `ldaps://10.1.4.111:636`

LDAPS requires that all AD Domain Controllers have properly installed SSL certificates.

- LDAP/S Global Catalog non-SSL (Port 3268), when the configuration is multiple domain, single forest, and not using SSL:

`ldap://globalcatalog.example.com:3268` or `ldap://10.1.4.111:3268`

- LDAP/S Global Catalog SSL/TLS encrypted (Port 3269), when the configuration is multiple domain, single forest, and using SSL:

`ldaps://globalcatalog.example.com:3269` or `ldaps://10.1.4.111:3269`

When the directory URL is set to the domain FQDN, LDAP authentication can fail if any directory server becomes unavailable. For high availability (HA), configure an LDAP HA cluster and specify the cluster virtual IP (VIP) address in the directory URL instead of the domain FQDN. For more information, see [KB-12568](#).

When constructing your LDAP/S URL to use a global catalog server, ensure that the domain control IP address or name belongs to a global catalog server within the domain that you are configuring. Otherwise, queries over port 3268 or 3269 might fail.

When querying the global catalog, the users `sAMAccountName` field must be unique across the Active Directory (AD) forest. If `sAMAccountName` field is not unique across subdomains, authentication might fail intermittently or consistently.

Cross-forest trust between multiple AD forests is not supported.

LDAPS support does not require custom certificates or certificate trust import.

For the complete list of required ports, see [Port Reference](#).

- **Search Type:** From the dropdown list, select either **Non Recursive (Default)** or **Recursive**.

When users are present in the first level group, select **Non Recursive (Default)**.

When you have multi-level group assignments, select **Recursive**. A recursive search performs a multi-level or nested search during authentication, which might cause slowness. If you experience slowness, select **Non-Recursive (Default)**.

- **Service Account Username:** Enter the service account user name in `user_name@domain.com` format.

Note: A service account is created to run only a particular service or application with the credentials specified for the account. According to the requirement of the service or application, the administrator can limit access to the service account. The service account is under the Managed Service Accounts in the AD server. An application or service uses the service account to interact with the operating system.

- **Service Account Password:** Enter the service account password.

Note: Be sure to update the service account credentials if the password changes or you use a different service account.

7. Click **Save**.

Configuring OpenLDAP Authentication in Prism Element

Configure OpenLDAP as the identity provider in Prism Element to enable OpenLDAP credential-based login for users.

Before you begin

Before you configure OpenLDAP in Prism Element, consider the following points:

- The service account for OpenLDAP must have full read permission on the directory service.
- Prism Element uses a service account to query OpenLDAP directories for user information and does not currently support certificate-based authentication with the OpenLDAP directory.
- The values for the **User Object Class**, **User Search Base**, **Username Attribute**, **Group Object Class**, **Group Search Base**, **Group Member Attribute**, and **Group Member Attribute** fields depend on your OpenLDAP configuration.
- Prism Element supports OpenLDAP with LDAP v2.

Important: Prism Element does not support insecure SSLv2 and SSLv3 ciphers. To prevent the SSL fallback situation and being denied access to Prism Element, configure the following settings in your browser:

- Disable SSLv2 and SSLv3
- Enable TLS

About this task

To configure OpenLDAP in Prism Element, follow these steps:

Procedure

1. Log in to Prism Element web console as an administrator.
2. From the main menu, click the gear icon.
3. On the left navigation bar, select **Authentication**.
4. Select the **Authentication Types** tab and select **Directory Service** checkbox.
5. Select the **Directory List** tab.

6. Click **+ New Directory** and enter the following information:

- **Directory Type:** From the dropdown list, select **OpenLDAP**.
- **Name:** Enter a name to identify the directory service on Prism Element.
- **Domain:** Enter the domain or sub-domain of the directory service.

For example nutanix.com

- **Directory URL:** Enter the URL address of the directory in ldap://host:ldap_port_num format.

host: The host value is either the IP address or fully qualified domain name.

ldap_port_num: The default LDAP port number is 389. Nutanix also supports ports 636 for LDAPS.

Examples:

- Default LDAP Port (389), when the configuration is single domain, single forest, and not using SSL:

`ldap://ldap.example.com` or `ldap://10.1.4.111:389`

- LDAPS (Port 636), when the configuration is single domain, single forest, and using SSL:

`ldaps://ldap.example.com` or `ldaps://10.1.4.111:636`

When the directory URL is set to the domain FQDN, LDAP authentication can fail if any directory server becomes unavailable. For high availability (HA), configure an LDAP HA cluster and specify the cluster virtual IP (VIP) address in the directory URL instead of the domain FQDN. For more information, see [KB-12568](#).

LDAPS support does not require custom certificates or certificate trust import.

For the complete list of required ports, see [Port Reference](#).

- **Search Type:** From the dropdown list, select either **Non Recursive(Default)** or **Recursive**.

When users are present in the first level group, you can select **Non Recursive(Default)**.

When you have multi-level group assignments, you can select **Recursive**. A recursive search performs a multi-level or nested search during authentication, which might cause slowness. If you experience slowness, select **Non-Recursive (Default)**.

- **User Object Class:** Enter the object class of users in the directory service.

For example: user, person, inetOrgPerson, organizationalPerson, or posixAccount

- **User Search Base:** Enter the base Distinguished Name (DN) to search the users.

For example:

`ou=users, dc=nutanix, dc=com`

`cn=users, dc=nutanix, dc=com`

- **Username Attribute:** Enter the attribute of the user object class which uniquely identifies a user in the directory.

For example: uid.

- **Group Object Class:** Enter the object class of groups in the directory service.

For example: posixGroup or groupOfNames

- **Group Search Base:** Enter the base Distinguished Name (DN) to search the user groups.

For example:

`cn=groups, dc=nutanix, dc=com`

ou=groups, dc=nutanix, dc=com

- **Group Member Attribute:** Enter the attribute of the group object that holds the user association.

For example: member or memberUid

- **Group Member Attribute Value:** Enter the attribute of the user object that is used to configure the membership in the group object.

For example: uid

- **Service Account Username:** Enter the service account user name in cn=username, dc=example, dc=com format.

For example cn=username, dc=nutanix, dc=com

Note: A service account is created to run only a particular service or application with the credentials specified for the account. According to the requirement of the service or application, the administrator can limit access to the service account. The service account is under the Managed Service Accounts in the OpenLDAP server. An application or service uses the service account to interact with the operating system.

- **Service Account Password:** Enter the service account password.

Note: Be sure to update the service account credentials if the service account password changes or a different service account is used.

7. Click **Save**.

Common Access Card Authentication with Client Chain Certificate in Prism Element

Prism Element uses client chain certificate authentication to enable the Common Access Card (CAC) login. Client chain certificate authentication is a certificate-based authentication mechanism where the user is prompted to select a certificate for login instead of a password.

Client chain certificate authentication provides enhanced security by requiring users to present a digital certificate instead of a password, reducing the risk of password-based attacks. In a one-way authentication process, Prism Element presents a certificate, and the user's browser verifies it. When the client chain certificate authentication is enabled, the process becomes a two-way authentication. During authentication, Prism Element also verifies the user's identity with a valid digital certificate to ensure that only authorized users can access the cluster. A user must provide a valid certificate when accessing Prism Element.

Client chain certificate authentication has the following workflow:

- **Certificate Installation:** A user installs the certificate on their local machine.

Note: The Certificate Authority (CA) must be the same for both the client chain certificate and the certificate on the local machine.

- **Certificate Presentation and Chain Verification:** When the user accesses Prism Element, the system prompts them to select their CAC certificate. Prism Element verifies the presented certificate against its trusted CA chain. The chain establishes trust by linking the client certificate to a trusted root CA. This chain verification step confirms the certificate's validity.
- **EDIPI Extraction and Account Verification:** Prism Element extracts the Electronically Data Interchange Personal Identifier (EDIPI) from the validated CAC certificate. It then queries Active Directory (AD) to verify that the user has a valid and active account associated with this EDIPI.
- **Authentication Method Selection:** Prism Element supports both certificate-based and basic authentication. When a certificate is present, the system prioritizes certificate-based authentication for the login. If a certificate is not present, the system uses basic authentication.
- **Login:** Prism Element logs in the user based on their validated certificate and verified Active Directory account.

Before you configure the CAC, consider the following points:

- Prism Element supports AD as directory service for CAC. OpenLDAP is not supported.
- When you enable CAC authentication, Prism Element disables all other directory services and local user login. Only the local admin user login is permitted in this case.
- When you authenticate Prism Element with client chain certificate, the **Subject name** field must be present. The subject name should match the UserPrincipalName (UPN) in the AD. The UPN is a username with domain address. For example user1@nutanix.com.
- If you map a Prism role to a CAC user and not to an AD group or organizational unit to which the user belongs, specify the EDIPI UPN or UPN of that user in the role mapping. A user who presents a CAC with a valid certificate is mapped to a role and taken directly to the web console home page. The web console login page is not displayed.
- By default, users with an admin role can log in to the Prism Element web console using CAC authentication. For other LDAP user roles, ensure the correct role mappings or role assignments are configured to enable CAC authentication.
- Client chain certificate must be PEM encoded.
- If you have logged in to Prism Element by using CAC authentication, to successfully log out of Prism Element, close the browser after you click Log Out.

Configuring Common Access Card Authentication in Prism Element

Configure Common Access Card (CAC) authentication in Prism Element.

Before you begin

Ensure that port 9441 is open in your firewall. After enabling the CAC authentication, the CAC login redirects the browser to use port 9441.

About this task

To configure the CAC authentication in Prism Element, follow these steps:

Procedure

1. Log in to Prism Element web console as an administrator.
2. From the main menu, click the gear icon.
3. On the left navigation bar, select **Authentication**.
4. Select the **Client** tab.
5. Select **Configure Client Chain Certificate** checkbox.
6. Click **Choose File** and select a client chain certificate from your local machine.
7. Click **Yes**.
Prism Element web console restarts and you are logged out automatically.
8. Select **Enable Client Authentication** checkbox.
9. Click **Yes**.
Prism Element web console restarts and you are logged out automatically.

10. Select **Configure Service Account** checkbox and enter the following information:

- **Directory:** From the dropdown list, select the directory.
- **Service Username:** Enter a service name in `user_name@domain.com` format.
- **Service Password:** Enter a service password.

11. Click **Save**.

12. Select **Enable CAC Authentication** checkbox.

13. Click **Yes**.

Prism Element web console restarts and you are logged out automatically.

Authentication Best Practices

Authentication best practices in Prism Element ensure secure access to Nutanix clusters.

Prism Element supports several authentication best practices to improve cluster security. You can use external identity services such as Active Directory to authenticate users and reserve the local admin account for emergency access. You can also change the default Controller VM password to meet complexity requirements across the cluster. Additionally, you can configure session idle timeouts to automatically log out inactive accounts and minimize the risk of unauthorized access.

Emergency Local Account Usage

Use the admin account as a local emergency account to ensure access when external services are unavailable.

The admin account ensures that both the Prism Element web console and the Controller VM are available when the external services such as Active Directory is unavailable.

For all the external authentication, you must configure the cluster to use an external IAM service such as Active Directory. You must create service accounts in the IAM and the accounts must have access grants to the cluster through Prism Element web console user account management configuration for authentication.

Note: The local emergency account usage does not support any external access mechanisms, specifically for the external application authentication or external REST API authentication.

Modifying the Default CVM Password

Change the default Controller VM (CVM) password for `nutanix` user account by adhering to the password complexity requirements.

Before you begin

Before you modify the default CVM password, consider the following points:

- Changing the password on one of the CVM is applied to all CVMs in the cluster.
- Nutanix recommends using the `admin` user account as the emergency administrative account.
- Console and SSH direct login are disabled for the `root` account.

About this task

To modify the default CVM password, follow these steps:

Procedure

1. SSH into any of the Controller VMs (CVMs) as a `nutanix` user:

```
$ ssh nutanix@cvm_ip_address
```

2. Change the `nutanix` user account password:

```
nutanix@cvm$ sudo passwd nutanix
```

3. Enter your new password in the command output:

```
Changing password for nutanix.  
New password:  
Retype new password:  
passwd: all authentication tokens updated successfully.
```

Ensure that your password meets the following requirements:

- At least eight characters long
- At least one lowercase letter
- At least one uppercase letter
- At least one digit
- At least one special character (allowed special characters are: "#\$%&'()*+,-./:;<=>@[^_`{|}~!\)
- At least four characters different from the old password
- Must not be among the last five passwords
- Must not have more than two consecutive occurrences of a character
- Must contain at least one character from each of the following four character classes: uppercase letters, lowercase letters, digits, and special characters
- Must not contain the words "nutanix", "ntnx", "password", or any simple dictionary words.

Note: Ensure that you preserve the updated `nutanix` user password. The local authentication (PAM) module requires the previous password of the `nutanix` user to successfully start the password reset process.

Setting Admin Session Timeout

By default, Prism Element logs out users after 15 minutes of inactivity. You can change the session timeout and configure overrides for non-admin users.

About this task

To configure admin session timeout, follow these steps:

Procedure

1. Log in to Prism Element web console as an administrator.
2. From the main menu, click the gear icon.
3. On the left navigation bar, select **UI Settings**.
4. From the **Default Session Idle Timeout For Non-Admin Users** dropdown list, select the default session idle timeout for all non-administrative users.

5. From the **Session Idle Override For Non-Admin Users** dropdown list, select the appropriate option to override the session timeout for non-administrative users.

Role Mapping

You can create and manage the role mapping for directory service users in Prism Element web console.

When you enable directory service for user authentication, by default no permissions are granted to the directory service users. You must grant permissions to the directory users by specifying the roles for the organizational units (OUs), groups and users within a directory.

Role Mapping Considerations

Before you configure the role mapping, consider the following points:

- If you are using Active Directory, you must also assign roles to entities or users, before upgrading from a previous AOS version.
- You can create a role map for each authorized directory.
- You can create multiple maps that apply to a single directory. When a directory has multiple maps, the most specific rule for a user is granted.

For example, if you assign a cluster admin role to a group, everyone in that group is granted a cluster admin access. However, you can override that access for specific individuals within the group. If you also assign a viewer role to certain users in the group, those specific users only have viewer access, while everyone else in the group has the cluster admin access.

- When role mapping is not defined for an authorized service directory, all users in that directory receive full administrator permissions. After you create a role map, users who are not explicitly assigned permissions through the mapping are denied access.

Configuring a Role Mapping

Configure a role mapping for directory service users.

About this task

To configure the role mapping, follow these steps:

Procedure

1. Log in to Prism Element web console as an administrator.
2. From the main menu, click the gear icon.
3. On the left navigation bar, select **Role Mapping**.

4. Click **+ New Mapping** and enter the following information:

- **Directory or Provider:** From the dropdown list, select the identity provider.
- **Type:** From the dropdown list, select the entity type.
- **Role:** Choose one of the following roles:
 - **Viewer:** Users are granted view-only access.
 - **User Admin:** Users can perform administrative tasks, view system information, and manage user accounts.
 - **Cluster Admin:** Users can perform administrative tasks and view system information, but cannot manage user accounts.
 - **Backup Admin:** Users can perform backup-related administrative tasks.

Note: After updating to AOS 6.0 or later, users and user groups with Active Directory (AD) **Backup Admin** role assigned cannot log in to Prism Element using their AD credentials or access v3 APIs. For more information, see [KB-14105](#).

- **Values:** Enter the entity names in a comma separated list without spaces.

Note: The entity names are case-sensitive.

The values represent the actual names of organizational units (OUs), groups, or users assigned to the specified role. If you specify an organizational unit, all users within the unit are included. For groups, all users of the group are included. For users, each named user is included.

For example, the system grants the cluster admin role to all users in both the admin-gp and support-gp groups when you select Group as the **Type**, Cluster Admin as the **Role**, and enter admin-gp,support-gp in the **Values** field.

Do not include the domain name in the **Values** field. For example, enter admin-gp not admin-gp@nutanix.com. However, users must include their domain when logging into the Prism Element web console.

The AD User Principal Name (UPN) must follow the `user@domain_name` format.

When configuring role mapping with Active Directory forest setup, administrators can map users with the same name from any domain within the forest. Nutanix recommends configuring role mapping with Active Directory setup that uses a specific domain.

5. Click **Save**.

Editing a Role Mapping

Edit a role mapping that you created for the directory service users.

About this task

To edit the role mapping, follow these steps:

Procedure

1. Log in to Prism Element web console as an administrator.
2. From the main menu, click the gear icon.
3. On the left navigation bar, select **Role Mapping**.

4. Click the pencil icon associated with the role.
5. Edit the fields in the **Update Role Mapping** window as needed and click **Save**.

Deleting a Role Mapping

Delete a role mapping that you created for the directory service users.

About this task

To delete a role mapping, follow these steps:

Procedure

1. Log in to Prism Element web console as an administrator.
2. From the main menu, click the gear icon.
3. On the left navigation bar, select **Role Mapping**.
4. Click the x icon associated with the role.
5. Click **OK**.

Local User Account in Prism Element

Prism Element allows you to create and manage local user accounts.

During the cluster deployment process, an **admin** user account is created by default. However, you can create additional local user accounts by granting specific permissions for accessing Prism Element.

Local User Account Considerations

Before creating a local user account, consider the following points:

- A local user account cannot SSH into CVM and PCVM. SSH access to CVM and PCVM is limited only to the built-in **nutanix** and **admin** user accounts.
- By default, no permissions are granted to a local user account. After creating a local user account, you must specify the custom or built-in roles and permissions.
- A local user account does not have password expiration policy.

Creating a Local User Account in Prism Element

You can create a local user account in Prism Central.

About this task

To create the local user account in Prism Element, follow these steps:

Procedure

1. Log in to Prism Element web console as an administrator.
2. From the main menu, click the gear icon.
3. On the left navigation bar, select **Local User Management**.

4. Click **+ New User** and enter the following information:

- **Username:** Enter a user name.
- **First Name:** Enter a first name.
- **Last Name:** Enter a last name.
- **Email:** Enter a valid user email address.

Note: The system uses the email address for client authentication and logging when the local user performs user and cluster tasks in the web console.

- **Password:** Enter a password.

Ensure that your password meets the following requirements:

- At least eight characters long
 - At least one lowercase letter
 - At least one uppercase letter
 - At least one digit
 - At least one special character (allowed special characters are: "\$%&'()*+,-./:;<=>@[\\]^_`{|}~!\\")
 - At least four characters different from the old password
 - Must not be among the last five passwords
 - Must not have more than two consecutive occurrences of a character
 - Must contain at least one character from each of the following four character classes: uppercase letters, lowercase letters, digits, and special characters
 - Must not contain the words "nutanix", "ntnx", "password", or any simple dictionary words.
- **Language:** From the dropdown list, select the language setting for the user.
For example, if you select **zh-CN**, the user interface displays in Simplified Chinese when the user logs in Prism Element.
 - **Roles:** Choose one of the following role checkboxes:

- **User Admin:** Users can perform administrative tasks, view system information, and manage local user accounts. The user admin role has the highest level of privilege among the other local user roles.

Note: When you select the **User Admin** checkbox, the system automatically selects the **Cluster Admin** and **Backup Admin** checkboxes.

- **Cluster Admin:** Users can perform administrative tasks and view system information, but cannot manage local user accounts.

Note: When you select the **Cluster Admin** checkbox, the system automatically selects the **Backup Admin** checkbox.

- **Backup Admin:** Users can perform Nutanix Mine-related administrative tasks.

The Backup Admin role supports Nutanix Mine integrations starting with AOS version 5.19 and offers minimal functionality in cluster management. For more information about the Backup Admin role and its limitations in Nutanix Mine, see [Backup Admin Role Capabilities](#) on page 101.

If you deselect all the role checkboxes, users are granted view-only access.

5. Click **Save**.

Backup Admin Role Capabilities

Understand the limited permissions and read-only access of Backup Admin users in Nutanix Mine clusters.

Nutanix introduced the Backup Admin role for Mine integrations starting with AOS 5.19 version. The Backup Admin role has minimal functionality in cluster management and restricted access to Nutanix Mine clusters.

Users with the Backup Admin role have the following limited functionality:

- View-only access:
 - **Health, Analysis, and Tasks:** View in read-only mode.
 - **Cluster, Hardware, and Networking:** View configuration details but cannot expand the cluster, remove hosts, or modify network settings.
 - **Alerts and Events:** View all alerts and events but cannot acknowledge or resolve them.
- Available Actions:
 - **VMs:** Power VMs on and off.
 - **Image Management:** Upload new images from the **Settings** page.
- Unavailable features and actions:
 - Creating VMs or configuring their networks.
 - Accessing File Server and Data Protection options.
 - Configuring Alert Policies or Email settings.
 - Accessing the Cluster Registration widget.

Editing a Local User Account in Prism Element

You can edit a local user account settings in Prism Element.

About this task

To edit the local user account, follow these steps:

Procedure

1. Log in to Prism Element web console as an administrator.
2. From the main menu, click the gear icon.
3. On the left navigation bar, select **Local User Management**.
4. Click the pencil icon next to the local user account.
5. Edit the fields in the **Update User** window as needed and click **Save**.

Deleting a Local User Account in Prism Element

You can delete a local user account in Prism Element.

About this task

To delete the local user account, follow these steps:

Procedure

1. Log in to Prism Element web console as an administrator.
2. From the main menu, click the gear icon.
3. On the left navigation bar, select **Local User Management**.
4. Click the remove icon next to the local user account.
5. Click **Delete**.

Disabling Login Access to a Local User Account in Prism Element

By default, login access is enabled for a local user account. You can disable login access to the local user account in Prism Element.

About this task

To disable login access to the local user account, follow these steps:

Procedure

1. Log in to Prism Element web console as an administrator.
2. From the main menu, click the gear icon.
3. On the left navigation bar, select **Local User Management**.
4. Under the **Enabled** column, click the **Yes** toggle button.
The toggle button displays **No**, indicating that the local user account login access is disabled.

Resetting a Local User Account Password Using nCLI

Reset a local user account password using the Nutanix command-line interface (nCLI).

About this task

To reset the local user account password using nCLI, follow these steps:

Procedure

1. Do one of the following:
 - » SSH into any of the Controller VMs (CVMs) as a **nutanix** user:

```
$ ssh nutanix@cvm_ip_address
```
 - » SSH into Prism Central VM (PCVM) as a **nutanix** user:

```
$ ssh nutanix@pcvm_ip_address
```

2. Reset the local user account password:

```
» nutanix@cvm$ ncli user reset-password user-name='user_name'
password='new_password'
```

```
» nutanix@pcvm$ ncli user reset-password user-name='user_name'
password='new_password'
```

- Replace `'user_name'` with the local user account name.
- Replace `'new_password'` with the new password.

Ensure that your password meets the following requirements:

- At least eight characters long
- At least one lowercase letter
- At least one uppercase letter
- At least one digit
- At least one special character (allowed special characters are: "\$%&'()*+,-./:;<=>@[!^_`{|}~!\)
- At least four characters different from the old password
- Must not be among the last five passwords
- Must not have more than two consecutive occurrences of a character
- Must contain at least one character from each of the following four character classes: uppercase letters, lowercase letters, digits, and special characters
- Must not contain the words "nutanix", "ntnx", "password", or any simple dictionary words.

Configuring Your Profile

Configure your profile information in Prism Element.

In the Prism Element web console, you can manage your profile settings by updating personal information and credentials. You can modify your first name, last name, email address, and preferred language. You can also change your password to strengthen security, keeping your account details accurate and ensuring reliable access within the environment.

Updating Your Profile

You can update your profile information in Prism Element.

About this task

To update your profile information in Prism Element, follow these steps:

Procedure

1. Log in to Prism Element web console as an administrator.
2. Click **User Menu > Update Profile**

3. On the **Update Profile** window, enter the following information:
 - **First Name:** Enter a first name.
 - **Last Name:** Enter a last name.
 - **Email Address:** Enter a valid user email address.
 - **Language:** From the dropdown list, select the language setting.
4. Click **Save**.

Changing Your Password

You can change your profile's password in Prism Element.

About this task

To change your profile's password in Prism Element, follow these steps:

Procedure

1. Log in to Prism Element web console as an administrator.
2. Click **User Menu > Change Password**
3. On the **Change Password** window, enter the following information:
 - **Current Password:** Enter the current password.
 - **New Password:** Enter the new password.
 - **Confirm Password:** Re-enter the new password.
4. Click **Save**.

Exporting an SSL Certificate for Third-Party Backup Applications

You can export an SSL certificate from a CVM to use it with third-party backup applications.

About this task

To export the SSL certificate from a CVM, follow these steps:

Procedure

1. SSH into any of the Controller VMs (CVMs) as a **nutanix** user:

```
$ ssh nutanix@cvm_ip_address
```

2. Run the following command to obtain the virtual IP address of the cluster:

```
nutanix@cvm$ ncli cluster info
```

Output:

```
Cluster Id       : 0001ab12-abcd-efgh-0123-012345678m89::123456
Cluster Uuid     : 0001ab12-abcd-efgh-0123-012345678m89
Cluster Name     : three
Cluster Version  : 6.0
Cluster Full Version : el7.3-release-fraser-6.0-a0b1cd...
External IP address : 10.10.10.10
Node Count       : 3
```

```
Block Count      : 1
. . . . .
```

The external IP address in the output is the virtual IP address of the cluster.

- Run the following command to enter into the Python prompt:

```
nutanix@cvm$ python
```

- Run the following command to import the SSL library:

```
>>> import ssl
```

- Run the following command to print the SSL certificate:

```
>>> print(ssl.get_server_certificate(('virtual_IP_address',9440),
ssl_version=ssl.PROTOCOL_TLSv1_2))
```

Example:

```
>>> print(ssl.get_server_certificate(('10.10.10.10', 9440),
ssl_version=ssl.PROTOCOL_TLSv1_2))
```

Example Output:

[illegible]

SSL Certificate Management in Prism Element

Manage SSL certificates in Prism Element to ensure secure access.

Prism Element web console supports SSL certificate-based authentication for console access. To enable secure communication with a cluster, Prism Element web console includes a default self-signed SSL certificate. You can replace the default self-signed SSL certificate with your own self-signed SSL certificate or a certificate authority (CA) signed SSL certificate.

For production purposes, Nutanix recommends that you replace the default self-signed certificate with a CA signed SSL certificate.

Note: You can import only a cluster-wide SSL certificate in Prism Element web console. The SSL certificate cannot be customized for an individual controller VM (CVM). Nutanix recommends that you check for the validity of the certificate periodically and replace the certificate if it is invalid.

Generating a Self-signed SSL Certificate with Subject Alternative Name

Generate a self-signed SSL certificate with Subject Alternative Name (SAN) using OpenSSL to secure Prism Element.

Before you begin

- The certificate import process validates that the key and certificate pair uses the correct signature algorithm to comply with NIST SP800-131a and RFC 6460 (NSA Suite B) standards. Use the correct key types, sizes, curves, and signature algorithms. For more information, see [Supported Key Configurations for SSL Certificates](#) on page 116.
- Nutanix recommends including a DNS name for all CVMs in the self-signed SSL certificate using the SAN extension to avoid SSL certificate errors when accessing a CVM through its DNS name instead of the shared cluster IP address.

About this task

To generate a self-signed SSL certificate with SAN, follow these steps:

Procedure

1. SSH into any of the Controller VMs (CVMs) as a `nutanix` user:

```
$ ssh nutanix@cvm_ip_address
```

2. Generate a private key:

- RSA private key with a bit length of 2048:

```
nutanix@cvm$ openssl genrsa -out my_key_name.key 2048
```

- RSA private key with a bit length of 4096:

```
nutanix@cvm$ openssl genrsa -out my_key_name.key 4096
```

- ECDSA private key using the prime256v1 curve:

```
nutanix@cvm$ openssl ecparam -name prime256v1 -genkey -out my_key_name.pem
```

Important: While you are generating the private key, ensure that the private key is not password protected.

3. Generate the Certificate Signing Request (CSR):

- For RSA 2048 and RSA 4096 private keys:

```
nutanix@cvm$ openssl req -new -nodes -key my_key_name.key -signature_algorithm -  
out my_csr_name.csr
```

signature_algorithm: Specify sha256 or sha384 or sha512.

For example, to generate a CSR for RSA 2048 private key using SHA-256 signature algorithm:

```
nutanix@cvm$ openssl req -new -nodes -key my_key_name.key -sha256 -out  
my_csr_name.csr
```

- For ECDSA 256 private key:

```
nutanix@cvm$ openssl req -new -nodes -key my_key_name.pem -sha256 -out  
my_csr_name.csr
```

For example, to generate a CSR for ECDSA 256 private key using SHA-256 signature algorithm:

```
nutanix@cvm$ openssl req -new -nodes -key my_key_name.pem -sha256 -out  
my_csr_name.csr
```

4. Enter the information in the command output to incorporate into your certificate request:

```
You are about to be asked to enter information that will be incorporated  
into your certificate request.  
What you are about to enter is what is called a Distinguished Name or a DN.  
There are quite a few fields but you can leave some blank  
For some fields there will be a default value,  
If you enter '.', the field will be left blank.  
-----  
Country Name (2 letter code) []:# Enter the 2 letter country code  
State or Province Name (full name) []:# Enter the full name of the state or province  
Locality Name (eg, city) []:# Enter the name of the city or locality  
Organization Name (eg, company) []: # Enter the legal name of your organization or  
company  
Organizational Unit Name (eg, section) []: # Enter the department or business unit  
Common Name (eg, fully qualified host name) []:# Enter the fully qualified domain  
name (FQDN) of the server  
Email Address []:# Enter a valid email address  
Please enter the following 'extra' attributes  
to be sent with your certificate request  
A challenge password []: # (Optional) Enter your password
```

5. Create a configuration file in your home directory with your preferred text editor named *san.cnf* that contains the following text:

```
[req]  
distinguished_name = req_distinguished_name  
req_extensions = v3_req  
  
[req_distinguished_name]  
[v3_req]  
basicConstraints = CA:FALSE  
keyUsage = nonRepudiation, digitalSignature, keyEncipherment  
subjectAltName = @alt_names  
  
[alt_names]  
DNS.0 = example1.domain.com # Primary domain or fully qualified domain name (FQDN) of  
the server  
DNS.1 = example2.domain.com # Secondary domain or alias for the server  
DNS.2 = example3.domain.com # Additional domain name that should be trusted
```

```
DNS.3 = *.domain.com # Wildcard entry to match any subdomain under domain.com
IP.0 = x.x.x.x # IP address of the CVM
IP.1 = y.y.y.y # Additional IP address of the CVM
```

[*alt_names*]

Specify your DNS and IP addresses. If you have a range of hosts, use wildcards (*) to match any subdomain of the domain name.

6. Generate a self-signed certificate:

```
nutanix@cvm$ openssl x509 -req -days number_of_days -in my_csr_name.csr -signkey
my_key_name.key -out my_crt_name.crt -signature_algorithm -extensions v3_req -
extfile san.cnf
```

number_of_days: Specify the number of days until a newly generated certificate expires.

signature_algorithm: Specify sha256, sha384, or sha512. Ensure that you use the same signature algorithm that was used to generate the CSR.

Example:

```
nutanix@cvm$ openssl x509 -req -days 1460 -in my_csr_name.csr -signkey
my_key_name.key -out my_crt_name.crt -sha256 -extensions v3_req -extfile san.cnf
```

7. Copy *my_key_name.key* and *my_crt_name.crt* from the CVM to your local machine:

```
nutanix@cvm$ scp my_key_name.key my_crt_name.crt username@local-machine:/
local_file_path/
```

8. Log out of the CVM.

What to do next

After generating the self-signed certificate with a private key, follow the procedure described in [Importing a Self-Signed SSL Certificate in Prism Element](#) on page 112 to replace the default certificate with your self-signed SSL certificate. The following table lists certificate components and its corresponding file type to choose when SSL certificate window prompts:

Table 7: SSL Certificate Import Files

Certificate Components	File type
Private Key	my_key_name.key
Public Certificate	my_crt_name.crt
CA Certificate/Chain	my_crt_name.crt

Generating a Certificate Signing Request with Subject Alternative Name for Submission to Certificate Authority

Generate a Certificate Signing Request (CSR) with Subject Alternative Name (SAN) using OpenSSL for Certificate Authority (CA) submission.

Before you begin

- The certificate import process validates that the key and certificate pair uses the correct signature algorithm to comply with NIST SP800-131a and RFC 6460 (NSA Suite B) standards. Use the correct key types, sizes, curves, and signature algorithms. For more information, see [Supported Key Configurations for SSL Certificates](#) on page 116.

- Nutanix recommends including a DNS name for all CVMs in the CSR using the SAN extension to avoid SSL certificate errors when accessing a CVM through its DNS name instead of the shared cluster IP address.

About this task

To generate a CSR with SAN, follow these steps:

Procedure

1. SSH into any of the Controller VMs (CVMs) as a `nutanix` user:

```
$ ssh nutanix@cvm_ip_address
```

2. Create a configuration file in your home directory with your preferred text editor named `ssl.cnf` that contains the following text:

```
[req]
distinguished_name = req_distinguished_name
req_extensions = v3_req
prompt = no

[req_distinguished_name]
countryName = # Country Name (2 letter code)
stateOrProvinceName = # State or Province Name (full name)
localityName = # Locality Name (eg, city)
organizationName = # Organization Name (eg, company)
organizationalUnitName = # Organizational Unit Name (eg, BU)
commonName = # Common Name (e.g. server FQDN or YOUR name)
emailAddress = # Email Address

[v3_req]
subjectAltName = @alt_names

[alt_names]
DNS.0 = example1.domain.com # Primary domain or fully qualified domain name (FQDN) of
the server
DNS.1 = example2.domain.com # Secondary domain or alias for the server
DNS.2 = example3.domain.com # Additional domain name that should be trusted
DNS.3 = *.domain.com # Wildcard entry to match any subdomain under domain.com
IP.0 = x.x.x.x # IP address of the CVM
IP.1 = y.y.y.y # Additional IP address of the CVM
```

[alt_names] - Specify your DNS and IP addresses. If you have a range of hosts, use wildcards (*) to match any subdomain of the domain name.

3. Generate a private key:

- RSA private key with a bit length of 2048:

```
nutanix@cvm$ openssl genrsa -out my_key_name.key 2048
```

- RSA private key with a bit length of 4096:

```
nutanix@cvm$ openssl genrsa -out my_key_name.key 4096
```

- ECDSA private key using the prime256v1 curve:

```
nutanix@cvm$ openssl ecparam -name prime256v1 -genkey -out my_key_name.pem
```

Important: While you are generating the private key, ensure that the private key is not password protected.

4. Generate a CSR:

- For RSA 2048 and RSA 4096 private keys:

```
nutanix@cvm$ openssl req -new -nodes -key my_key_name.key -signature_algorithm -  
out my_csr_name.csr -config ssl.cnf
```

signature_algorithm: Specify sha256 or sha384 or sha512.

For example, to generate a CSR for RSA 2048 private key using SHA-256 signature algorithm:

```
nutanix@cvm$ openssl req -new -nodes -key my_key_name.key -sha256 -out  
my_csr_name.csr -config ssl.cnf
```

- For ECDSA 256 private key:

```
nutanix@cvm$ openssl req -new -nodes -key my_key_name.pem -sha256 -out  
my_csr_name.csr -config ssl.cnf
```

For example, to generate a CSR for ECDSA 256 private key using SHA-256 signature algorithm:

```
nutanix@cvm$ openssl req -new -nodes -key my_key_name.pem -sha256 -out  
my_csr_name.csr -config ssl.cnf
```

5. Copy `my_key_name.key` and `my_csr_name.csr` from the CVM to your local machine:

```
nutanix@cvm$ scp my_key_name.key my_csr_name.csr username@local-machine:/  
local_file_path/
```

6. Log out of the CVM.

7. Send your CSR file to the Certificate Authority (CA) of your choice.

After receiving your CSR, the CA sends the following files:

- CA signed public certificate
- CA's public certificate
- Root CA public certificate (if the CA is intermediate)

The issuing CA validates the public certificate. If the issuing CA is intermediate, the root CA validates the issuing CA certificate. The system validates the certificate chain to establish trust.

8. Download all the certificate files received from CA to the local file directory.

9. (Optional) If the CA chain certificate provided by the certificate authority is not in a single file, run the following command to concatenate the list of CA certificates into a chain file:

```
$ cat intermediateCAcert.crt rootCAcert.crt > ca_chain_certs.crt
```

Start the chain with the signer's certificate and end it with the root CA certificate.

Ensure that the chain file only has the root and intermediate certificates. Prism Element fails to import the chain file if it contains public or private certificates.

What to do next

Follow [Importing a Self-Signed SSL Certificate in Prism Element](#) on page 112 section to replace the default certificate with a CA signed certificate. The following table lists certificate components and its corresponding file type to choose when SSL certificate window prompts:

Table 8: SSL Certificate Import Files

Certificate Components	File type
Private Key	my_key_name.key
Public Certificate	ca_signed_public_cert.cer
CA Certificate/Chain	ca_public_cert.crt or ca_chain_certs.crt

Verifying the Certificate Generation Request

Run the following commands to verify the certificate generation request.

- Verify that the CA certificate chain is valid:

```
nutanix@cvm$ openssl verify -CAfile ca_chain_certs.crt myPublicCert.cer
```

Example output:

```
myPublicCert.cer: OK
```

- Verify the private key and signature algorithm details:

```
nutanix@cvm$ openssl x509 -in my_cert_name.crt -text -noout | grep -i 'rsa\|ecdsa\|Public'
```

Example output:

```
Signature Algorithm: ecdsa-with-SHA256
      Subject Public Key Info:
        Public Key Algorithm: id-ecPublicKey
        Public-Key: (256 bit)
        Signature Algorithm: ecdsa-with-SHA256
```

- Verify that the CA certificate chain uses SHA 256 as the signature algorithm:

```
nutanix@cvm$ openssl crl2pkcs7 -nocrl -certfile ca_chain_certs.crt | openssl pkcs7 -print_certs -noout -text | grep -Ew '(Subject|Issuer|Signature Algorithm):' | grep -C1 Issuer
```

Troubleshooting the Certificate Generation Request

The following troubleshooting tips can help you resolve common issues that can occur when generating certificates.

Chain certificate format

If your chain certificate file has public or private certificates, it will fail to import in Prism Element web console. Ensure that the chain certificate file only has the root and intermediate certificates.

For example, if a public certificate is present in a chain file, you can remove it by opening your chain file in your preferred text editor. Ensure that there are no extra white spaces at the bottom of the file.

DER-encoded certificate issue

If the certificate is DER encoded, it fails to import in Prism Element web console. You can resolve the issue by converting it to PEM-encoded ASCII format.

- Ensure that the certificate is DER encoded:

```
nutanix@cvm$ openssl x509 -in cert.crt -inform der -text -noout
```

- If the certificate is DER encoded, run the following command to convert the certificate from DER to PEM-encoded ASCII format:

```
nutanix@cvm$ openssl x509 -in certDER.crt -inform der -outform pem -out cert.crt
```

Certificate format

Ensure that all the certificates do not have any extra data (or custom attributes) before the beginning (-----BEGIN CERTIFICATE-----) or after the end (-----END CERTIFICATE-----) of the block.

Importing a Self-Signed SSL Certificate in Prism Element

Import a self-signed SSL certificate into Prism Element.

Before you begin

Ensure that you generate a self-signed SSL certificate. For information, see [Generating a Self-signed SSL Certificate with Subject Alternative Name](#) on page 106.

About this task

To import a self-signed SSL certificate into Prism Element, follow these steps:

Procedure

1. Log in to Prism Element web console as an administrator.
2. From the main menu, click the gear icon.
3. On the left navigation bar, select **SSL Certificate**.
4. Click **Replace Certificate**.
5. Select **Import Key and Certificate** and click **Next**.

6. On the SSL certificate window, provide the following information:

- **Private Key Type:** From the dropdown list, select the appropriate private key type for the self-signed certificate.

Note: EC DSA 521 and EC DSA 384 bit private keys are not supported.

- **Private Key:** Click **Choose file** and select the private key.

Note: The private key that you import must be unencrypted. Password-protected private keys are not supported.

- **Public Certificate:** Click **Choose file** and select the self-signed certificate corresponding to the private key.
- **CA Certificate/Chain:** Click **Choose file** select the self-signed certificate corresponding to the private key.

SSL Certificate ?

Note: Any change made to a certificate will restart the prism session and the user will be logged out automatically.

Guidelines for RSA: Please use a SHA-256, SHA-384, or SHA-512 signature algorithm with RSA certificates for security and performance.

Private Key Type

RSA 2048 bit

Private Key ?

Choose file my_key_name.key

Public Certificate

Choose file my crt_name.crt

CA Certificate/Chain

Choose file my crt_name.crt

Cancel Import Files

Figure 10: Importing self-signed certificate

The following table lists certificate components and its corresponding file type to choose when SSL certificate window prompts:

Certificate Components	File type
Private Key	my_key_name.key
Public Certificate	my crt_name.crt
CA Certificate/Chain	my crt_name.crt

7. Click **Import Files**.

Note: Prism Element stores only one custom SSL certificate. When you upload a new certificate, Prism Element replaces the existing certificate.

After you import the new certificate, the Prism Element web console restarts. If the certificate and credentials are valid, Prism Element uses the new certificate immediately. All open browser sessions becomes invalid until you reload the page and accept the new certificate. If the certificate is invalid due to corrupted file or wrong certificate type, Prism Element discards the new certificate and reverts to the default certificate provided by Nutanix.

Importing CA-Signed SSL Certificate in Prism Element

Import a certificate authority (CA) signed SSL certificate into Prism Element.

Before you begin

Ensure that you generate a Certificate Signing Request (CSR) for submission to a CA. For more information, see [Generating a Certificate Signing Request with Subject Alternative Name for Submission to Certificate Authority](#) on page 108.

About this task

To import a CA-signed SSL certificate into Prism Element, follow these steps:

Procedure

1. Log in to Prism Element web console as an administrator.
2. From the main menu, click the gear icon.
3. On the left navigation bar, select **SSL Certificate**.
4. Click **Replace Certificate**.
5. Select **Import Key and Certificate** and click **Next**.

6. On the SSL certificate window, provide the following information:

- **Private Key Type:** From the dropdown list, select the appropriate private key type for the CA signed certificate.

Note: Prism Central does not support EC DSA 521 and EC DSA 384 bit private keys.

- **Private Key:** Click **Choose file** and select the private key.

Note: The private key that you import must be unencrypted. Password-protected private keys are not supported.

- **Public Certificate:** Click **Choose file** and select the CA signed public portion of the certificate corresponding to the private key.

- **CA Certificate/Chain:** Click **Choose file** and select the certificate or chain of the signing authority for the public certificate.

Note: For more information on how to create a chain file from the list of CA certificates, see [Generating a Certificate Signing Request with Subject Alternative Name for Submission to Certificate Authority](#) on page 63.

SSL Certificate ?

Note: Any change made to a certificate will restart the prism session and the user will be logged out automatically.

Guidelines for RSA: Please use a SHA-256, SHA-384, or SHA-512 signature algorithm with RSA certificates for security and performance.

Private Key Type

RSA 2048 bit

Private Key ⓘ

Choose file my_key_name.key

Public Certificate

Choose file ca_signed_public_cert.cer

CA Certificate/Chain

Choose file ca_public_cert.crt

Cancel Import Files

Figure 11: Importing CA-signed certificate

The following table lists certificate components and its corresponding file type to choose when SSL certificate window prompts:

Table 9: CA Signed Certificate File Type

Certificate Components	File type
Private Key	my_key_name.key
Public Certificate	ca_signed_public_cert.cer
CA Certificate/Chain	ca_public_cert.crt or ca_chain_certs.crt

7. Click **Import Files**.

Note: Prism Element stores only one custom SSL certificate. When you upload a new certificate, Prism Element replaces the existing certificate.

After you import the new certificate, the Prism Element web console restarts. If the certificate and credentials are valid, Prism Element uses the new certificate immediately. All open browser sessions becomes invalid until you reload the page and accept the new certificate. If the certificate is invalid due to corrupted file or wrong certificate type, Prism Element discards the new certificate and reverts to the default certificate provided by Nutanix.

Regenerating Self-Signed Certificate in Prism Element

Regenerate a self-signed SSL certificate in Prism Element.

About this task

To regenerate a self-signed certificate in Prism Element, follow these steps:

Procedure

1. Log in to Prism Element web console as an administrator.
2. From the main menu, click the gear icon.
3. On the left navigation bar, select **SSL Certificate**.
4. Click **Replace Certificate**.
5. Select **Regenerate Self Signed Certificate** and click **Apply**.
6. Click **OK**.
Prism Element generates and applies a new RSA 2048 bit self-signed certificate.

Supported Key Configurations for SSL Certificates

Nutanix supports only the following key types, sizes or curves, and signature algorithms for SSL certificates:

Table 10: SSL Certificate Key Type Options

Key Type	Size or Curve	Signature Algorithm
RSA	4096	SHA-256, SHA-384 or SHA512
RSA	2048	SHA-256, SHA-384 or SHA512
EC DSA 256	prime256v1	ecdsa-with-sha256

In Prism Central versions prior to pc.2024.3, uploading an RSA 4096-bit certificate might cause some issues. For more information, see [KB 12775](#).

Availability Zone (AZ) pairing between Prism Central clusters supports RSA 2048-bit certificates only. RSA 4096-bit certificates are not supported.

Nutanix does not support SHA-1 certificates, including root CAs.

Client and CAC authentication support only RSA 2048-bit certificates.

Cluster Lockdown in Prism Element

Nutanix supports key-based SSH access to Prism Element.

Cluster lockdown in Prism Element enhances security by disabling the password-based SSH login. Instead, authorized user SSH keys provide access to Prism Element, which improves the overall security posture.

Nutanix supports the following key-based SSH encryption algorithms:

- AES128-CTR
- AES192-CTR
- AES256-CTR

Nutanix supports the following key types:

- RSA
- ECDSA

Cluster Lockdown Considerations

Before you configure the cluster lockdown in Prism Element, consider the following points:

- Generate the public key using the `ssh-keygen` command on Mac or Linux, or the PuTTY application on Windows. For more information, see [KB-1895](#).
- When you enable the cluster lockdown, Prism Element does not store the passwords for both the CVM and host, and you cannot change these passwords to access the cluster resources.
- Adding a user SSH key enables SSH access for both the `nutanix` and `admin` accounts on the CVM and host.
- Disabling remote login and deleting all user SSH keys locks down cluster SSH access.
- You can configure multiple user SSH keys.
- For additional security, you can configure SSH security level for the CVM. For more information, see [CVM Security Hardening](#) on page 153.

Configuring Cluster Lockdown in Prism Element

Configure cluster lockdown to enable SSH key-based access to your cluster.

Before you begin

Ensure that you generate the public key using `ssh-gen` command on Mac or Linux, or the PuTTY application on Windows. For more information, see [KB-1895](#).

About this task

To configure the cluster lockdown in Prism Element, follow these steps:

Procedure

1. Log in to Prism Element web console as an administrator.
2. From the main menu, click the gear icon.
3. On the left navigation bar, select **Cluster Lockdown**.
4. Deselect the **Enable Remote Login with Password** checkbox.
5. Click **+ New user SSH Key** and enter the following information:
 - **Name:** Enter a key name.
 - **Key:** Paste the public key value.
6. Click **Save**.

Deleting a User SSH Key in Prism Element

You can disable the SSH key-based access to your cluster.

About this task

Caution: Disabling remote login and deleting all user SSH keys locks down the cluster SSH access. Ensure to select the **Enable Remote Login with Password** checkbox before deleting all the user SSH keys.

To remove the user SSH key, follow these steps:

Procedure

1. Log in to Prism Element web console as an administrator.
2. From the main menu, click the gear icon.
3. On the left navigation bar, select **Cluster Lockdown**.
4. Select the **Enable Remote Login with Password** checkbox.
5. Click the **X** remove icon.
6. Select **OK**.

Data-at-Rest Encryption

Nutanix provides an option to secure data while it is at rest using either self-encrypted drives or software-only encryption and key-based access management (cluster's native or external KMS for software-only encryption).

Encryption Methods

Nutanix provides you with the following options to secure your data.

- **Self Encrypting Drives (SED) Encryption** - You can use a combination of SEDs and an external KMS to secure your data while it is at rest.
- **Software-only Encryption** - Nutanix AOS uses the AES-256 encryption standard to encrypt your data. Once enabled, software-only data-at-rest encryption cannot be disabled, thus protecting against accidental data leaks due to human errors. Software-only encryption supports both Nutanix Native Key Manager (local and remote) and External KMS to secure your keys.

Note the following points regarding data-at-rest encryption.

- Encryption is supported for AHV, ESXi, and Hyper-V.
 - For ESXi and Hyper-V, software-only encryption can be implemented at a cluster level or container level.
 - For AHV, encryption can be implemented at the cluster level, VM level, or VG level. For more information about VM or VG level encryption, see [Storage Policy Based Encryption](#).
- Nutanix recommends using cluster-level encryption. With the cluster-level encryption, the administrative overhead of selecting different containers for the data storage gets eliminated.
- Encryption cannot be disabled once it is enabled at a cluster level or container level.
- Encryption can be implemented on an existing cluster with data that exists. If encryption is enabled on an existing cluster (AHV, ESXi, or Hyper-V), the unencrypted data is transformed into an encrypted format in a low priority background task that is designed not to interfere with other workload running in the cluster.
- Data-at-rest encryption choices can be implemented at the entity level using storage policies in Prism Central. Deployments can continue to have data-at-rest encryption capabilities scoped for the entire cluster. Storage policy provides the additional option to control the encryption scope decisions at the entity (VM or VG) level. For more information, see [Storage Policies Based Encryption](#) in the *Prism Central Infrastructure Guide*
- Data can be encrypted using either self-encrypted drives (SEDs) or software-only encryption. You can change the encryption method from SEDs to software-only. You can perform the following configurations.
 - For ESXi and Hyper-V clusters, you can switch from SEDs and External Key Management (EKM) combination to software-only encryption and EKM combination. First, you must disable the encryption in the cluster where you want to change the encryption method. Then, select the cluster and enable encryption to transform the unencrypted data into an encrypted format in the background.
 - For AHV, background encryption is supported.
- Once the task to encrypt a cluster begins, you cannot cancel the operation. Even if you stop and restart the cluster, the system resumes the operation.
- In the case of mixed clusters with ESXi and AHV nodes, where the AHV nodes are used for storage only, the encryption policies consider the cluster as an ESXi cluster. So, the cluster-level and container-level encryption are available.
- You can use a combination of SED and non-SED drives in a cluster. After you encrypt a cluster using the software-only encryption, all the drives are considered as unencrypted drives. In case you switch from the SED encryption to the software-only encryption, you can add SED or non-SED drives to the cluster.
- Data-at-rest encryption is configured per cluster. Data is encrypted as part of the write operation and decrypted as part of the read operation. During the replication process, the source cluster reads and decrypts the data before sending it to the destination cluster, where the data is encrypted at rest only if the destination cluster also has data-at-rest encryption enabled. Therefore, you must enable encryption on each cluster where data must be encrypted at rest. Starting with AOS 7.3, AOS encrypts replication traffic between clusters by default. For more information, see [Replication Traffic Encryption](#) in the *Data Protection and Recovery with Prism Element Guide*.
- Software-only encryption does not impact most of data efficiency features such as deduplication, compression, zero block suppression, and so on. The software encryption is the last data transformation performed. For example, during the write operation, compression is performed first, followed by encryption.
- Software-only encryption requires additional space if erasure coding (EC) is enabled on the cluster (or container). The additional space requirement is temporary, and EC space savings are restored once the encryption process is complete.
- Enabling encryption on EC-enabled clusters involves completely decoding EC-encoded data and re-encoding the data for EC. This process of decoding and re-encoding requires additional space on the cluster. Encryption

is initiated only if the aggregate of EC-based space saving and current space usage is less than 85 percent of the cluster capacity. Additionally, the cluster continuously checks if the aggregate space usage stays below 85 percent for the encryption to progress.

Key Management

Nutanix supports a Native Key Management Server, also called Local Key Manager (LKM), thus avoiding the dependency on an External Key Manager (EKM). Cluster localised Key Management Service support requires a minimum of 3-node in a cluster and is supported only for software-only encryption. So, 1-node and 2-node clusters can use either the Native KMS (remote) option or an EKM. .

The following types of keys are used for encryption.

- Data Encryption Key (DEK): A symmetric key, such as AES-256, that is used to encrypt the data.
- Key Encryption Key (KEK): This key is used to encrypt or decrypt the DEK.

Note the following points regarding the key management.

- Nutanix does not support the use of the Local Key Manager with a third party External Key Manager.
- Dual encryption (both SED and software-only encryption) requires an EKM. For more information, see [Configuring Dual Encryption](#) on page 144.
- You can switch from an EKM to LKM, and inversely. For more information, see [Switching between Native Key Manager and External Key Manager](#) on page 137.
- Rekey of keys stored in the Native KMS is supported for the Master Keys. For more information, see [Changing Key Encryption Keys \(SEDs\)](#) on page 127 and [Changing Key Encryption Keys \(Software Only\)](#) on page 143.
- You must back up the keys stored in the Native KMS. For more information, see [Backing up Keys](#) on page 145.
- You must backup the encryption keys whenever you create a new container or remove an existing container. Nutanix Cluster Check (NCC) checks the status of the backup and sends an alert if you do not take a backup at the time of creating or removing a container.

Data-at-Rest Encryption (SEDs)

For customers who require enhanced data security, Nutanix provides a data-at-rest security option using Self Encrypting Drives (SEDs) included in the Ultimate license.

Note: Starting with AOS 7.3, Nutanix supports NVMe self-encrypting drives (SEDs) for data-at-rest encryption. The support of any NVMe SEDs with a specific platform requires qualification with AOS version 7.3 or later. If you are running the AOS Pro License on G6 platforms and later, you can use SED encryption by installing an add-on license.

Following features are supported:

- Data is encrypted on all drives at all times.
- Data is inaccessible in the event of drive or node theft.
- Data on a drive can be securely destroyed.
- A key authorization method allows password rotation at arbitrary times.
- Protection can be enabled or disabled at any time.
- No performance penalty is incurred despite encrypting all data.
- Re-key of the master encryption key (MEK) at arbitrary times is supported.

Note: If an SED cluster is present, then while executing the data-at-rest encryption, you will get an option to either select data-at-rest encryption using SEDs or data-at-rest encryption using AOS.

Encryption Type

Encrypt the cluster using:

☒ Drive-based encryption

Use the cluster's self-encrypting drives (SEDs) to encrypt all cluster data.

☐ Software-based encryption

Figure 12: SED and AOS Options

This solution provides enhanced security for data on a drive, but it does not secure data in transit.

Data Encryption Model

To accomplish these goals, Nutanix implements a data security configuration that uses SEDs with keys maintained through a separate key management device. Nutanix uses open standards (TCG and KMIP protocols) and FIPS validated SED drives for interoperability and strong security.



Figure 13: Cluster Protection Overview

This configuration involves the following workflow:

1. The security implementation begins by installing SEDs for all data drives in a cluster.

The drives are FIPS 140-2 validated and use FIPS 140-2 validated cryptographic modules.

For OEM platforms, contact the respective OEM partners to confirm FIPS certification on SED qualified drives for AOS.

Creating a new cluster that includes SEDs only is straightforward, but an existing cluster can be converted to support data-at-rest encryption by replacing the existing drives with SEDs (after migrating all the VMs/vDisks off of the cluster while the drives are being replaced).

Note: Contact Nutanix customer support for assistance before attempting to convert an existing cluster. A non-protected cluster can contain both SED and standard drives, but Nutanix does not support a mixed cluster when protection is enabled. All the disks in a protected cluster must be SED drives.

2. Data on the drives is always encrypted but read or write access to that data is open. By default, the access to data on the drives is protected by the in-built manufacturer key. However, when data protection for the cluster is enabled, the Controller VM must provide the proper key to access data on a SED. The Controller VM

communicates with the SEDs through a Trusted Computing Group (TCG) Security Subsystem Class (SSC) Enterprise protocol.

A symmetric data encryption key (DEK) such as AES 256 is applied to all data being written to or read from the disk. The key is known only to the drive controller and never leaves the physical subsystem, so there is no way to access the data directly from the drive.

Another key, known as a key encryption key (KEK), is used to encrypt/decrypt the DEK and authenticate to the drive. (Some vendors call this the authentication key or PIN.)

Each drive has a separate KEK that is generated through the FIPS compliant random number generator present in the drive controller. The KEK is 32 bytes long to resist brute force attacks. The KEKs are sent to the key management server for secure storage and later retrieval; they are not stored locally on the node (even though they are generated locally).

In addition to the above, the master encryption key (MEK) is used to encrypt the KEKs.

Each node maintains a set of certificates and keys in order to establish a secure connection with the external key management server.

3. Keys are stored in a key management server that is outside the cluster, and the Controller VM communicates with the key management server using the Key Management Interoperability Protocol (KMIP) to upload and retrieve drive keys.

Only one key management server device is required, but it is recommended that multiple devices are employed so the key management server is not a potential single point of failure. Configure the key manager server devices to work in clustered mode so they can be added to the cluster configuration as a single entity that is resilient to a single failure.

4. When a node experiences a full power off and power on (and cluster protection is enabled), the controller VM retrieves the drive keys from the key management server and uses them to unlock the drives.

If the Controller VM cannot get the correct keys from the key management server, it cannot access data on the drives.

If a drive is re-seated, it becomes locked.

If a drive is stolen, the data is inaccessible without the KEK (which cannot be obtained from the drive). If a node is stolen, the key management server can revoke the node certificates to ensure they cannot be used to access data on any of the drives.

Preparing for Data-at-Rest Encryption (External KMS for SEDs and Software Only)

About this task

Caution: Do not host a key management server vm on the encrypted cluster that is using it. Doing so could result in complete data loss if there is a problem with the VM while it is hosted in that cluster.

If you are using an external KMS for encryption using AOS, preparation steps outside the web console are required. The information in this section is applicable if you choose to use an external KMS for configuring encryption.

You must install the license of the external key manager for all nodes in the cluster. For a complete list of the supported key management servers, see [Compatibility and Interoperability Matrix](#). For instructions on how to configure a key management server, refer to the documentation from the appropriate vendor.

The system accesses the EKM under the following conditions:

- Starting a cluster
- Regenerating a key (key regeneration occurs automatically every year by default)
- Adding or removing a node (only when Self Encrypting Drives is used for encryption)
- Switching between Native to EKM or EKM to Native

- Starting and restarting a service (only if Software-based encryption is used)
- Upgrading AOS (only if Software-based encryption is used)
- NCC heartbeat check if EKM is alive

Procedure

1. Configure a key management server.

The key management server devices must be configured into the network so the cluster has access to those devices. For redundant protection, it is recommended that you employ at least two key management server devices, either in active-active cluster mode or stand-alone.

Note: The key management server must support KMIP version 1.0 or later.

» SafeNet

Ensure that **Security > High Security > Key Security > Disable Creation and Use of Global Keys** is checked.

» Vormetric

Set the appliance to compatibility mode. Suite B mode causes the SSL handshake to fail.

2. Generate a certificate signing request (CSR) for each node in the cluster.

- The Common Name field of the CSR is populated automatically with `unique_node_identifier.nutanix.com` to identify the node associated with the certificate.
- (Optional) After generating the CSR from Prism, to change the domain name portion of the Common Name (CN) to your own domain, run the following command:

```
nutanix@cvm$ ncli data-at-rest-encryption-certificate update-csr-information
domain-name=abcd.test.com
```

Replace `abcd.test.com` with the actual domain name.

- (Optional) To change the entire CN, contact Nutanix Support.
- A UID field is populated with a value of **Nutanix**. This can be useful when configuring a Nutanix group for access control within a key management server, because it is based on fields within the client certificates.

Note: Some vendors when doing client certificate authentication expect the client username to be a field in the CSR. While the CN and UID are pre-generated, many of the user populated fields can be used instead if desired. If a node-unique field such as CN is chosen, users must be created on a per node basis for access control. If a cluster-unique field is chosen, customers must create a user for each cluster.

3. Send the CSRs to a certificate authority (CA) and get them signed.

» Safenet

The SafeNet KeySecure key management server includes a local CA option to generate signed certificates, or you can use other third-party vendors to create the signed certificates.

To enable FIPS compliance, add user `nutanix` to the CA that signed the CSR. Under **Security > High Security > FIPS Compliance** click **Set FIPS Compliant**.

Note: Some CAs strip the UID field when returning a signed certificate.

To comply with FIPS, Nutanix does not support the creation of global keys.

In the SafeNet KeySecure management console, go to **Device > Key Server > Key Server > KMIP Properties > Authentication Settings**.

Then do the following:

- Set the **Username Field in Client Certificate** option to **UID (User ID)**.
- Set the **Client Certificate Authentication** option to **Used for SSL session and username**.

If you do not perform these settings, the KMS creates global keys and fails to encrypt the clusters or containers using the software only method.

4. Upload the signed SSL certificates (one for each node) and the certificate for the CA to the cluster. These certificates are used to authenticate with the key management server.
5. Generate keys (KEKs) for the SED drives and upload those keys to the key management server.

Configuring Data-at-Rest Encryption (SEDs)

Nutanix offers an option to use self-encrypting drives (SEDs) to store data in a cluster. When SEDs are used, there are several configuration steps that must be performed to support data-at-rest encryption in the cluster.

Before you begin

A separate key management server is required to store the keys outside of the cluster. Each key management server device must be configured and addressable through the network. It is recommended that multiple key manager server devices be configured to work in clustered mode so they can be added to the cluster configuration as a single entity (see step 5) that is resilient to a single failure.

About this task

To configure cluster encryption, do the following:

Procedure

1. Log in to Prism Element as an administrator.
2. Click the gear icon in the main menu and then select **Data at Rest Encryption** in the **Settings** page. The **Data at Rest Encryption** dialog box appears. Initially, encryption is not configured, and a message to that effect appears.
3. Click the **Create Configuration** button.
Clicking the **Continue Configuration** button, **configure it** link, or **Edit Config** button does the same thing, which is display the Data-at-Rest Encryption configuration page.
4. Select the Encryption Type as **Drive-based Encryption**. This option is displayed only when SEDs are detected.

5. In the **Certificate Signing Request Information** section, do the following:

- a. Enter appropriate credentials for your organization in the **Email**, **Organization**, **Organizational Unit**, **Country Code**, **City**, and **State** fields and then click the **Save CSR Info** button.

The entered information is saved and is used when creating a certificate signing request (CSR). To specify more than one **Organization Unit** name, enter a comma separated list.

Note: You can update this information until an SSL certificate for a node is uploaded to the cluster, at which point the information cannot be changed (the fields become read only) without first deleting the uploaded certificates.

- b. Click the **Download CSRs** button, and then in the new screen click the **Download CSRs for all nodes** to download a file with CSRs for all the nodes or click a **Download** link to download a file with the CSR for that node.

- c. Send the files with the CSRs to the desired certificate authority.

The certificate authority creates the signed certificates and returns them to you. Store the returned SSL certificates and the CA certificate where you can retrieve them for step 6.

- The certificates must be X.509 format. (DER, PKCS, and PFX formats are not supported.)
- The certificate and the private key should be in separate files.

6. In the **Key Management Server** section, do the following:

- a. Click the **Add New Key Management Server** button.

- b. In the **Add a New Key Management Server** screen, enter a name, IP address, and port number for the key management server in the appropriate fields.

The port is where the key management server is configured to listen for the KMIP protocol. The default port number is 5696. For the complete list of required ports, see [Port Reference](#).

- » If you have configured multiple key management servers in cluster mode, click the **Add Address** button to provide the addresses for each key management server device in the cluster.
- » If you have stand-alone key management servers, click the **Save** button. Repeat this step (**Add New Key Management Server** button) for each key management server device to add.

If your key management servers are configured into a leader/follower (active/passive) relationship and the architecture is such that the follower cannot accept write requests, do not add the follower into this configuration. The system sends requests (read or write) to any configured key management server, so both read and write access is needed for key management servers added here.

To prevent potential configuration problems, always use the **Add Address** button for key management servers configured into cluster mode. Only a stand-alone key management server should be added as a new server.

- c. To edit any settings, click the pencil icon for that entry in the key management server list to redisplay the add page and then click the **Save** button after making the change. To delete an entry, click the X icon.

7. In the **Add a New Certificate Authority** section, enter a name for the CA, click the **Upload CA Certificate** button, and select the certificate for the CA used to sign your node certificates (see step 4c). Repeat this step for all CAs that were used in the signing process.

8. Go to the **Key Management Server** section (see step 5) and do the following:

- Click the **Manage Certificates** button for a key management server.
- In the **Manage Signed Certificates** screen, upload the node certificates either by clicking the **Upload Files** button to upload all the certificates in one step or by clicking the **Upload** link (not shown in the figure) for each node individually.
- Test that the certificates are correct either by clicking the **Test all nodes** button to test the certificates for all nodes in one step or by clicking the **Test CS** (or **Re-Test CS**) link for each node individually. A status of **Verified** indicates the test was successful for that node.

Note: Before removing a drive or node from an SED cluster, ensure that the testing is successful and the status is **Verified**. Otherwise, the drive or node will be locked.

- Repeat this step for each key management server.

Note: Before removing a drive or node from an SED cluster, ensure that the testing is successful and the status is **Verified**. Otherwise, the drive or node will be locked.

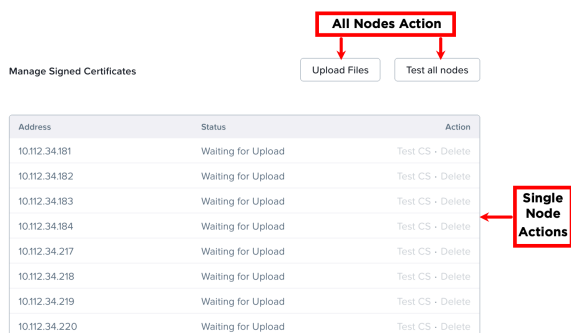


Figure 14: Upload Signed Certificates Screen

9. When the configuration is complete, click the **Protect** button on the opening page to enable encryption protection for the cluster.

A clear key icon appears on the page.

The key turns gold when cluster encryption is enabled.

Note: If changes are made to the configuration after protection has been enabled, such as adding a new key management server, you must rekey the disks for the modification to take full effect (see [Changing Key Encryption Keys \(SEDs\)](#) on page 127).

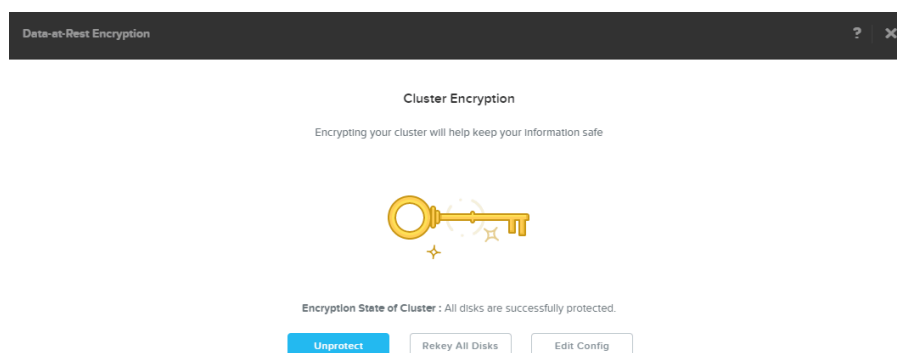


Figure 15: Data-at-Rest Encryption Screen (protected)

Enabling/Disabling Encryption (SEDs)

Data on a self encrypting drive (SED) is always encrypted, but enabling/disabling data-at-rest encryption for the cluster determines whether a separate (and secured) key is required to access that data.

About this task

To enable or disable data-at-rest encryption after it has been configured for the cluster (see [Configuring Data-at-Rest Encryption \(SEDs\)](#) on page 124), do the following:

Note: The key management server must be accessible to disable encryption.

Procedure

1. Log in to Prism Element as an administrator.
2. Click the gear icon in the main menu and then select **Data at Rest Encryption** in the **Settings** page.
3. In the **Cluster Encryption** page, do one of the following:
 - » If cluster encryption is enabled currently, click the **Unprotect** button to disable it.
 - » If cluster encryption is disabled currently, click the **Protect** button to enable it.

Enabling cluster encryption enforces the use of secured keys to access data on the SEDs in the cluster; disabling cluster encryption means the data can be accessed without providing a key.

Changing Key Encryption Keys (SEDs)

The key encryption key (KEK) can be changed at any time. This can be useful as a periodic password rotation security precaution or when a key management server or node becomes compromised. If the key management server is compromised, only the KEK needs to be changed, because the KEK is independent of the drive encryption key (DEK). There is no need to re-encrypt any data, just to re-encrypt the DEK.

About this task

To change the KEKs for a cluster, do the following:

Procedure

1. Log in to Prism Element as an administrator.
2. Click the gear icon in the main menu and then select **Data at Rest Encryption** in the **Settings** page.
3. In the **Cluster Encryption** page, select **Manage Keys** and click the **Rekey All Disks** button under **Hardware Encryption**.

Rekeying a cluster under heavy workloads might result in higher-than-normal IO latency, and some data might become temporarily unavailable. To continue with the rekey operation, click **Confirm Rekey**.

This step resets the KEKs for all the self encrypting disks in the cluster.

Note: The **Rekey All Disks** button appears only when cluster protection is active. If the cluster is already protected and a new key management server is added, you must press the **Rekey All Disks** button to use this new key management server for storing secrets.

Destroying Data (SEDs)

Data on a self encrypting drive (SED) is always encrypted, and the data encryption key (DEK) used to read the encrypted data is known only to the drive controller. All data on the drive can effectively be destroyed (that is, become permanently unreadable) by having the controller change the DEK. This is known as a crypto-erase.

About this task

To crypto-erase a SED, follow these steps:

Procedure

1. In the Prism Element web console, go to the Hardware dashboard and select the **Diagram** tab.

2. Select the target disk in the diagram (upper section of screen) and then click the **Remove Disk** button (at the bottom right of the following diagram).

As part of the disk removal process, the DEK for that disk is automatically cycled on the drive controller. The previous DEK is lost and all new disk reads are indecipherable. The key encryption key (KEK) is unchanged, and the new DEK is protected using the current KEK.

When a node is removed, all SEDs in that node are crypto-erased automatically as part of the node removal process.

When you run the cluster destroy command to decommission your entire cluster, the command automatically performs a crypto erase on the SED as the final step.

When you destroy a cluster that uses SEDs, Nutanix permanently deletes the KEK from the system. Any residual data on the drives remains encrypted. The KEK is no longer available in hardware or software, and therefore, the data cannot be decrypted or recovered.

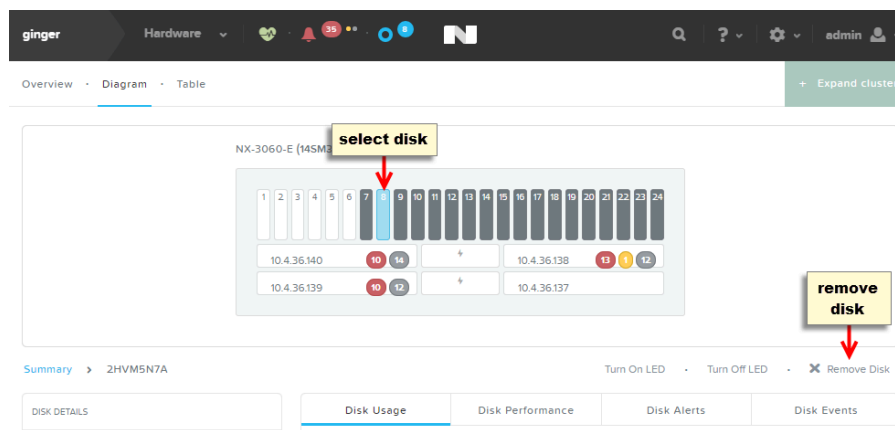


Figure 16: Removing a Disk

Data-at-Rest Encryption (Software Only)

For customers who require enhanced data security, Nutanix provides a software-only encryption option for data-at-rest security (SEDs not required).

Starting with AOS version 7.3, software-based data-at-rest encryption and native key management service (KMS) capabilities are included with the Pro and Ultimate editions of the Nutanix Cloud Infrastructure (NCI) license. For earlier AOS versions, these features require either an Ultimate license or a Pro add-on license. Software encryption supports the following features:

- For AHV, data can be encrypted at the cluster level. This applies to both empty clusters and clusters with existing data. In addition to cluster-level encryption, you can configure storage policies in Prism Central to encrypt data at the container or VM level. For more information, see [Storage Policy-Based Encryption](#).
- For ESXi and Hyper-V, the data can be encrypted on a cluster or container level. The cluster or container can be empty or contain existing data. Consider the following points for container level encryption.
 - Once you enable container level encryption, you can not change the encryption type to cluster level encryption later.
 - After the encryption is enabled, the administrator needs to enable encryption for every new container.
- Data is encrypted at all times.
- Data is inaccessible in the event of drive or node theft.
- Data on a drive can be securely destroyed.

- Re-key of the master encryption key at arbitrary times is supported.
- Cluster's native KMS is supported.

Note: In case of mixed hypervisors, only the following combinations are supported.

- ESXi and AHV
- Hyper-V and AHV

This solution provides enhanced security for data on a drive, but it does not secure data in transit.

Data Encryption Model

To accomplish the above mentioned goals, Nutanix implements a data security configuration that uses AOS functionality along with the cluster's native or an external key management server. Nutanix uses open standards (KMIP protocols) for interoperability and strong security.

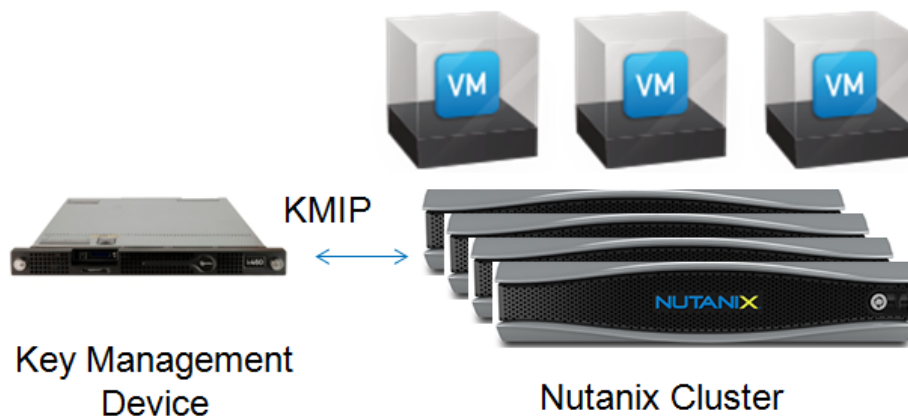


Figure 17: Cluster Protection Overview

This configuration involves the following workflow:

- For software encryption, data protection must be enabled for the cluster before any data is encrypted. Also, the Controller VM must provide the proper key to access the data.
- A symmetric data encryption key (DEK) such as AES 256 is applied to all data being written to or read from the disk. The key is known only to AOS, so there is no way to access the data directly from the drive.
- In case of an external KMS:

Each node maintains a set of certificates and keys in order to establish a secure connection with the key management server.

Only one key management server device is required, but it is recommended that multiple devices are employed so the key management server is not a potential single point of failure. Configure the key manager server devices to work in clustered mode so they can be added to the cluster configuration as a single entity that is resilient to a single failure.

Configuring Data-at-Rest Encryption (Software Only)

Nutanix offers a software-only option to perform data-at-rest encryption in a cluster or container.

Before you begin

- You must review [Key Management Server \(KMS\) Considerations](#) on page 136.
- Starting with AOS version 7.3, software-based data-at-rest encryption and native key management service (KMS) capabilities are included with the Pro and Ultimate editions of the Nutanix Cloud Infrastructure (NCI) license. For earlier AOS versions, these features require either an Ultimate license or a Pro add-on license.

Caution: For security, you cannot disable software-only data-at-rest encryption once it is enabled.

About this task

To configure cluster or container encryption, follow these steps:

Procedure

1. Log in to Prism Element as an administrator.
2. Click the gear icon in the main menu and then select **Data at Rest Encryption** in the **Settings** menu. The **Data at Rest Encryption** dialog box appears. Initially, encryption is not configured, and a message to that effect appears.
3. Click the **Create Configuration** button.
Clicking the **Continue Configuration** button, **configure it** link, or **Edit Config** button does the same thing, which is display to the Data-at-Rest Encryption configuration page

4. Select the Encryption Type as **Encrypt the entire cluster** or **Encrypt storage containers**. Then click **Save Encryption Type**.

Caution: You can enable encryption for the entire cluster or just the container. However, if you enable encryption on a container; and you experience any encryption key issues like loss of encryption key, you can encounter the following:

- The entire cluster data is affected, not just the encrypted container.
- All the user VMs of the cluster cannot access the data.

The hardware option is displayed only when SEDs are detected. Else, software based encryption type will be used by default.

For ESXi and Hyper-V, the data can be encrypted on a cluster or container level. The cluster or container can be empty or contain existing data. Consider the following points for container level encryption.

- Once you enable container level encryption, you cannot change the encryption type to cluster level encryption later.
- After the encryption is enabled, the administrator needs to enable encryption for every new container.

To enable encryption for every new storage container, follow these steps:

- a. In the web console, select **Storage** from the pull-down main menu (upper left of screen) and then select the **Table** and **Storage Container** tabs.
- b. To enable encryption, select the target storage container and then click the **Update** link. The **Update Storage Container** window appears.
- c. In the **Advanced Settings** area, select the **Enable** check box to enable encryption for the storage container you selected.

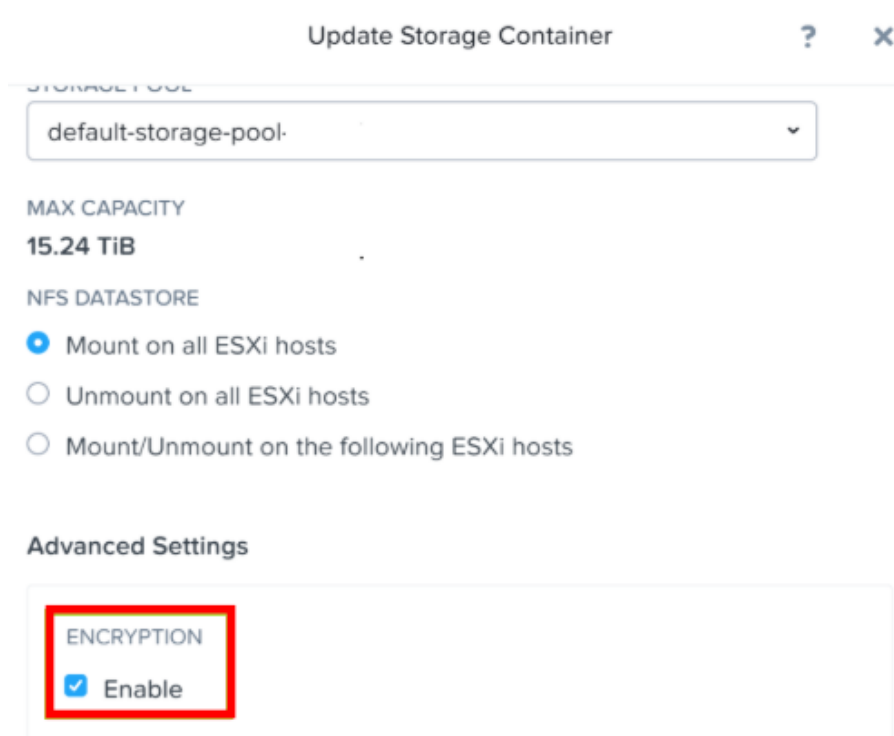


Figure 18: Update storage container

- d. Click **Save** to complete.
5. Select the Key Management Service.
 - » Select **Native KMS (local)** or **Native KMS (remote)** and click **Save KMS type**. If you select Native KMS, skip to step 10 to enable encryption.
 - » Select the **External KMS** option and click **Save KMS type**. Continue to step 6 to configure certificates and key management servers.
 - **Native KMS (local)** requires a minimum of 3-node cluster. 1-node and 2-node clusters are not supported.
 - **Native KMS (remote)** is for ROBO environments (typically, 1 or 2 node clusters) and provides enhanced security for software based encryption of ROBO clusters managed by Prism Central. This option is available only if the cluster is registered to Prism Central.

Note: You can switch between the KMS types at a later stage if the specific KMS prerequisites are met, see [Switching between Native Key Manager and External Key Manager](#) on page 137.

6. (External KMS only) In the **Certificate Signing Request Information** section, follow these steps:
 - a. Enter appropriate credentials for your organization in the **Email**, **Organization**, **Organizational Unit**, **Country Code**, **City**, and **State** fields and then click the **Save CSR Info** button.

The entered information is saved and is used when creating a certificate signing request (CSR). To specify more than one **Organization Unit** name, enter a comma separated list.
 - b. Click the **Download CSRs** button, and then in the new screen click the **Download CSRs for all nodes** to download a file with CSRs for all the nodes or click a **Download** link to download a file with the CSR for that node.
 - c. Send the files with the CSRs to the desired certificate authority.

The certificate authority creates the signed certificates and returns them to you. Store the returned SSL certificates and the CA certificate where you can retrieve them in step 8.

 - The certificates must be X.509 format. (DER, PKCS, and PFX formats are not supported.)
 - The certificate and the private key must be in separate files.

7. (External KMS only) In the **Key Management Server** section, follow these steps:
 - a. Click the **Add New Key Management Server** button.
 - b. In the **Add a New Key Management Server** screen, enter a name, IP address, and port number for the key management server in the appropriate fields.

The port is where the key management server is configured to listen for the KMIP protocol. The default port number is 5696. For the complete list of required ports, see [Port Reference](#).

 - » If you have configured multiple key management servers in cluster mode, click the **Add Address** button to provide the addresses for each key management server device in the cluster.
 - » If you have stand-alone key management servers, click the **Save** button. Repeat this step (**Add New Key Management Server** button) for each key management server device to add.

If your key management servers are configured into a master/slave (active/passive) relationship and the architecture is such that the follower cannot accept write requests, do not add the follower into this configuration. The system sends requests (read or write) to any configured key management server, so both read and write access is needed for key management servers added here.

To prevent potential configuration problems, always use the **Add Address** button for key management servers configured into cluster mode. Only a stand-alone key management server must be added as a new server.
 - c. To edit any settings, click the pencil icon for that entry in the key management server list to redisplay the add page and then click the **Save** button after making the change. To delete an entry, click the X icon.
8. (External KMS only) In the **Add a New Certificate Authority** section, enter a name for the CA, click the **Upload CA Certificate** button, and select the certificate for the CA used to sign your node certificates (see step 6c). Repeat this step for all CAs that were used in the signing process.

9. (External KMS only) Go to the **Key Management Server** section (see step 7) and follow these steps:
 - a. Click the **Manage Certificates** button for a key management server.
 - b. In the **Manage Signed Certificates** screen, upload the node certificates either by clicking the **Upload Files** button to upload all the certificates in one step or by clicking the **Upload** link (not shown in the figure) for each node individually.
 - c. Test that the certificates are correct either by clicking the **Test all nodes** button to test the certificates for all nodes in one step or by clicking the **Test CS** (or **Re-Test CS**) link for each node individually. A status of **Verified** indicates the test was successful for that node.
 - d. Repeat this step for each key management server.

Note: Before removing a drive or node from an SED cluster, ensure that the testing is successful and the status is **Verified**. Otherwise, the drive or node will be locked.

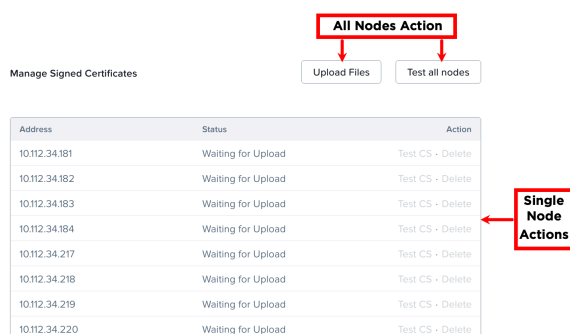


Figure 19: Upload Signed Certificates Screen

10. Click the **Enable Encryption** button.
Enable Encryption window is displayed.



Figure 20: Data-at-Rest Encryption Screen (unprotected)

Caution: To help ensure that your data is secure, you cannot disable software-only data-at-rest encryption once it is enabled. Nutanix recommends regularly backing up your data, encryption keys, and key management server.

11. Enter **ENCRYPT**.

12. Click the **Encrypt** button.

The data-at-rest encryption is enabled. To view the status of the encrypted cluster or container, go to **Data at Rest Encryption** in the **Settings** menu.

When you enable encryption, a low priority background task runs to encrypt all the unencrypted data. This task is designed to take advantage of any available CPU space to encrypt the unencrypted data within a reasonable time. If the system is occupied with other workloads, the background task consumes less CPU space. Depending on the amount of data in the cluster, the background task can take 24 to 36 hours to complete.

If changes are made to the configuration after protection has been enabled, such as adding a new key management server, you must do the rekey operation for the modification to take full effect. In case of EKM, rekey to change the KEKs stored in the EKM. In case of LKM, rekey to change the master key used by native key manager, see [Changing Key Encryption Keys \(Software Only\)](#) on page 143 for details.

Once the task to encrypt a cluster begins, you cannot cancel the operation. Even if you stop and restart the cluster, the system resumes the operation.

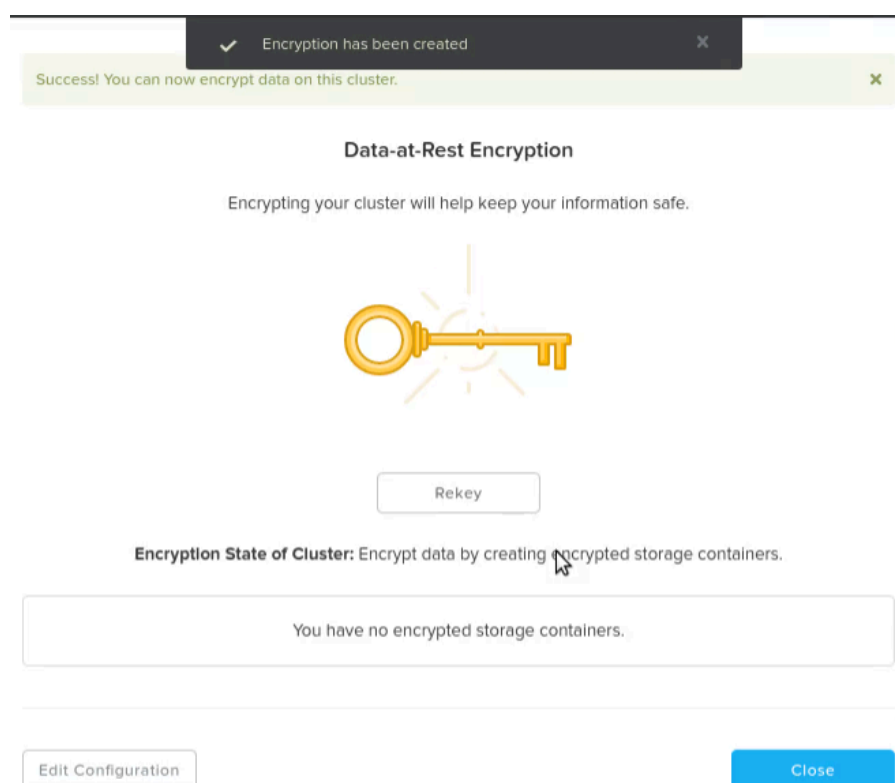


Figure 21: Data-at-Rest Encryption Screen (protected)

Key Management Server (KMS) Considerations

Before configuring software data-at-rest encryption, ensure that you review the following KMS considerations and requirements:

- Nutanix provides the option to choose the KMS type as the Native KMS (local), Native KMS (remote), External KMS, and [Cloud KMS](#).
- Cluster Localised Key Management Service (Native KMS (local)) requires a minimum of 3-node cluster. 1-node and 2-node clusters are not supported.

- Software encryption using Native KMS is supported for remote office/branch office (ROBO) deployments using the Native KMS (remote) KMS type.
- For external KMS, a separate key management server is required to store the keys outside of the cluster. Each key management server device must be configured and addressable through the network. It is recommended that multiple key manager server devices be configured to work in clustered mode so they can be added to the cluster configuration as a single entity that is resilient to a single failure.

Caution: Do not host a key management server VM on the encrypted cluster that is using it. Doing so could result in complete data loss if there is a problem with the VM while it is hosted in that cluster.

You must install the license of the external key manager for all nodes in the cluster. See [Compatibility and Interoperability Matrix](#) for a complete list of the supported key management servers. For instructions on how to configure a key management server, refer to the documentation from the appropriate vendor.

Switching between Native Key Manager and External Key Manager

After software encryption has been established, Nutanix supports the ability to switch the KMS type from the external key manager to the native key manager, from the native key manager to an external key manager, or from the remote KMS to the native KMS or an external KMS, without any down time.

For external KMS, a separate key management server is required to store the keys outside of the cluster. Each key management server device must be configured and addressable through the network. It is recommended that multiple key manager server devices be configured to work in clustered mode so they can be added to the cluster configuration as a single entity that is resilient to a single failure.

The Native KMS operates as an integrated solution and requires a minimum three-node cluster.

Nutanix recommends that you backup and save the encryption keys with identifiable names before and after changing the KMS type. For more information, see [Backing up Keys](#) on page 145.

To migrate from one external KMS to another, see [Changing External KMS](#) on page 138.

To change the KMS type, change the KMS selection by editing the encryption configuration. For details, see step 5 on page 133 in [Configuring Data-at-Rest Encryption \(Software Only\)](#) on page 130 section.

Select Key Management Server (KMS)

The KMS manages the encryption keys used to encrypt data.

☐ Native KMS (local)

Cluster Localised Key Management Service is not possible with 2 or less nodes

☒ Native KMS (remote)

Keep your keys secured with a registered remote PC. This can be used for software encryption with any cluster size.

☐ An external KMS

Configure Key Management Servers and upload SVM certificates. You will manually download and upload certificates to validate the KMS. Nutanix recommends having two or more Key Management Servers for redundancy.

Figure 22: Select KMS type

Note: This operation completes in a few minutes, depending on the number of encrypted objects and network speed.

Changing External KMS

Recommended workflow to migrate from one external key management server (KMS) to another external KMS.

About this task

To migrate from one external KMS to another external KMS for software data-at-rest encryption, follow these steps:

Procedure

1. Take a backup of the encryption keys.
For more information, see [Backing up Keys](#) on page 145.
2. Switch from the external KMS to either native KMS (local) or native KMS (remote), depending on your environment.
For more information, see [Switching between Native Key Manager and External Key Manager](#) on page 137.
3. Remove the existing external KMS configuration and configure the new external KMS.
Ensure that you upload the required certificates for the new external KMS. For more information on updating the software data-at-rest encryption settings, see [Configuring Data-at-Rest Encryption \(Software Only\)](#) on page 130.
4. Switch back from the native KMS (local or remote) used in Step 2 on page 138 to the new external KMS.

Switching Between Local and Remote Native Key Management Servers

Switch the KMS type between a local native KMS and a remote native KMS for software data-at-rest encryption.

Before you begin

Before switching the KMS, ensure that you meet the following prerequisites:

- To switch to a local native KMS, the cluster must have a minimum of three nodes.
- To switch to a remote native KMS, the cluster must be registered to Prism Central.
- To switch to a remote native KMS, the cluster must not be configured as a backup target for Prism Central Backup and Restore.

For more information on KMS requirements, see [Key Management Server \(KMS\) Considerations](#) on page 136.

Nutanix recommends that you back up and save the encryption keys before and after changing the KMS type. For more information, see [Backing up Keys](#) on page 145.

About this task

To switch between a local native KMS and a remote native KMS after you establish software encryption, follow these steps:

Procedure

1. Log in to Prism Element as an administrator.
2. In the main menu, click the **Settings** icon.

3. On the **Settings** page, select **Data-at-rest Encryption**.
4. Click **Edit Configuration**.
5. In the **Select Key Management Server (KMS)** section, select **Native KMS (local)** or **Native KMS (remote)**.
6. Click **Save KMS type**.

Cloud Key Management Server

Cloud Key Management Server (Cloud KMS) enables secure management of encryption keys through the KMS service provided by a cloud platform.

For [Software data-at-rest encryption](#), Nutanix provides the option to choose the KMS type as the Native KMS (local), Native KMS (remote), External KMS, and Cloud KMS (PC-only). Use the Cloud KMS option to securely manage encryption keys using your cloud provider's KMS using Prism Central. Enabling Cloud KMS provides the ability to manage all KMS-related operations, including changing or updating the KMS configuration. Cloud KMS offers a simplified and secure method to manage data-at-rest encryption for on-premises Nutanix clusters, NC2 on Azure and NC2 on AWS, provided the NC2 cluster can communicate with the Azure Key Vault endpoint.

Note: Cloud KMS support is available for Microsoft Azure Key Vault only.

Cloud KMS Requirements and Considerations

The following requirements and considerations apply to setting up and using Cloud KMS:

- Prism Central integrates with Microsoft Azure Key Vault. To set up Cloud KMS, you must provide specific Azure configurations, including the Key Vault URI, client ID, tenant ID, and a client secret, which authorizes Prism Central to use the RSA keys stored in Azure. For more information, see [Setting Up an Azure Key Vault for Cloud Key Management Server](#) on page 141.
- The Azure Key Vault URI must be accessible from the NC2 cluster. This often involves placing the Key Vault in the same Azure subscription or another subscription under the same Azure Active Directory (Entra ID) tenant, but this placement is not required as long as the NC2 cluster can reach the Key Vault URI.
- Cloud KMS is supported for RSA keys only.
- Cloud KMS supports RSA key sizes of 2048, 3072, and 4096 bits.
- Cloud KMS supports both Standard and Premium SKUs of Azure Key Vault. For details on FIPS compliance levels and differences between Azure Key Vault SKUs, refer to Microsoft's official documentation.
- Cloud KMS is supported on cluster running AOS version 7.0 or later.
- Cloud KMS is supported on Prism Central version pc.2024.3 or later.
- After enabling Cloud KMS, all KMS management, any future changes to the KMS type (such as switching back to local KMS) must be done using Prism Central. You cannot manage KMS configurations directly through Prism Element.

Configuring Cloud Key Management Server

Configure Azure Cloud Key Management Server (KMS) for software based data-at-rest encryption.

Before you begin

Before you configure Cloud KMS in Prism Central, ensure the following prerequisites are met:

- Understand the Cloud KMS requirements and considerations. For more information, see [Cloud KMS Requirements and Considerations](#) on page 139.
- Ensure that an Azure Key Vault is configured and accessible. For more information, see [Setting Up an Azure Key Vault for Cloud Key Management Server](#) on page 141.

About this task

To configure Cloud KMS for software data-at-rest-encryption, follow these steps:

Procedure

1. SSH into the Prism Central VM (PCVM) as a `nutanix` user:

```
$ ssh nutanix@pcvm_ip_address
```

2. Register Azure key management server:

```
nutanix@PCVM$ mantle_cli kms azure create --name KMS Name --url Key Vault URI --key_id Key ID --tenant_id Tenant ID --client_id Client ID --expiry_date Client Secret Expiry Date
```

Replace *KMS Name*, *Key Vault URI*, *Key ID*, *Tenant ID*, *Client ID*, and *Client Secret Expiry Date* with the relevant values for your environment.

For more information on Azure Key Vault configuration, see [Setting Up an Azure Key Vault for Cloud Key Management Server](#) on page 141.

3. At the client secret prompt, enter and confirm the client secret.
4. (Optional) Ensure that changes to KMS configuration propagate fully through the Nutanix environment before enabling data-at-rest encryption to avoid failures.
The KMS configuration can take up to 10 minutes to propagate. You can verify the configuration by running the following command on the Controller VM:

```
nutanix@cvm$ mantle_ops kms_info
```

If the configuration has not yet propagated, the command output displays **Information about KMS servers on IDF:** with no details listed. When propagation is complete, the output includes KMS details such as the KMS UUID, name, type, and access information under this section.

5. Log in to Prism Central as an administrator.
6. From the [Application Switcher Function](#), select the **Infrastructure** application and from the navigation bar, select **Hardware > Clusters**.
7. Select the checkbox associated with the cluster.
8. Click **Actions > Enable Data-at-Rest Encryption**.
9. Choose one of the following encryption types:
 - » **Encrypt the entire cluster**
 - » **Encrypt storage containers**
10. Click **Save Encryption Type**.
11. Select **External KMS (Cloud)** as the key management server.
12. Click **Enable Encryption**.

13. Type ENCRYPT.

Caution: To help ensure that your data is secure, you cannot disable software-only data-at-rest encryption after it is enabled. Nutanix recommends regularly backing up your data, encryption keys, and key management server.

For more information on configuring software data-at-rest encryption, see [Configuring Data-at-Rest Encryption \(Software Only\)](#).

Setting Up an Azure Key Vault for Cloud Key Management Server

Azure Key Vault details are required for Cloud key management server (KMS) configuration in the Prism Central VM.

About this task

To set up an Azure Key Vault in your Azure environment, follow these steps:

Procedure

1. Create an Azure Key Vault.
2. Generate or import RSA keys.
3. Configure an App Registration in Azure.
4. Set up the key vault access policy to allow the App Registration to perform key management, cryptographic, and rotation policy operations.
5. Retrieve the following details from your Azure configuration to use in the [Cloud KMS configuration](#).
 - KMS Name: Name for the Key Management Service. This name does not need to be unique.
 - Client Secret Expiry Date: Expiration date of the client secret.
 - Tenant ID: A unique identifier for your organization in Azure Active Directory.
 - Client ID: A unique identifier for the App Registration that allows Prism Central to connect to the Azure Key Vault.
 - Key ID: Name of the encryption key in the Azure Key Vault instance used for Cloud KMS. The key must be of type RSA.
 - Key Vault URI: URI of the Azure Key Vault instance that contains the encryption key.

For more information on configuring the Azure Key Vault, see *Microsoft Azure documentation*.

Updating Cloud KMS Configuration

Update the existing Azure Cloud KMS configuration.

About this task

To update the Azure Cloud KMS configuration, follow these steps:

Procedure

1. SSH into the Prism Central VM (PCVM) as a `nutanix` user:

```
$ ssh nutanix@pcvm_ip_address
```

2. Update the KMS configuration:

```
nutanix@PCVM$ mantle_cli kms azure update --kms_uuid KMS UUID --name New Key Management Server Name --client_id New Client ID --tenant_id New Tenant ID --expiry_date New Client Secret Expiry Date
```

Replace *KMS UUID*, *New Key Management Server Name*, *New Client ID*, *New Tenant ID*, and *New Client Secret Expiry Date* with the relevant values for your environment.

Note: The *KMS UUID* parameter is mandatory and identifies the KMS instance to update.

3. At the client secret prompt, enter and confirm the client secret.
4. Verify the updated configuration:

```
nutanix@PCVM$ mantle_cli kms azure list
```

Deleting Cloud KMS Configuration

Remove the existing Azure Cloud KMS configuration.

About this task

To delete the Cloud KMS (Azure) configuration, follow these steps:

Note: You cannot delete the Cloud KMS configuration that is currently being used for software data-at-rest encryption.

Procedure

1. SSH into the Prism Central VM (PCVM) as a `nutanix` user:

```
$ ssh nutanix@pcvm_ip_address
```

2. Delete the KMS configuration:

```
nutanix@PCVM$ mantle_cli kms azure delete --kms_uuid KMS UUID
```

Replace *KMS UUID* with the relevant value for your environment.

3. Verify the updated configuration:

```
nutanix@PCVM$ mantle_cli kms azure list
```

Changing the KMS Type Using Prism Central

Change the KMS Type from Cloud KMS to another supported KMS Type.

About this task

Using Prism Central, you can change the KMS Type from Cloud KMS to another supported KMS Type, such as Native KMS (Local). Refer to the [Key Management Server \(KMS\) Considerations](#) on page 136 before proceeding. To change the KMS Type, follow these steps:

Procedure

1. Log on to Prism Central as an administrator.
2. In the Application Switcher, select **Infrastructure**.
3. Go to **Hardware > Clusters**, and from the **List** view, select the encrypted cluster.

4. Click **Actions** and select **Manage KMS Type**.
5. Select the new KMS Type.
6. Click **Change KMS Type**.

Changing Key Encryption Keys (Software Only)

The key encryption key (KEK) can be changed at any time. This can be useful as a periodic password rotation security precaution or when a key management server or node becomes compromised. If the key management server is compromised, only the KEK needs to be changed, because the KEK is independent of the drive encryption key (DEK). There is no need to re-encrypt any data, just to re-encrypt the DEK.

About this task

To change the KEKs for a cluster, do the following:

Procedure

1. Log in to Prism Element as an administrator.
2. Click the gear icon in the main menu and then select **Data at Rest Encryption** in the **Settings** page.
3. In the **Cluster Encryption** page, select **Manage Keys** and click the **Rekey** button under **Software Encryption**.

Note: The **Rekey** button appears only when cluster protection is active. If the cluster is already protected and a new key management server is added, you must press the **Rekey** button to use this new key management server for storing secrets.

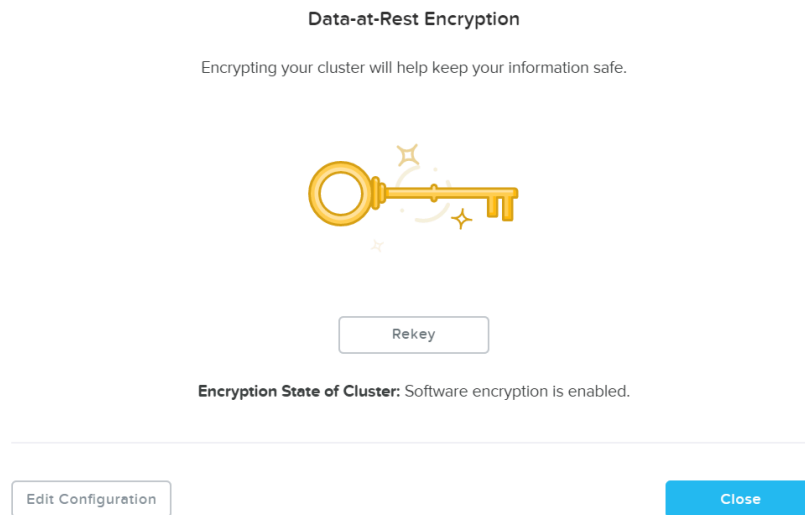


Figure 23: Cluster Encryption Screen

For Native KMS, a master encryption key (MEK) is used to encrypt the key encryption key (KEK). The system automatically regenerates the MEK yearly.

Destroying Data (Software Only)

Data on the AOS cluster is always encrypted, and the data encryption key (DEK) used to read the encrypted data is known only to the AOS. All data on the drive can effectively be destroyed (that is, become permanently unreadable) by deleting the container or cluster. This is known as a crypto-erase.

About this task

Note: To help ensure that your data is secure, you cannot disable software-only data-at-rest encryption once it is enabled. Nutanix recommends regularly backing up your data, encryption keys, and key management server.

To crypto-erase the container or cluster, do the following:

Procedure

1. Delete the storage container or destroy the cluster.

- For information on how to delete a storage container, see [Modifying a Storage Container](#) in the *Prism Element Web Console Guide*.
- For information on how to destroy a cluster, see [Destroying a Cluster](#) in the *Acropolis Advanced Administration Guide*.

When you delete a storage container, the Curator scans and deletes the DEK and KEK keys automatically.

When you destroy a cluster, then:

- Native Key Manager (local) destroys the master key shares and the encrypted DEKs/KEKs.
- Native Key Manager (remote) retains the root key on the Prism Central if the cluster is still registered to a Prism Central when it is destroyed. You must unregister a cluster from the Prism Central and then destroy the cluster to delete the root key.
- External Key Manager deletes the encrypted DEKs. However, the KEKs remain on the EKM. You must use an external key manager UI to delete the KEKs.

2. Delete the key backup files, if any.

Switching from SED-EKM to Software-LKM

This section describes the steps to switch from SED and External KMS combination to software-only and LKM combination.

About this task

To switch from SED-EKM to Software-LKM, do the following.

Procedure

1. Perform the steps for the software-only encryption with External KMS. For more information, see [Configuring Data-at-Rest Encryption \(Software Only\)](#) on page 130.

After the background task completes, all the data gets encrypted by the software. The time taken to complete the task depends on the amount of data and foreground I/O operations in the cluster.

2. Disable the SED encryption. Ensure that all the disks are unprotected.

For more information, see [Enabling/Disabling Encryption \(SEDs\)](#) on page 127.

3. Switch the key management server from the External KMS to Local Key Manager. For more information, see [Switching between Native Key Manager and External Key Manager](#) on page 137.

Configuring Dual Encryption

About this task

Dual Encryption protects the data on the clusters using both SED and software-only encryption. An external key manager is used to store the keys for dual encryption, the Native KMS is not supported.

To configure dual encryption, do the following:

Procedure

1. Log in to Prism Element as an administrator.
2. Click the gear icon in the main menu and then select **Data at Rest Encryption** in the **Settings** page.
3. In the Cluster Encryption page, check to enable both **Drive-based** and **Software-based** encryption.
4. Click **Save Encryption Type**.
5. Continue with the rest of the encryption configuration, see:
 - [Configuring Data-at-Rest Encryption \(Software Only\)](#) on page 130
 - [Configuring Data-at-Rest Encryption \(SEDs\)](#) on page 124

Backing up Keys

About this task

You can take a backup of encryption keys:

- When you enable Software-only Encryption for the first time
- After you regenerate the keys

Backing up encryption keys is critical in the very unlikely situation in which keys get corrupted.

You can download key backup file for a cluster on a Prism Element or all clusters on a Prism Central. To download key backup file for all clusters, see [Taking a Consolidated Backup of Keys \(Prism Central\)](#).

To download the key backup file for a cluster, do the following:

Procedure

1. Log in to Prism Element as an administrator.
2. Click the gear icon in the main menu and then select **Data at Rest Encryption** in the **Settings** page.
3. In the Cluster Encryption page, select **Manage Keys**.
4. Enter and confirm the password.
5. Click the **Download Key Backup** button.

The backup file is saved in the default download location on your local machine. Ensure you move the backup key file to a safe location.

If vTPM is enabled, the downloaded key backup includes vTPM keys, in addition to the encryption keys.

Taking a Consolidated Backup of Keys (Prism Central)

If you are using the Native KMS option with software encryption for your clusters, you can take a consolidated backup of all the keys from Prism Central.

About this task

To take a consolidated backup of keys for software encryption-enabled clusters (Native KMS-only), do the following:

Procedure

1. Log in to Prism Central as an administrator.
2. Click the hamburger icon, then select **Clusters > List** view.
3. Select a cluster, go to **Actions**, then select **Manage & Backup Keys**.
4. Download the backup keys:
 - a. In **Password**, enter your password.
 - b. In **Confirm Password**, reenter your password.
 - c. To change the encryption key, select the **Rekey Encryption Key (KEK)** box .
 - d. To download the backup key, click **Backup Key**.

Ensure that you move the backup key file to a safe location.

If vTPM is enabled, the downloaded key backup includes vTPM keys, in addition to the encryption keys.

Importing Keys

You can import the encryption keys from backup. You must note the specific commands in this topic if you backed up your keys to an external key manager (EKM).

About this task

Note: Nutanix recommends that you contact Nutanix Support for this operation. Extended cluster downtime might result if you perform this task incorrectly.

Procedure

1. SSH into any of the Controller VMs (CVMs) as a nutanix user:

```
$ ssh nutanix@cvm_ip_address
```

2. Retrieve the encryption keys stored on the cluster and verify that all the keys you want to retrieve are listed.
In this example, the password is Nutanix.123. *date* is the timestamp portion of the backup file name.

```
nutanix@cvm$ mantle_recovery_util --backup_file_path=/home/nutanix/  
encryption_key_backup_date \  
--password=Nutanix.123 --list_key_ids=true
```

3. Import the keys into the cluster.

```
nutanix@cvm$ mantle_recovery_util --backup_file_path=/home/nutanix/key_backup \  
--password=Nutanix.123 --interactive_mode
```

4. If you are using an external key manager such as IBM Security Key Lifecycle Manager, Gemalto Safenet, or Vormetric Data Security Manager, use the `--store_kek_remotely` option to import the keys into the cluster.
In this example, *date* is the timestamp portion of the backup file name.

```
nutanix@cvm$ mantle_recovery_util --backup_file_path path/encryption_key_backup_date \  
\  
--password key_password --store_kek_remotely
```

Tip: The imported key backup includes vTPM keys (if vTPM is enabled during backup), in addition to the encryption keys.

Data-at-Rest Encryption with Linux Unified Key Setup

Describes how Nutanix uses Linux Unified Key Setup (LUKS) to encrypt configuration data at rest for AOS and AHV environments.

Nutanix provides two types of data-at-rest encryption, each addressing a different aspect of data protection.

- AOS data-at-rest encryption. For more information, see [Data-at-Rest Encryption \(Software Only\)](#) on page 129.
- Linux Unified Key Setup (LUKS)-based data-at-rest encryption.

Nutanix uses LUKS to provide data-at-rest encryption for configuration data associated with AOS and AHV. This encryption protects data stored on the underlying operating system. The configuration data includes Nutanix cluster configuration information and VM configuration metadata. It does not include customer workload data stored inside VM disks. AOS data-at-rest encryption protects workload data stored in AOS. LUKS-based encryption is additive and extends encryption coverage to Nutanix configuration data. LUKS does not replace AOS data-at-rest encryption.

You can use LUKS encryption in deployments that require protection of configuration data at rest, such as environments subject to compliance standards.

Nutanix recommends using AOS data-at-rest encryption for workload data and using LUKS encryption only in strict environments where configuration data must be encrypted.

You can enable LUKS during cluster provisioning or while imaging a node by using Nutanix Foundation. For more information, see the [Field Installation Guide](#).

LUKS Encryption Requirements and Considerations

Review the following requirements and considerations before enabling LUKS encryption.

- LUKS encryption is supported for AHV only.
- LUKS encryption is supported only on nodes with physical TPM enabled.
- LUKS encryption is supported for the following software versions:
 - AOS 7.3 or later
 - AHV 10.3 or later
 - Foundation 5.9 or later
- Enabling LUKS-based encryption increases CPU overhead and can reduce workload throughput. Nutanix has observed approximately 20 to 30 percent additional CPU usage in some environments, depending on workload characteristics and I/O patterns.

Internationalization (i18n)

The following table lists all the supported and unsupported entities in UTF-8 encoding.

Table 11: Internationalization Support

Supported Entities	Unsupported Entities
Cluster name	Acropolis file server
Storage Container name	Share path
Storage pool	Internationalized domain names

Supported Entities	Unsupported Entities
VM name	E-mail IDs
Snapshot name	Hostnames
Volume group name	Integers
Protection domain name	Password fields
Remote site name	Any Hardware related names (for example, vSwitch, iSCSI initiator, vLAN name)
User management	
Chart name	

Caution: The creation of none of the above entities are supported on Hyper-V because of the DR limitations.

Entities Support (ASCII or non-ASCII) for the Active Directory Server

- In the New Directory Configuration, **Name** field is supported in non-ASCII.
- In the New Directory Configuration, **Domain** field is not supported in non-ASCII.
- In Role mapping, **Values** field is supported in non-ASCII.
- User names and group names are supported in non-ASCII.

Log Forwarding

Configure cluster-wide log forwarding and fingerprinting to ensure log integrity and meet compliance requirements.

The Controller VM (CVM) supports a cluster-wide log forwarding configuration to ensure log integrity by sending all logs to a central log host. Because of the CVM's appliance-based design, system and audit logs do not support local retention periods. Retaining logs locally increases the risk of a distributed denial-of-service (DDoS) attack through excessive log traffic.

Nutanix recommends deploying a central log host within the management enclave to meet compliance and internal policy requirements. This setup supports compliance with internal policies and regulatory requirements. In the event of a system compromise, the central log host helps preserve log integrity.

By default, the CVM uses the audisp plugin to forward audit logs to the rsyslog daemon, which stores in the /home/log/messages path. You can search for audispd on the central log host to retrieve the full audit log content. The audit daemon uses a rules engine that complies with the Operating System Security Requirements Guide (OS SRG) and is embedded in the CVM STIG.

Use the `rsyslog-config` command to enable log forwarding for system, audit, AIDE, and SCMA logs across all CVMs in a cluster at the required log level. For more information, see [Send Logs to Remote Syslog Server](#) in the *Acropolis Advanced Administration Guide*.

Documenting the Log Fingerprint

Document log fingerprints to support forensic analysis and ensure non-repudiation.

About this task

To document the log fingerprint, follow these steps:

Procedure

1. SSH into any of the Controller VMs (CVMs) as a `nutanix` user:

```
$ ssh nutanix@cvm_ip_address
```

2. Document the fingerprint for each public key assigned to an individual admin:

```
nutanix@cvm$ ssh-keygen -lf /file-path/id_rsa.pub
```

The fingerprint compares to the SSH daemon log entries and forwards to the central log host path `/home/log/secure` in the CVM.

Note: After completion of the SSH public key inclusion in Prism and verification of connectivity, disable the password authentication for all the CVMs and AHV hosts. For more information, see [Configuring Cluster Lockdown in Prism Element](#) on page 117.

Admin Account Password Retry Lockout

For enhanced security, Prism Central and Prism Element locks out the default admin account for a period of 15 minutes after three unsuccessful login attempts. Once the account is locked out, the following message is displayed at the login screen.

```
Account locked due to too many failed attempts
```

You can attempt entering the password after the 15 minutes lockout period, or contact Nutanix Support in case you have forgotten your password.

Note: You cannot modify the default 15 minutes lock out period.

Portal Proxy Connection in Prism Element

The portal proxy connection enables secure communication between Prism Element and Nutanix Support Portal using API keys.

The portal proxy connection provides the following capabilities:

- Direct support case creation from the Prism Element web console
- Automatic updates to the cluster's license status, which eliminates the manual downloading and uploading of the licensing files

Portal Proxy Connection Considerations

Before you configure the portal proxy connection, consider the following points:

- You can configure only one API key in the portal proxy connection.
- When you disable the portal proxy connection, the associated API key becomes invalid. You cannot reuse the same API key for a new portal proxy connection.

Configuring the Portal Proxy Connection in Prism Element

Configure the portal proxy connection in Prism Element.

About this task

To configure the portal proxy connection in Prism Element web console, follow these steps:

Procedure

1. Log in to Prism Element web console as an administrator.
2. From the main menu, click the gear icon.
3. On the left navigation bar, select the **Portal Connection**.
4. On the **Portal Connection** page, enter the following information:
 - a. **API Key**: Enter the API key that you generated from the My Nutanix portal.
 - b. (Optional) **Public Key**: To upload the optional SSL key that you downloaded from the My Nutanix portal, click **Choose File**.
5. Click **Save**.

Updating the Portal Proxy Connection in Prism Element

Update the portal proxy connection in Prism Element.

About this task

To update the portal proxy connection in Prism Element, follow these steps:

Procedure

1. Log in to Prism Element web console as an administrator.
2. From the main menu, click the gear icon.
3. On the left navigation bar, select the **Portal Connection**.
4. Click **Update**.
5. Update the fields in the **Portal Connection** window as needed and click **Save**.
For more information on the fields available in the **Portal Connection** window, see [Configuring the Portal Proxy Connection in Prism Element](#) on page 149.

Disabling the Portal Proxy Connection in Prism Element

Disable the portal proxy connection in Prism Element web console.

Before you begin

You cannot reuse the disabled API key for a new portal proxy connection.

About this task

To disable the portal proxy connection in Prism Element web console, follow these steps:

Procedure

1. Log in to Prism Element web console as an administrator.
2. From the main menu, click the gear icon.
3. On the left navigation bar, select the **Portal Connection**.
4. Click **Disable**.
5. Click **Yes**.

HARDENING INSTRUCTIONS USING NCLI

Implement security hardening features for Nutanix AHV, Controller VM, and Prism Central VM.

You can use nCLI to apply security hardening settings for AHV hosts, Controller VMs, and Prism Central VMs. The hardening parameters cover enabling advanced intrusion detection environment (AIDE) for integrity monitoring, enforcing high-strength passwords, configuring SSH login banners, and controlling core dumps. You can also set SSH security levels, apply IP-based restrictions, enforce lock status, and configure IP set-based firewalls for intra-cluster communication.

AHV Security Hardening

AHV security hardening configurations allow you to mitigate vulnerabilities and strengthen the overall security posture of your cluster.

Note: For the complete list of cluster security parameters, see *Edit the hypervisor security compliance config of a Cluster* in the [Command Reference](#) guide.

To configure AHV security hardening by modifying the key parameters, follow these steps:

- SSH into any of the Controller VMs (CVMs) as a nutanix user:

```
$ ssh nutanix@cvm_ip_address
```

Replace `cvm_ip_address` with the IP address of the CVM.

- View the hypervisor security configuration of the cluster:

```
nutanix@cvm$ ncli cluster get-hypervisor-security-config
```

Output:

```
Enable Aide           : false
Enable Core           : false
Enable High Strength P... : false
Enable Banner         : false
Schedule              : DAILY
Enable iTLB Multihit M... : false
Enable Retbleed Mitiga... : false
Enable Memory Poison   : false
Enable user core dump ... : true
Enable Fapolicy        : false
Enable Kernel Core     : true
```

- AIDE:** The Advanced Intrusion Detection Environment (AIDE) setting is a file and directory integrity checker. To enable AIDE, run the following command:

```
nutanix@cvm$ ncli cluster edit-hypervisor-security-params enable-aide=true
```

- Core:** The core dump setting consists of the recorded state of the working memory of a program at a specific time, generally when the program crashes or terminates abnormally. You can use core dumps to assist in diagnosing or debugging errors. You must always disable core dumps to prevent AHV from generating stack traces that might contain sensitive data.

Caution: Nutanix recommends that you do not set the core setting to true unless instructed by the Nutanix support team to ensure data privacy.

- High Strength Password:** The high strength password setting enforces stronger password policies on the AHV host. It requires users to have a minimum password length of 15 characters (`minlen=15`), mandates that at least 8

characters must differ from the old password (difok=8), limits the use of the same character class consecutively to a maximum of 4 characters (maxclassrepeat=4), checks passwords against a user-specific list of prohibited passwords (dictcheck=1), and reduces the maximum number of failed login attempts before account lockout from 5 to 3. To enable the high strength password setting, run the following command:

```
nutanix@cvm$ ncli cluster edit-hypervisor-security-params \
enable-high-strength-password=true
```

- **Banner:** The banner setting allows you to display a banner message upon SSH login. To enable the banner setting, run the following commands on each AHV host in a cluster:

- SSH into the AHV host as a root user:

```
$ ssh root@ahv_ip_address
```

- Backup the DoD banner file on the AHV host:

```
[root@AHV-host ~]# cp -a /etc/puppet/modules/kvm/files/issue.DoD \
/etc/puppet/modules/kvm/files/issue.DoD.bak
```

Important: The system does not preserve any changes in the banner file across upgrades.

- Modify the DoD banner file on the AHV host:

```
[root@AHV-host ~]# vi /etc/issue.DoD
```

- SSH into any of the Controller VMs (CVMs) as a **nutanix** user:

```
$ ssh nutanix@cvm_ip_address
```

- Enable the banner setting:

```
nutanix@cvm$ ncli cluster edit-hypervisor-security-params enable-banner=true
```

- **iTLB Multihit Mitigation:** The iTLB multihit mitigation setting mitigates CVE-2018-12207 (iTLB Multihit), a vulnerability in Intel CPUs that allows a user virtual machine (UVM) to perform a denial-of-service (DoS) attack on other co-hosted UVMs, potentially causing a host crash due to a Machine Check Exception (MCE). The iTLB multihit mitigation setting protects UVM against the DoS attacks but might reduce performance. To enable the iTLB multihit mitigation, run the following command:

```
nutanix@cvm$ ncli cluster edit-hypervisor-security-params enable-itlb-multihit-
mitigation=True
```

- **Retbleed Mitigation:** The Retbleed mitigation setting addresses the Retbleed vulnerability (CVE-2022-29900, CVE-2022-29901). The Retbleed is a speculative execution vulnerability affecting processors using return instructions. Enabling the Retbleed mitigation protects against these attacks by restricting the use of indirect branches and returns, which might reduce performance depending on the workload. To enable the Retbleed mitigation, run the following command:

```
nutanix@cvm$ ncli cluster edit-hypervisor-security-params enable-retbleed-
mitigation=True
```

- **Memory Poison:** The memory poison technique overwrites freed memory with a specific value (often non-zero) to help detect use-after-free vulnerabilities. When a program tries to access freed and poisoned memory, it might crash, exposing the vulnerability. Without memory poisoning, the program might continue operating with corrupted data, leading to unpredictable behavior and potential security exploits. To enable the memory poison, run the following command:

```
nutanix@cvm$ ncli cluster edit-hypervisor-security-params enable-memory-poison=True
```

- **Fapolicy:** Fapolicy is a RHEL 8 security framework for managing access control.

Note: Nutanix recommends that you do not set the Fapolicy setting to true unless there is a strict organisation policy to enable it. Enabling the FaPolicy setting might reduce performance.

- **Kernel Core:** The kernel core dump is a snapshot of the program's memory at the time of a crash. Enabling the kernel core setting allows capturing a core dump of the kernel if it crashes, which can be helpful in debugging kernel-related issues. To enable kernel core, run the following command:

```
nutanix@cvm$ ncli cluster edit-hypervisor-security-params enable-kernel-core=true
```

CVM Security Hardening

The CVM security hardening configurations allow you to mitigate vulnerabilities and strengthen the overall security posture of your cluster.

Note: For the complete list of cluster security parameters, see *Edit the security params of a Cluster* in the [Command Reference](#) guide.

To configure the CVM security hardening by modifying the key parameters, follow these steps:

- SSH into any of the Controller VMs (CVMs) as a **nutanix** user:

```
$ ssh nutanix@cvm_ip_address
```

Replace `cvm_ip_address` with the IP address of the CVM.

- View the cluster-wide security configuration:

```
nutanix@cvm$ ncli cluster get-cvm-security-config
```

Output:

```
Enable Aide           : false
Enable Core           : false
Enable High Strength P... : false
Enable Banner         : false
Schedule              : DAILY
Enable Kernel Core    : true
Enable Page Poison    : false
Enable Slub Debug     : false
SSH Security Level    : DEFAULT
Enable Lock Status    : false
IP Restriction State  : NORMAL
Enable DoDin Additiona... : false
Enable Fapolicy       : false
Enable Processor Mitig... : false
SSH whitelisted addres... :
```

- **AIDE:** The Advanced Intrusion Detection Environment (AIDE) setting is a file and directory integrity checker. To enable AIDE, run the following command:

```
nutanix@cvm$ ncli cluster edit-cvm-security-params enable-aide=true
```

- **Core:** The core dump setting consists of the recorded state of the working memory of a program at a specific time, generally when the program is crashed or terminated abnormally. You can use core dumps to assist in diagnosing

or debugging errors. You must always disable core dumps to prevent the CVM from generating stack traces that might contain sensitive data.

Caution: Nutanix recommends that you do not set the core setting to true unless instructed by the Nutanix support team to ensure data privacy.

- **High Strength Password:** The high strength password setting enforces stronger password policies on the CVM. It requires users to have a minimum password length of 15 characters (minlen=15), mandates that at least 8 characters must differ from the old password (difok=8), limits the use of the same character class consecutively to a maximum of 4 characters (maxclassrepeat=4), checks passwords against a user-specific list of prohibited passwords (dictcheck=1), and reduces the maximum number of failed login attempts before account lockout from 5 to 3. To enable the high strength password setting, run the following command:

```
nutanix@cvm$ ncli cluster edit-cvm-security-params enable-high-strength-password=true
```

- **Banner:** The banner setting allows you to display a banner message upon SSH login. To enable the banner setting, run the following commands on each CVM in a cluster:

- Backup the DoD banner file:

```
nutanix@cvm$ sudo cp -a /srv/salt/security/CVM/sshd/DODbanner \
/srv/salt/security/CVM/sshd/DODbannerbak
```

- Modify the DoD banner file:

```
nutanix@cvm$ sudo vi /srv/salt/security/CVM/sshd/DODbanner
```

- Enable the banner setting:

```
nutanix@cvm$ ncli cluster edit-cvm-security-params enable-banner=true
```

- **Kernel Core:** The kernel core dump is a snapshot of the program's memory at the time of a crash. Enabling the kernel core setting allows capturing a core dump of the kernel if it crashes, which can be helpful in debugging kernel-related issues. To enable kernel core, run the following command:

```
nutanix@cvm$ ncli cluster edit-cvm-security-params enable-kernel-core=true
```

- **Page Poison:** The page poison setting is a technique that overwrites freed memory with a specific value to help detect use-after-free vulnerabilities. Enabling page poison mitigation helps prevent use-after-free attacks but might reduce performance. To enable page poison, run the following command:

```
nutanix@cvm$ ncli cluster edit-cvm-security-params enable-page-poison=true
```

- **Slub Debug:** The slub debug setting is a memory debugging technique that can help detect memory corruption issues. Enabling slub debug mitigation helps prevent memory corruption but might reduce performance. To enable slub debug, run the following command:

```
nutanix@cvm$ ncli cluster edit-cvm-security-params enable-slub-debug=true
```

- **SSH Security Level:** The SSH security level setting allows you to enable different security levels for the `nutanix` user for accessing the cluster using SSH login. You can configure the security levels with one of the following options:

- **Default:** The `nutanix` user can start an SSH session with either a password or an SSH key without any change to the account privileges.
- **Limited:** For SSH sessions started with a password, the `nutanix` user is switched to an admin user that has lower operational privileges. The SSH key-based logins do not change the `nutanix` user's privileges.
- **Restricted:** For all SSH sessions, whether started with a password or an SSH key, the `nutanix` user is switched to an admin user that has lower operational privileges.

To configure the SSH security level, run the following command:

```
nutanix@cvm$ ncli cluster edit-cvm-security-params ssh-security-level=default
```

The admin user, while having lower privileges than the `nutanix` user, is designed to have sufficient privileges for administrative tasks. For more information, see [Controller VM Access](#) in the AHV Administration Guide.

In addition to configuring the SSH security level, you can also consider cluster lockdown to disable password-based SSH authentication by adding SSH keys. For more information, see [Configuring Cluster Lockdown in Prism Element](#) on page 117.

- **IP Restriction State:** The IP restriction state setting controls the different levels of IP restriction for the `nutanix` user to access the cluster using SSH login. You can set it to normal or restricted depending on the desired level of restriction. When you set the IP restriction state to restricted, only the IPs in the SSH whitelisted addresses can access the system using SSH. To configure the IP Restriction State, run the following command:

```
nutanix@cvm$ ncli cluster edit-cvm-security-params ip-restriction=normal
```

Note: You can use the IP restriction state setting when the SSH client IP addresses are known and stable.

- **DoDin Additional:** The DoDin additional setting provides enhanced options for maintaining DISA compliance. It allows permanent account locking after an incorrect password is entered, as opposed to implementing a temporary 15-minute timeout period. Also, it makes AIDE send email messages to the account associated with NCC and runs AIDE more frequently. To enable DoDin, run the following command:

```
nutanix@cvm$ ncli cluster edit-cvm-security-params enable-dodin-opts=true
```

- **Fapolicy:** Fapolicy is a RHEL 8 security framework for managing access control.

Note: Nutanix recommends that you do not set the Fapolicy setting to true unless there is a strict organisation policy to enable it. Enabling the FaPolicy setting might reduce performance.

- **Processor Mitigation:** The processor mitigation setting enables mitigations against processor-based vulnerabilities. Enabling processor mitigation protects against certain types of attacks but might reduce performance. To enable processor mitigation, run the following command:

```
nutanix@cvm$ ncli cluster edit-cvm-security-params enable-processor-mitigations=true
```

- **SSH Whitelisted Address:** Displays the list of IP addresses added to the CVM SSH whitelist. To add an IP address to the SSH whitelisted addresses, run the following command:

```
nutanix@cvm$ ncli cluster edit-cvm-security-params add-to-ssh-whitelist=x.x.x.x
```

Replace `x.x.x.x` with the IP address.

Use the SSH Whitelisted Address setting to add the IP addresses of trusted machines, such as jump boxes, that connect to the CVM using SSH.

Avoid whitelisting broad IP ranges. Allow only the minimal set of required IP addresses, such as the smallest subnet or specific jump box IP addresses.

If a whitelisted IP address is misconfigured or changes without being re-added to the whitelist, SSH access to the CVM is blocked. To restore access, disable the restriction using the local console or serial access.

- **Lock Status:** The lock status setting freezes the security configuration and prevents from making any changes to the security configuration through nCLI or API calls. To enable the lock status, run the following command:

```
nutanix@cvm$ ncli cluster edit-cvm-security-params enable-lock-status=true
```

Note: You must contact Nutanix Support to unlock the lock status setting.

PCVM Security Hardening

PCVM security hardening configurations allow you to mitigate vulnerabilities and strengthen the overall security posture of your cluster.

Note: For the complete list of cluster security parameters, see *Edit the security params of a Cluster* in the [Command Reference](#) guide.

To configure PCVM security hardening by modifying the key parameters, follow these steps:

- SSH into any of the Prism Central VM (PCVM) as a `nutanix` user:

```
$ ssh nutanix@pcvm_ip_address
```

Replace `pcvm_ip_address` with the IP address of the PCVM.

- View the cluster-wide security configuration:

```
nutanix@pcvm$ ncli cluster get-pcvm-security-config
```

Output:

```
Enable Aide           : false
Enable Core           : false
Enable High Strength P... : false
Enable Banner         : false
Schedule              : DAILY
Enable Kernel Core    : false
Enable Page Poison    : false
Enable Slub Debug     : false
SSH Security Level    : DEFAULT
Enable Lock Status    : false
IP Restriction State  : NORMAL
Enable DoDin Additiona... : false
Enable Fapolicy       : false
Enable Processor Mitig... : false
SSH whitelisted addres... :
```

- **AIDE:** The Advanced Intrusion Detection Environment (AIDE) setting is a file and directory integrity checker. To enable AIDE, run the following command:

```
nutanix@pcvm$ ncli cluster edit-pcvm-security-params enable-aide=true
```

- **Core:** The core dump setting consists of the recorded state of the working memory of a program at a specific time, generally when the program is crashed or terminated abnormally. You can use core dumps to assist in diagnosing or debugging errors. You must always disable core dumps to prevent the PCVM from generating stack traces that might contain sensitive data.

Caution: Nutanix recommends that you do not set the core setting to true unless instructed by the Nutanix support team to ensure data privacy.

- **High Strength Password:** The high strength password setting enforces stronger password policies on the PCVM. It requires users to have a minimum password length of 15 characters (minlen=15), mandates that at least 8 characters must differ from the old password (difok=8), limits the use of the same character class consecutively to a maximum of 4 characters (maxclassrepeat=4), checks passwords against a user-specific list of prohibited passwords (dictcheck=1), and reduces the maximum number of failed login attempts before account lockout from 5 to 3. To enable the high strength password setting, run the following command:

```
nutanix@pcvm$ ncli cluster edit-pcvm-security-params enable-high-strength-password=true
```

- **Banner:** The banner setting allows you to display a banner message upon SSH login. To enable the banner setting, run the following commands on each PCVM in a cluster:

- Backup the DoD banner file:

```
nutanix@pcvm$ sudo cp -a /srv/salt/security/PC/sshd/DODbanner \
/srv/salt/security/PC/sshd/DODbannerbak
```

- Modify the DoD banner file:

```
nutanix@pcvm$ sudo vi /srv/salt/security/PC/sshd/DODbanner
```

- Enable the banner setting:

```
nutanix@pcvm$ ncli cluster edit-pcvm-security-params enable-banner=true
```

- **Kernel Core:** The kernel core dump is a snapshot of the program's memory at the time of a crash. Enabling the kernel core setting allows capturing a core dump of the kernel if it crashes, which can be helpful in debugging kernel-related issues. To enable kernel core, run the following command:

```
nutanix@pcvm$ ncli cluster edit-pcvm-security-params enable-kernel-core=true
```

- **Page Poison:** The page poison setting is a technique that overwrites freed memory with a specific value to help detect use-after-free vulnerabilities. Enabling page poison mitigation helps prevent use-after-free attacks but might reduce performance. To enable page poison, run the following command:

```
nutanix@pcvm$ ncli cluster edit-pcvm-security-params enable-page-poison=true
```

- **Slub Debug:** The slub debug setting is a memory debugging technique that can help detect memory corruption issues. Enabling slub debug mitigation helps prevent memory corruption but might reduce performance. To enable slub debug, run the following command:

```
nutanix@pcvm$ ncli cluster edit-pcvm-security-params enable-slub-debug=true
```

- **SSH Security Level:** The SSH security level setting allows you to enable different security levels for the `nutanix` user for accessing the cluster using SSH login. You can configure the security levels with one of the following options:
 - **Default:** The `nutanix` user can start an SSH session with either a password or an SSH key without any change to the account privileges.
 - **Limited:** For SSH sessions started with a password, the `nutanix` user is switched to an admin user that has lower operational privileges. The SSH key-based logins do not change the `nutanix` user's privileges.
 - **Restricted:** For all SSH sessions, whether started with a password or an SSH key, the `nutanix` user is switched to an admin user that has lower operational privileges.

To configure the SSH security level, run the following command:

```
nutanix@pcvm$ ncli cluster edit-pcvm-security-params ssh-security-level=DEFAULT
```

The admin user, while having lower privileges than the `nutanix` user, is designed to have sufficient privileges for administrative tasks. For more information, see [Controller VM Access](#) in the AHV Administration Guide.

In addition to configuring the SSH security level, you can also consider cluster lockdown to disable password-based SSH authentication by adding SSH keys. For more information, see [Configuring Cluster Lockdown in Prism Element](#) on page 117.

- **DoDin Additional:** The DoDin additional setting provides enhanced options for maintaining DISA compliance. It allows permanent account locking after an incorrect password is entered, as opposed to implementing a temporary 15-minute timeout period. Also, it makes AIDE send email messages to the account associated with NCC and runs AIDE more frequently. To enable DoDin, run the following command:

```
nutanix@pcvm$ ncli cluster edit-pcvm-security-params enable-dodin-opts=true
```

- **Fapolicy:** Fapolicy is a RHEL 8 security framework for managing access control.

Note: Nutanix recommends that you do not set the Fapolicy setting to true unless there is a strict organisation policy to enable it. Enabling the FaPolicy setting might reduce performance.

- **Processor Mitigation:** The processor mitigation setting enables mitigations against processor-based vulnerabilities. Enabling processor mitigation protects against certain types of attacks but might reduce performance. To enable processor mitigation, run the following command:

```
nutanix@pcvm$ ncli cluster edit-pcvm-security-params enable-processor-mitigations=true
```

- **Lock Status:** The lock status setting freezes the security configuration and prevents from making any changes to the security configuration through nCLI or API calls. To enable the lock status, run the following command:

```
nutanix@pcvm$ ncli cluster edit-pcvm-security-params enable-lock-status=true
```

Note: You must contact Nutanix Support to unlock the lock status setting.

IP Set Based Firewall

The IP set-based firewall feature implements strict firewall rules for intra cluster communication. This feature enables individual IP address filtering for intra cluster communication for various internal services. Prism considers only individual Controller VM and hypervisor IP addresses as trusted sources to receive RPC and API request from other CVMs in the same cluster.

The IP set-based firewall feature is enabled by default in AOS version 6.8.

ELIMINATE THE DEFAULT PASSWORDS DURING A CLUSTER CREATION

Enhance cluster security during creation by eliminating default passwords for the `nutanix` user and CVMs.

Hackers often exploit default passwords, which are a well-known security vulnerability. Removing default passwords and providing mechanisms for more secure SSH access configurations, strengthens the overall security posture of your Nutanix clusters.

During cluster creation, you can configure the following security settings to eliminate the default passwords:

- Applying **Password Lockdown Mode** to the cluster restricts password-based SSH access to `nutanix` and `admin` accounts.
- Applying **Lockdown Mode** to the cluster restricts both password-based and public SSH key based authentication.
- Adding public SSH keys from trusted systems further secures remote access, without relying on password-based authentication.

The system maintains the new password and public SSH keys during the cluster management workflow, such as adding a node to a cluster, removing a node from the cluster, or performing a break-fix operation.

Note: Elimination of default passwords supported only for `nutanix` user and CVM.

Removing the Default SSH Password of a CVM

During a cluster creation, you can remove the default SSH password for the Controller VMs (CVMs) `nutanix` user for enhanced security.

About this task

To remove the default SSH password of a CVM during the cluster creation, follow these steps:

Procedure

1. SSH into any of the Controller VMs (CVMs) as a `nutanix` user:

```
$ ssh nutanix@cvm_ip_address
```

2. Create a cluster and set the SSH access mode:

```
nutanix@cvm$ cluster --cluster_create_password_enforcement=true --  
cluster_create_security_opts=true --encrypted_password="encrypted_string" create
```

Replace `encrypted_string` with the encrypted password to be applied on the cluster. For more information, see [Controller VM Password Complexity Requirements](#).

Enabling SSH Access to a CVM Using a Public Key

During a cluster creation, you can enable SSH access to a Controller VM (CVM) using a public key and remove the default SSH password of a CVM for enhanced security.

About this task

To enable SSH access to a CVM using a public key and remove the default SSH password of a CVM during the cluster creation, follow these steps:

Procedure

1. SSH into any of the Controller VMs (CVMs) as a `nutanix` user:

```
$ ssh nutanix@cvm_ip_address
```

2. Create a cluster and set the SSH access mode:

```
nutanix@cvm$ cluster --password_lockdown_mode=true --  
cluster_create_password_enforcement=true --cluster_create_security_opts=true --  
encrypted_password="encrypted_string" --external_access_keys="public_key" create
```

Replace `encrypted_string` with the encrypted password to be applied on the cluster. For more information on password complexity requirements, see [Controller VM Password Complexity Requirements](#).

Replace `public_key` with the public SSH key value of the trusted system to allow SSH access to the cluster.

The cluster creation begins by securing the CVM with the following system configuration:

- Replaces the CVM's default password with the that password you entered.
- Restricts SSH access to the CVM using a password.
- Maintains the new password and public SSH key during cluster management workflows, such as adding a node, removing a node, or performing a breakfix operation.

3. (Optional) To create a cluster with multiple public SSH key based access:

```
nutanix@cvm$ cluster --password_lockdown_mode=true --  
cluster_create_password_enforcement=true --cluster_create_security_opts=true --  
external_access_keys="public_key_1", "public_key_2" create
```

Enabling Cluster Lockdown Mode

During a cluster creation, you can enable cluster lockdown mode to disable both password-based and public SSH key-based authentication to a cluster. This method provides the highest level of security for the Controller VM, along with eliminating default password usage.

About this task

To enable cluster lockdown mode during the cluster creation, follow these steps:

Procedure

1. SSH into any of the Controller VMs (CVMs) as a `nutanix` user:

```
$ ssh nutanix@cvm_ip_address
```

2. Create a cluster with lockdown mode enabled:

```
nutanix@cvm$ cluster --lockdown_mode=true --cluster_create_password_enforcement=true  
--cluster_create_security_opts=true create
```

The cluster creation starts and the system restricts the password-based and SSH key based authentication.

FIREWALL BEST PRACTICES

Protect Nutanix components with a stateful firewall that allows traffic only from trusted management endpoints.

The [Ports and Protocols](#) page provides detailed information about ports used by Nutanix products and services including transfer protocol, description, source, destination, and associated service.

Nutanix recommends that the networks for the hypervisor, CVM, Prism Central, and other Nutanix components are protected behind a stateful firewall in the network environment. This firewall should allow traffic sources only from known management endpoints.

SECURING TRAFFIC THROUGH NETWORK SEGMENTATION

Network segmentation enhances security, resilience, and cluster performance by isolating a subset of traffic to its own network.

You can achieve traffic isolation in one or more of the following ways:

Isolating Backplane Traffic by using VLANs (Logical Segmentation)

You can separate management traffic from storage replication (or backplane) traffic by creating a separate network segment (LAN) for storage replication. For more information about the types of traffic seen on the management plane and the backplane, see [Traffic Types In a Segmented Network](#) on page 163.

To enable the CVMs in a cluster to communicate over these separated networks, the CVMs are multihomed. Multihoming is facilitated by the addition of a virtual network interface card (vNIC) to the Controller VM and placing the new interface on the backplane network. Additionally, the hypervisor is assigned an interface on the backplane network.

The traffic associated with the CVM interfaces and host interfaces on the backplane network can be secured further by placing those interfaces on a separate VLAN.

In this type of segmentation, both network segments continue to use the same external bridge and therefore use the same set of physical uplinks. For more information, see [Physically Isolating the Backplane Traffic on an AHV Cluster](#) on page 176.

Isolating backplane traffic from management traffic requires minimal configuration through the Prism Element web console. No manual host (hypervisor) configuration steps are required.

For more information, see [Isolating the Backplane Traffic Logically on an Existing Cluster](#) on page 174.

Isolating Backplane Traffic Physically (Physical Segmentation)

You can physically isolate the backplane traffic (intra-cluster traffic) from the management traffic (Prism Element, SSH, SNMP) in to a separate vNIC on the CVM and using a dedicated virtual network that has its own physical NICs. This type of segmentation offers true physical separation of backplane traffic from management traffic.

You can use Prism Element to configure the vNIC on the CVM and configure backplane traffic to communicate over the dedicated virtual network. However, you must first manually configure the virtual network on the hosts and associate it with the physical NICs that is required for true traffic isolation.

For more information, see [Physically Isolating the Backplane Traffic on an AHV Cluster](#) on page 176.

Isolating service-specific traffic

You can also secure traffic associated with a service (for example, Nutanix Volumes) by confining its traffic to a separate vNIC on the CVM and using a dedicated virtual network that has its own physical NICs. This type of segmentation offers true physical separation for service-specific traffic.

You can use Prism Element to create the vNIC on the CVM and configure the service to communicate over the dedicated virtual network. However, you must first manually configure the virtual network on the hosts and associate it with the physical NICs that is required for true traffic isolation. You need one virtual network for each service that you plan to isolate. For a list of the services whose traffic you can isolate in the current release, see [Cluster Services That Support Traffic Isolation](#) on page 171.

For more information, see [Isolating Service-Specific Traffic](#) on page 189.

Isolating Stargate-to-Stargate traffic over RDMA

Some Nutanix platforms support remote direct memory access (RDMA) for Stargate-to-Stargate service communication. You can create a separate virtual network for RDMA-enabled NICs. If a node has RDMA-enabled NICs, Foundation passes the NICs through to the CVMs during imaging. The CVMs use only the first of the two RDMA-enabled NICs for Stargate-to-Stargate communications. The virtual NIC on the CVM is named `rdma0`. Foundation does not configure the RDMA LAN. After creating a cluster, you need to enable RDMA by creating an RDMA LAN from Prism Central. For more information, see [Remote Direct Memory Access](#) in the *NX Series Hardware Administration Guide*.

For more information, see [Isolating the Backplane Traffic on an Existing RDMA Cluster](#) on page 200.

Traffic Types In a Segmented Network

Backplane and management traffic are two types of traffic entering and leaving a Nutanix cluster in a segmented network:

Backplane traffic

Backplane traffic is intra-cluster traffic that is necessary for the cluster to function, and it comprises traffic between CVMs and traffic between CVMs and hosts for functions such as storage RF replication, host management, high availability, and so on. This traffic uses `eth2` on the CVM. In AHV, VM live migration traffic is also backplane, and uses the AHV backplane interface, VLAN, and virtual switch when configured. For nodes that have RDMA-enabled NICs, CVMs use a separate RDMA LAN for Stargate-to-Stargate communications.

Management traffic

Management traffic refers to administrative traffic or traffic associated with Prism Element and SSH connections, remote logging, SNMP, and so on. The current implementation simplifies the definition of management traffic to include any traffic that is not on the backplane network and includes communications between user VMs and CVMs. This traffic uses `eth0` on the CVM.

You can further isolate traffic on the management plane based on specific services or features. For example, the traffic that the cluster receives from an external iSCSI initiators (such as Nutanix Volumes iSCSI traffic) is categorized as management traffic. For a list of services supported in the current release, see [Cluster Services That Support Traffic Isolation](#) on page 171.

Segmented and Unsegmented Networks

Configurations for segmented and unsegmented networks for Nutanix clusters (both ESXi and AHV).

In a default unsegmented network configuration for Nutanix clusters (both ESXi and AHV), the Controller VM (CVM) has two virtual NICs: `eth0` and `eth1`.

Interface `eth0` is connected to a default external virtual switch, which is in turn connected to the external network through a bond or NIC team that contains the host physical uplinks.

Interface `eth1` is connected to an internal network that enables the CVM to communicate with the hypervisor.

The figures *Unsegmented Network - ESXi Cluster*, and *Unsegmented Network - AHV Cluster* shows an unsegmented network, and all external CVM traffic, whether backplane or management traffic, uses interface `eth0`. These interfaces are on the default VLAN on the default virtual switch.

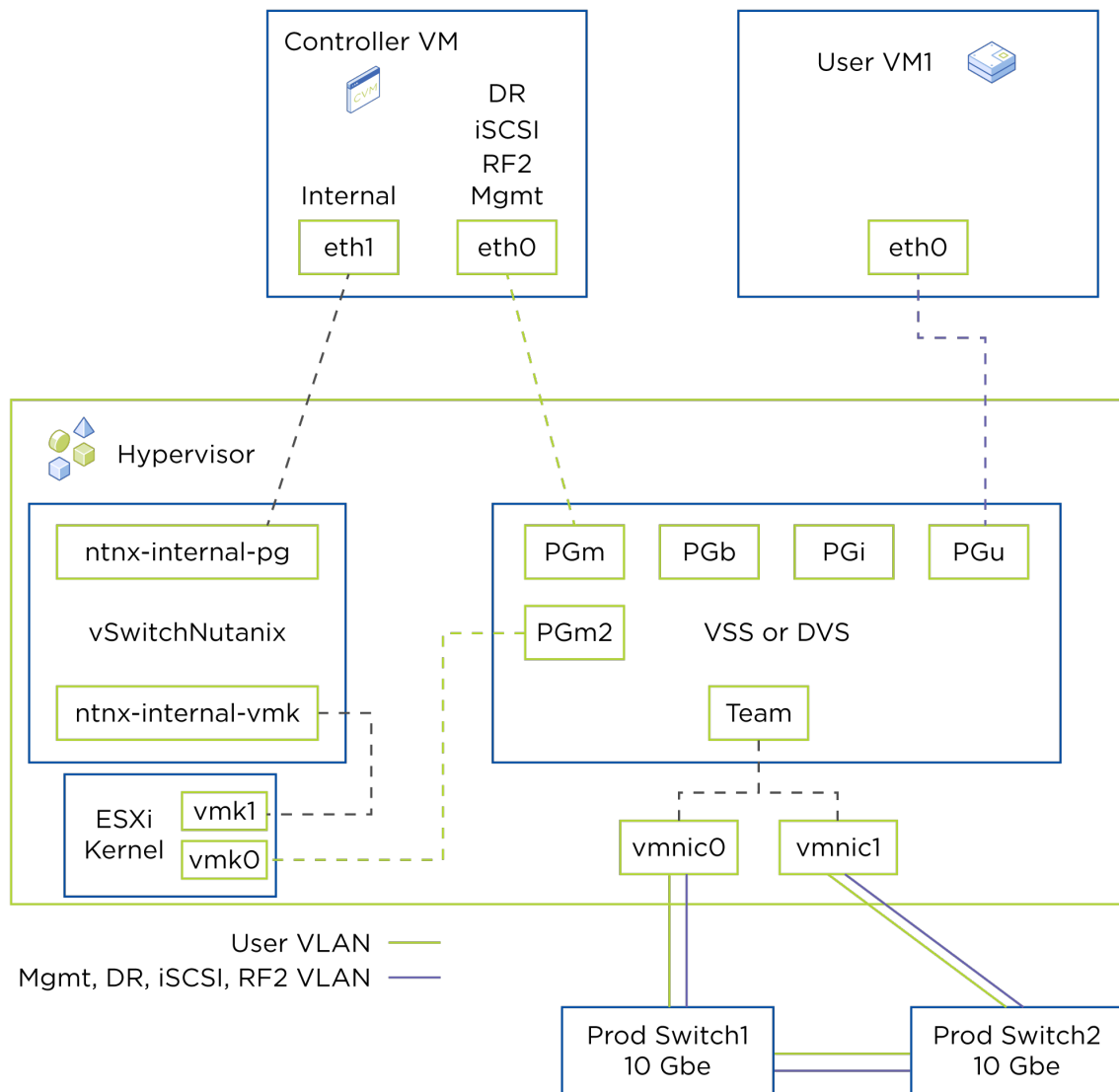


Figure 24: Unsegmented Network- ESXi Cluster

The following figure *Unsegmented Network- AHV Cluster* shows an unsegmented network on an AHV cluster.

In AHV, VM live migration traffic is also backplane and uses the AHV backplane interface, VLAN, and virtual switch when configured.

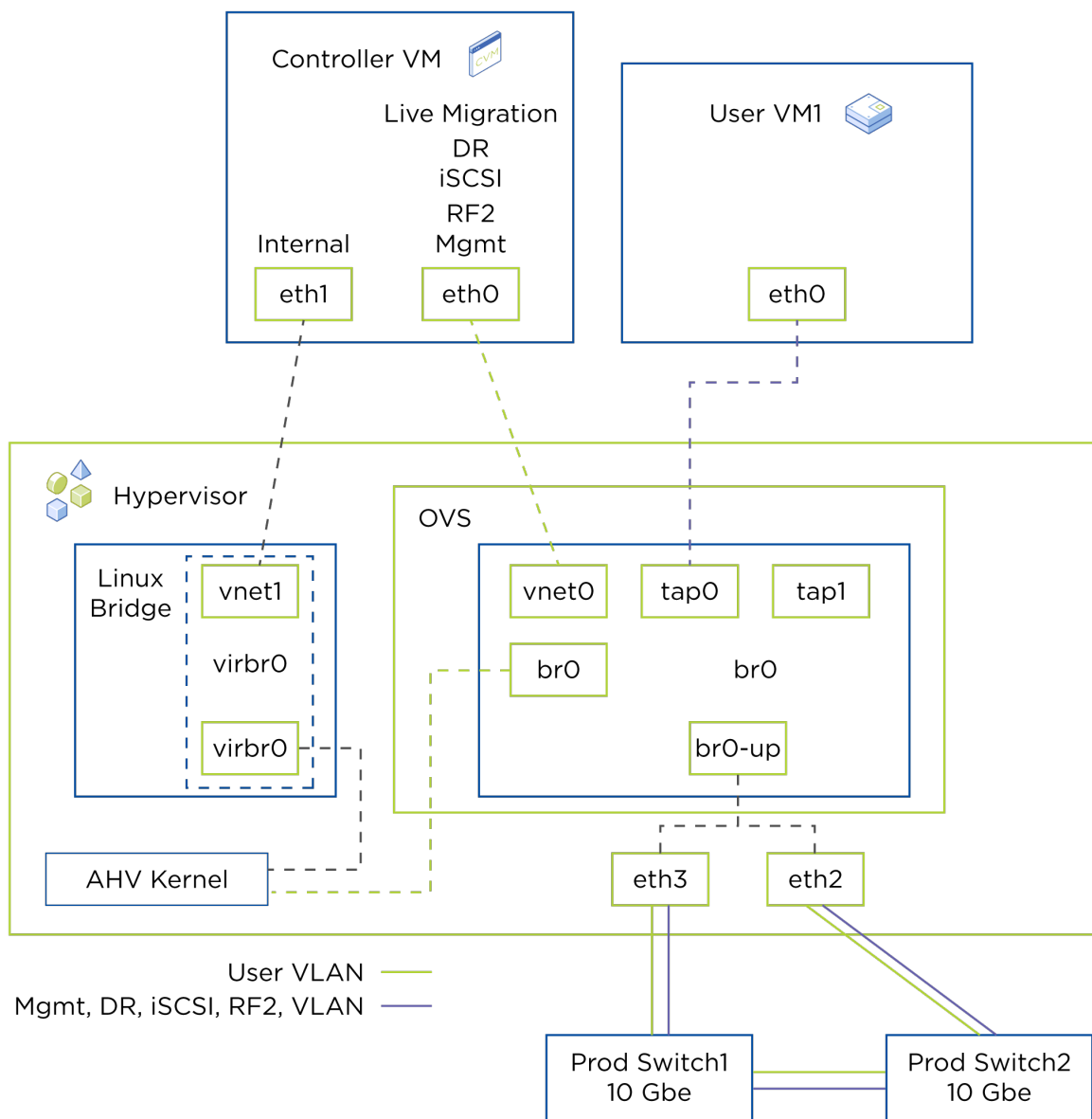


Figure 25: Unsegmented Network- AHV Cluster

If you further isolate service-specific traffic, additional vNICs are created on the CVM. Each service requiring isolation is assigned a dedicated virtual NIC on the CVM. The NICs are named `ntnx0`, `ntnx1`, and so on. Each service-specific NIC is placed on a configurable existing or new virtual network (vSwitch or bridge) and a VLAN and IP subnet are specified.

Network with Segmentation

In a segmented network, management traffic uses CVM interface `eth0` and additional services can be isolated to different VLANs or virtual switches. In backplane segmentation, the backplane traffic uses interface `eth2`. The backplane network uses either the default VLAN or a separate VLAN that you specify when segmenting the network. In ESXi, you must select a port group for the new vmkernel interface. In AHV, this internal interface is created automatically in the selected virtual switch. For physical separation of the backplane network, create this new port group on a separate virtual switch in ESXi or select a desired virtual switch in the AHV GUI.

To isolate service-specific traffic such as Volumes or Disaster Recovery as well as backplane traffic, additional vNICs are needed on the CVM, but no new vmkernel adapters or internal interfaces are required. AOS creates additional vNICs on the CVM, then assigns each service that requires isolation a dedicated vNIC on the CVM. The NICs are named `ntnx0`, `ntnx1`, and so on. AOS places each service-specific NIC on a configurable existing or new virtual network (vSwitch or bridge) and specifies a VLAN and IP subnet.

You can choose to perform backplane segmentation alone with no other forms of segmentation. You can also choose to use one or more types of service specific-segmentation with or without backplane segmentation. In all of these cases, you can choose to segment any service to either the existing virtual switch or a new virtual switch for further physical traffic isolation. The combination you select is driven by the security and networking requirements of your deployment. In most cases, Nutanix recommends the default configuration with no segmentation of any kind due to simplicity and ease of deployment.

The following figure shows an implementation scenario where the backplane and service-specific segmentation are configured with two vSwitches on ESXi hypervisors.

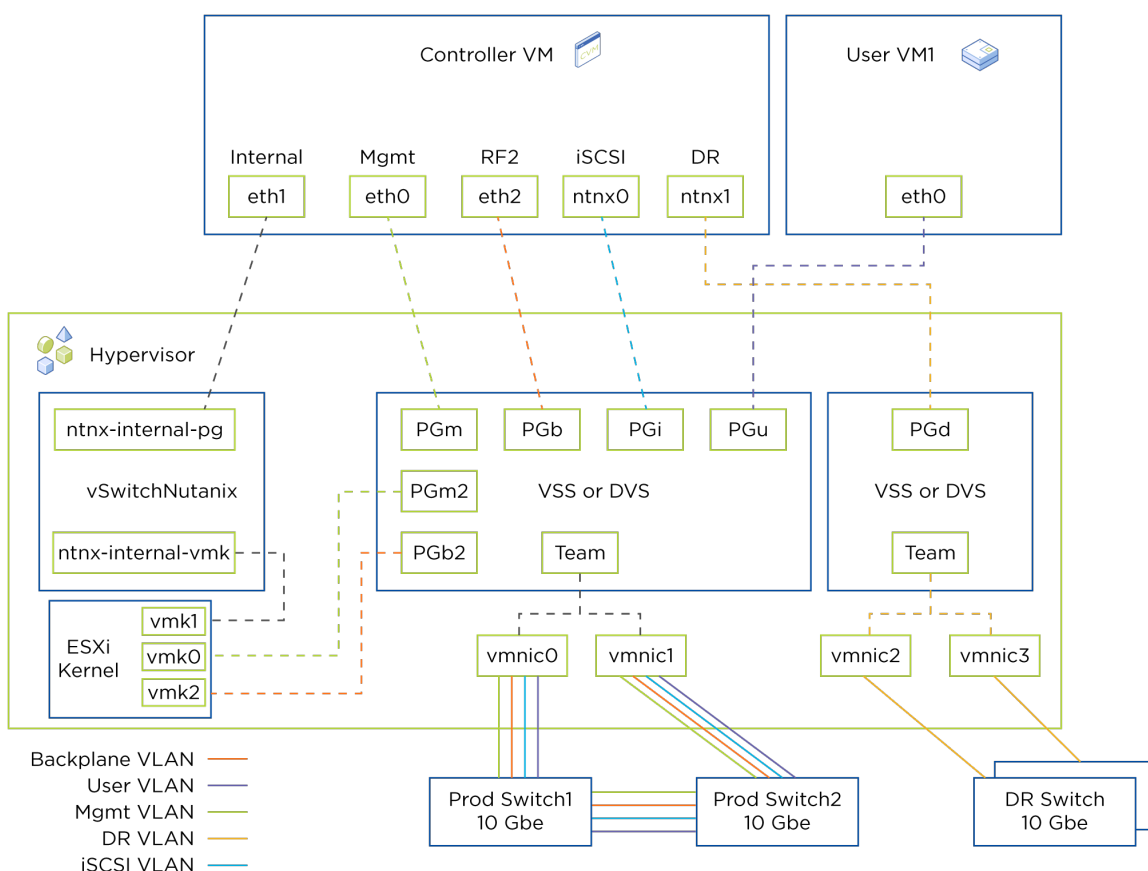


Figure 26: Backplane and Service-specific Segmentation Configured with two vSwitches on an ESXi Cluster

The following are the CVM to ESXi hypervisor connection details:

- The `eth0` vNIC on the CVM and `vmk0` on the host are carrying management traffic and connected to the hypervisor through the existing PGm (portgroup) on vSwitch0.
- The `eth2` vNIC on the CVM and `vmk2` on the host are carrying backplane traffic and connected to the hypervisor through a new user created PGb on the existing vSwitch.

- The **ntnx0** vNIC on the CVM is carrying iSCSI traffic and connected to the hypervisor through **PGi** on the vSwitch1. No new vmkernel adapter is required.
- The **ntnx1** vNIC on the CVM is carrying DR traffic and connected to the hypervisor through **PGd** on the vSwitch2. No new vmkernel adapter is required.

The following figure shows an implementation scenario where the backplane and service-specific segmentation are configured with two vSwitches on an AHV hypervisors. In the figure, the interface name **br0-bp** is read as **br0-backplane**.

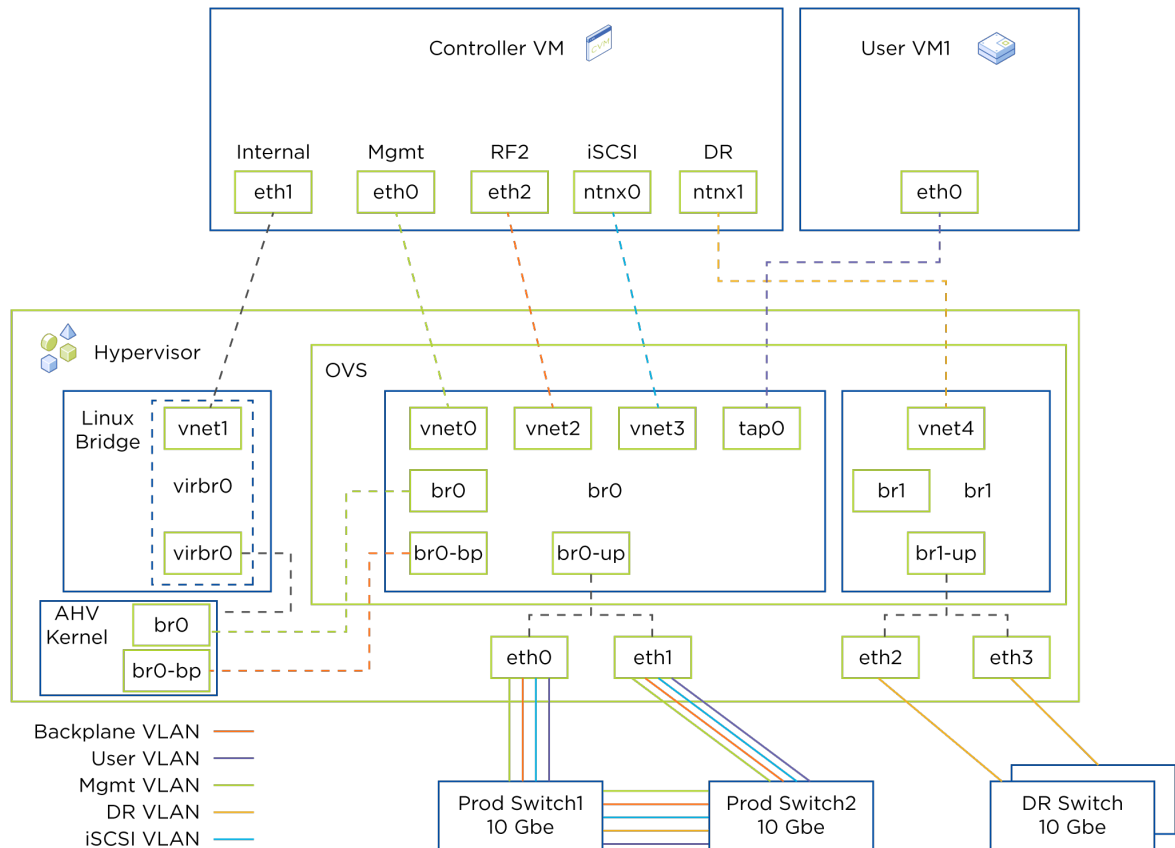


Figure 27: Backplane and Service Specific Segmentation Configured with two vSwitches on an AHV Cluster

The following are the CVM to AHV hypervisor connection details:

- The **eth0** vNIC on the CVM is carrying management traffic and connected to the hypervisor through the existing **vnet0**.
- Other vNICs such as **eth2**, **ntnx0**, and **ntnx1** are connected to the hypervisor through the auto created interfaces on either the existing or new vSwitch.

The following table describes the vNIC, port group (PG), VM kernel (VMK), virtual network (VNET) and virtual switch connections for CVM and hypervisor in different implementation scenarios. The tables capture information for ESXi and AHV hypervisors:

Table 12:

Implementation Scenario	vNIC on CVM	Connected to ESXi	Connected to AHV
Backplane segmentation with one vSwitch	eth0: DR, iSCSI, and Management traffic	vmk0 via existing PGm on vSwitch	Existing vnet0
	eth2: Backplane traffic	New vmk2 via PGb on vSwitch0 CVM vNIC via PGb on vSwitch0	Auto created interfaces on bridge br0
Backplane segmentation with two vSwitches	eth0: Management traffic	vmk0 via existing PGm on vSwitch0	Existing vnet0
	eth2: Backplane traffic	New vmk2 via PGb on new vSwitch CVM vNIC via PGb on new vSwitch	Auto created interfaces on new virtual switch
Service-specific segmentation for Volumes with one vSwitch	eth0: DR, Backplane, and Management traffic	vmk0 via existing PGm on vSwitch0	Existing vnet0
	ntnx0: iSCSI (Volumes) traffic	CVM vNIC via PGi on vSwitch0	Auto created interface on existing br0
Service-specific segmentation for Volumes with two vSwitches	eth0: DR, Backplane, and Management traffic	vmk0 via existing PGm on vSwitch0	Existing vnet0
	ntnx0: iSCSI (Volumes) traffic	CVM vNIC via PGi on new vSwitch	Auto created interface on new virtual switch
Service-specific segmentation for DR with one vSwitch	eth0: iSCSI, Backplane, and Management traffic	vmk0 via existing PGm on vSwitch0	Existing vnet0
	ntnx1: DR traffic	CVM vNIC via PGd on vSwitch0	Auto created interface on existing br0
Service Specific Segmentation for DR with two vSwitches	eth0: iSCSI, Backplane, and Management traffic	vmk0 via existing PGm on vSwitch0	Existing vnet0
	ntnx1: DR traffic	CVM vNIC via PGd on new vSwitch	Auto created interface on new virtual switch

Implementation Scenario	vNIC on CVM	Connected to ESXi	Connected to AHV
Backplane and Service-specific segmentation with one vSwitch	eth0: Management traffic	vmk0 via existing PGm on vSwitch0	Existing vnet0
	eth2: Backplane traffic	New vmk2 via PGb on vSwitch0 CVM vNIC via PGb on vSwitch0	Auto created interfaces on br0
	ntnx0: iSCSI traffic	CVM vNIC via PGi on vSwitch0	Auto created interface on br0
	ntnx1: DR traffic	CVM vNIC via PGd on vSwitch0	Auto created interface on br0
Backplane and Service-specific segmentation with two vSwitches	eth0: Management traffic	vmk0 via existing PGm on vSwitch0	Existing vnet0
	eth2: Backplane traffic	New vmk2 via PGb on new vSwitch CVM vNIC via PGb on new vSwitch	Auto created interfaces on new virtual switch
	ntnx0: iSCSI traffic	CVM vNIC via PGi on vSwitch1 No new user defined vmkernel adapter is required.	Auto created interface on new virtual switch
	ntnx1: DR traffic	CVM vNIC via PGd on vSwitch2. No new user defined vmkernel adapter is required.	Auto created interface in new virtual switch

Supported Environment

AOS and hypervisor compatibility information to support network segmentation.

The following are the hypervisor compatibility for network segmentation:

Segmentation Type	Supported Hypervisor
Network segmentation by traffic type where you can separate backplane traffic from management traffic	AHV, ESXi, Hyper-V
Service-specific traffic isolation	AHV, ESXi

The following are the AOS requirements for network segmentation:

Segmentation Type	AOS Version
Logical network segmentation	5.5 or later
Physical network segmentation	5.11 or later
Service-specific traffic isolation	5.11 or later

RDMA Requirements for Network Segmentation

The following is the RDMA requirement to support network segmentation:

Network segmentation is supported with RDMA for AHV and ESXi hypervisors only. For more information about RDMA, see [Remote Direct Memory Access](#) in the *NX Series Hardware Administration Guide*.

Prerequisites

Ensure that the following prerequisites are met before you configure network segmentation:

- Ensure that proxy ARP is disabled within the Nutanix VLAN before configuring network segmentation.
- If the cluster is registered to Prism Central and currently uses a segmented Data Services IP (DSIP) located in a secondary subnet, you must configure an additional DSIP within the original subnet as Prism Central. This is required because:
 - Prism Central cannot perform upgrade operations using a segmented DSIP.
 - The segmented DSIP and the cluster DSIP are distinct entities.
 - A DSIP in the same subnet ensures direct communication between Prism Central and the cluster during the upgrade process.

Failure to configure a DSIP in the appropriate subnet will prevent Prism Central from initiating or completing cluster upgrades.

Nutanix Volumes

Stargate does not monitor the health of a segmented network. If physical network segmentation is configured, network failures or connectivity issues are not tolerated. To overcome this issue, configure redundancy in the network. That is, use two or more uplinks in a fault tolerant configuration, connected to two separate physical switches.

Disaster Recovery

Ensure that the following prerequisites are met before you configure network segmentation for disaster recovery:

- Ensure that the VLAN and subnet that you plan to use for the network segment are routable.
- Ensure that you have a pool of IP addresses to specify when configuring segmentation. For each cluster, you need $n+1$ IP addresses, where n is the number of nodes in the cluster. The additional IP address is for the virtual IP address requirement.
- Ensure that network segmentation for disaster recovery is enabled at both sites (local and remote) before configuring remote sites at those sites.

Limitations

Nutanix Volumes

- When you enable network segmentation for Volumes, you cannot recover volume group attachments during VM recovery.

- Nutanix service VMs such as Objects worker nodes continue to communicate with the CVM `eth0` interface when using Volumes for iSCSI traffic. Other external clients, such as Files, use the new service-specific CVM interface.

Cluster Services That Support Traffic Isolation

You can isolate traffic associated with the following services to its own virtual network:

- Management (the default network that cannot be moved from CVM `eth0`)
- Backplane
- RDMA
- Service-specific disaster recovery
- Service-specific Volumes

Unsupported Configurations for Network Segmentation

The following cluster configurations do not support network segmentation:

- Clusters on which the CVMs have a manually created `eth2` interface.
- Clusters on which the `eth2` interface on one or more CVMs have manually assigned IP addresses. During an upgrade to an AOS release that supports network segmentation, AOS creates an `eth2` interface on each CVM in a cluster. Even though the cluster does not use these interfaces until you configure network segmentation, you must not manually configure these interfaces in any way.
- Clusters on which CVM interfaces are connected to port groups backed by NSX NVDS switches.

Caution:

Nutanix deprecated support for manual multi-homed CVM network interfaces from AOS version 5.15 and later. Such a manual configuration can lead to unexpected issues on these releases. If you configured an `eth2` interface on the CVM manually, see the [KB 9479](#) and Nutanix Field Advisory #78 for details on how to remove the `eth2` interface.

Troubleshooting Tips

This section provides information to assist troubleshooting network segmentation deployments.

The Failed to restart one or more services after Backplane was enabled error might occur while enabling network segmentation. In such cases, the network segmentation task gets completed, however, restarting one or more services fails to complete on time.

To ensure that the necessary services starts on time, log in to a CVM over SSH and run the following command:

```
nutanix@CVM:~$ cluster start
```

Verify the service status and ensure that all services are listed as UP.

Prepare the Network on an AHV Host

Host networking on AHV enables traffic segmentation and must be configured on both existing cluster nodes and new, unconfigured nodes before isolation.

You must configure host networking for physical and service-specific network segmentation on an AHV host. Configuring host networking is a prerequisite for physical and service-specific network segmentation.

Two scenarios for configuring host networking on an AHV host:

- Segmenting traffic on nodes that are already part of a cluster; for more information, see [Configuring the Network on Existing Nodes](#) on page 172.
- Segmenting traffic on an unconfigured node that is not part of a cluster; for more information, see [Configuring the Network on New Nodes](#) on page 172.

For information about the procedures to create, update and delete a virtual switch in Prism Element Web Console, see [Configuring a Virtual Network for Guest VMs](#) in the *Prism Element Web Console Guide*.

Note: The term *unconfigured node* in the *Configuring the Network on Existing Nodes* and *Configuring the Network on New Nodes* procedures refers to a node that is not part of a cluster and is being prepared for cluster expansion.

If you are configuring host networking on an ESXi host, you create vSwitches and port groups to achieve the same results. For more information, see the ESXi documentation.

Configuring the Network on Existing Nodes

Configure host networking for physical and service-specific network segmentation on existing nodes.

About this task

Perform the following procedure if you are segmenting traffic on nodes that are already part of a cluster:

Procedure

1. Remove the uplinks from the default virtual switch `vs0`.
2. Create a virtual switch for the backplane traffic or service whose traffic you plan to isolate.

For information about creating a new virtual switch, see [Creating or Updating a Virtual Switch](#) in the *Prism Web Console Guide*.

3. Add the uplinks to the new virtual switch.

What to do next

Prism Element can configure a VLAN only on AHV hosts. If the hypervisor is ESXi, in addition to configuring the VLAN on the physical switch, you must configure the VLAN on the port group.

If you are performing physical network segmentation, see [Physically Isolate the Backplane Traffic on an Existing Cluster](#) on page 175.

If you are performing service-specific traffic isolation, see [Service-Specific Traffic Isolation](#) on page 189.

Configuring the Network on New Nodes

Configure host networking for physical and service-specific network segmentation on new nodes.

About this task

Perform the following procedure if you are segmenting traffic on an unconfigured node (new host) that is not part of a cluster:

Procedure

1. Log into the new AHV host as a root user using SSH and create a directory:

```
$ ssh root@IP-address-host "mkdir -p /dev/shm/config/"
```

Replace `IP-address-host` with the IP address of the newly created AHV host.

2. Create a cluster_config file:

```
$ ssh root@IP-address-host "echo -e '{\n \"arp_ip\": \"ARP-internal-IP-address\", \n \"dhcp_ip\": \"IP-address-cvm\" \n}' > /dev/shm/config/cluster_config.0"
```

Replace:

- `IP-address-host` with the IP address of the newly created AHV host.
- `ARP-internal-IP-address` with an internal IP address from the subnet reserved for Kubernetes infrastructure network, for example `192.168.5.2`

The newly created node must include a cluster_config file for the `manage_ovs` commands to work.

3. Log into the new AHV host as a nutanix user using SSH and create a bridge for the backplane traffic or service:

```
nutanix@cvm$ manage_ovs --bridge_name bridge-name create_single_bridge
```

Replace the `bridge-name` with a name of the bridge, for example `br0`.

Perform this step only on a newly imaged or foundation setup.

Note: After you run the `manage_ovs --bridge_name` command, the system shows the Failed to fetch gflags. Acropolis service might be down error message. You can ignore the error and proceed.

4. From the default bridge `br0`, log in to the host CVM and keep only `eth0` and `eth1` in `br0`:

```
nutanix@cvm$ manage_ovs --bridge_name br0 --interfaces eth0,eth1 --bond_name bond_name --bond_mode bond_mode update_uplinks
```

Replace:

- `bond-name` with the name of the uplink port such as `br0-up` for which you want to set the bond mode.
- `bond_mode` with a valid bond mode such as `active-backup`.

5. Log in to the host CVM and add `eth2` and `eth3` to the uplink bond of `br1`:

```
nutanix@cvm$ manage_ovs --bridge_name br1 --interfaces eth2,eth3 --bond_name bond-name --bond_mode active-backup update_uplinks
```

Replace the `bond-name` with a name of the uplink port corresponding to the bridge name `br1`, for example `br1-up`.

Note: If this step is not done correctly, it creates a network loop that causes a network outage. Ensure that no other uplink interfaces exist on this bridge before adding the new interfaces, and always add interfaces into a bond.

What to do next

Prism Element can configure a VLAN only on AHV hosts. If the hypervisor is ESXi, in addition to configuring the VLAN on the physical switch, you must configure the VLAN on the port group.

If you are performing physical network segmentation, see [Physically Isolate the Backplane Traffic on an Existing Cluster](#) on page 175.

If you are performing service-specific traffic isolation, see [Service-Specific Traffic Isolation](#) on page 189.

Network Segmentation for Traffic Types- Backplane and Management

You can segment the network in a Nutanix cluster to isolate traffic types such as backplane and management in the following ways:

- **On an existing cluster:** You can segment the network on an existing cluster using the Prism Element web console.

- **During cluster creation:** Segment the network when creating a cluster using Nutanix Foundation 3.11.2 or higher versions.

For more information about segmenting the network when creating a cluster, see the [Field Installation Guide](#).

Avoid using the following IP addresses when selecting an IP Address range for network segmentation:

- **169.254.0.0/16** - Automatic private IP Addresses range (APIPA)
- **192.168.5.0/24** - Internal backplane communication
- **192.168.7.2/27** - Cross cluster TCP Proxy subnet
- **10.100.0.0/16** and **10.200.32.0/24** - Objects internal services

Isolate the Backplane Traffic Logically Using VLAN-Based Segmentation

Use VLAN-based segmentation to isolate backplane traffic on an existing clusters using the Prism Element web console.

You must configure a separate VLAN for the backplane network to achieve logical segmentation. The network segmentation process creates a separate network for backplane communications on the existing default virtual switch. The process then places the **eth2** interfaces (that the process creates on the CVMs during an upgrade process) and the host interfaces on the newly created network. This method allows you to achieve logical segmentation of traffic over the selected VLAN. From the specified subnet, assign IP addresses to each new interface. You need two IP addresses per node. When you specify the VLAN ID, AHV places the newly created interfaces on the specified VLAN.

In this method, logical segmentation (VLAN-based segmentation) is done on the default bridge for AHV nodes. The process creates the host backplane interface (VMkernel) on the backplane network port group on ESXi or **br0-backplane** (interface) on **br0** bridge in case of AHV. The **eth2** interface on the CVM is on the CVM backplane network by default.

You don't need to manually create the VMkernel adapter for backplane segmentation as the workflow creates it on the port group selected as the host port group or default backplane network port group if none was selected.

You need separate VLANs for the management and backplane networks. For example, configure VLAN 100 as the management network VLAN and VLAN 200 as the backplane network VLAN on the Ethernet links that connect the Nutanix nodes to the physical switch.

Isolating the Backplane Traffic Logically on an Existing Cluster

Configure VLAN-based segmentation for backplane traffic on an existing ESXi and Hyper-V clusters.

Before you begin

- Use the RDMA-specific procedure if your cluster includes RDMA-enabled NICs.

For more information, see [Isolating the Backplane Traffic on an Existing RDMA Cluster](#) on page 200.

- For ESXi clusters, you must create and manage port groups that networking uses for CVM and backplane networking. Ensure that you create port groups on the default virtual switch **vs0** for the ESXi hosts and CVMs.

The backplane traffic segmentation is logical and based on the VLAN that you tag for the port groups. When creating the port groups, ensure that you tag the new port groups created for the ESXi hosts and CVMs with the appropriate VLAN ID. Consult your networking team to acquire the necessary VLANs for use with Nutanix nodes.

Note: Nutanix does not control these VLAN IDs.

- For new backplane networks, you must specify a non-routable subnet. The interfaces on the backplane network are automatically assigned IP addresses from this subnet, so reserve the entire subnet for the backplane network

segmentation. See the [Configuring Backplane IP Pool](#) on page 183 topic to create an IP pool for backplane interfaces.

- To segment the network on an existing AHV cluster for a backplane LAN, follow the procedure described in the [Physically Isolating the Backplane Traffic on an AHV Cluster](#) on page 176 topic.

About this task

To segment the network on an existing ESXi and Hyper-V clusters for a backplane LAN, follow these steps:

Procedure

1. Log in to the Prism Element web console as an administrator.
2. From the main menu, click the gear icon.
3. In the left pane, click **Network Configuration**.
The **Network Configuration** dialog box appears.
4. In the **Internal Interfaces** > **Backplane LAN** row, click **Configure**.
The **Create Interface** dialog box appears.
5. In the **Create Interface** dialog box, provide the necessary information:
 1. In the **Subnet IP** field, specify a non-routable subnet.
Ensure that the subnet has sufficient IP addresses. The segmentation process requires two IP addresses per node. Reconfiguring the backplane to increase the size of the subnet involves cluster downtime, so you might also want to make sure that the subnet can accommodate new nodes in the future.
 2. In the **Netmask** field, specify the netmask.
 3. (Optional) To assign the interfaces on the network to a VLAN, specify the VLAN ID in the **VLAN ID** field.
Nutanix recommends that you use a VLAN. If you do not specify a VLAN ID, Prism Element uses the default VLAN on the virtual switch.
 4. In the **Host Port Group** list, select the port group that you created for the host.
 5. In the **CVM Port Group** list, select the port group that you created for the CVM.

Note: For VLAN-based segmentation, Nutanix recommends that you leave the Host Port Group and CVM Port Group fields blank. Prism Element selects the default port group from vSwitch0 when you do not provide the Host Port Group and CVM Port Group details.

6. Click **Verify and Save**.

The network segmentation process creates the backplane network if the network settings that you specified pass validation.

Physically Isolate the Backplane Traffic on an Existing Cluster

Use the Prism Element web console to configure the `eth2` interface on a separate virtual switch. This setup isolates backplane traffic on a dedicated physical network.

If you don't configure a separate virtual switch, backplane traffic uses a different VLAN on the default switch for logical isolation.

A virtual switch is referred to differently across hypervisors:

- AHV - Virtual switch
- VMware ESXi - vSwitch
- Microsoft Hyper-V - Hyper-V Virtual Switch

Network segmentation process creates a separate network for backplane communications on the new virtual switch. The segmentation process places the CVM `eth2` interfaces and the host interfaces on the newly created network. Specify a subnet with a network mask and, optionally, a VLAN ID. From the specified subnet or an IP pool assign IP addresses to each new interface in the new network. You require a minimum of two IP addresses per node.

If you specify the optional VLAN ID, the segmentation process places the newly created interfaces on VLAN.

Nutanix highly recommends using separate VLAN for the backplane network to achieve true segmentation.

Requirements

- Ensure that physical isolation of backplane traffic is supported by the AOS version deployed.
- Ensure that you configure the network (port groups or bridges) on the hosts and associate the network with the required physical NICs before you enable physical isolation of the backplane traffic.

For AHV, see [Configuring the Network on Existing Nodes](#) on page 172. For ESXi and Hyper-V, see VMware or Microsoft documentation.

Limitations

- On an ESXi cluster with a DVS switch and network segmentation enabled, the ESXi hypervisor might trigger HA events such as host isolation response and reboot all VMs when the management network goes down. To avoid this situation, configure multiple isolation response addresses using the `das.isolationaddress` Advance Option in VMware vCenter.
- Segmenting backplane traffic can involve up to two rolling reboots of the CVMs. The first rolling reboot moves the backplane interface (`eth2`) of the CVM to the selected port group, virtual switch or Hyper-V switch. This is done only for CVMs whose backplane interface is not already connected to the selected port group, virtual switch or Hyper-V switch. The second rolling reboot migrates the cluster services to the newly configured backplane interface.

Physically Isolating the Backplane Traffic on an AHV Cluster

Create a dedicated virtual switch and configure the backplane network to physically segment backplane traffic on an AHV-based Nutanix cluster.

Before you begin

Ensure that the network is configured on the AHV hosts. For more information, see [Configuring the Network on Existing Nodes](#) on page 172.

Note: Before you perform the following procedure, ensure that the uplinks you added to the virtual switch are in the UP state.

About this task

To physically segment the backplane traffic on an AHV cluster, follow these steps:

Procedure

1. Shut down all the guest VMs in the cluster from within the guest OS or use the Prism Element web console.

2. Place all AHV hypervisor nodes of a cluster into the maintenance mode. Do not place the CVM nodes into the maintenance mode:

- a. Use SSH to log in to a Controller VM in the cluster.
- b. Determine the IP address of all the AHV hypervisor nodes:

```
nutanix@cvm$ acli host.list
```

Note all the hypervisor IP addresses for the cluster.

- c. Put each node in the maintenance mode:

```
nutanix@cvm$ acli host.enter_maintenance_mode hypervisor-IP-address [wait="{ true  
| false }" ] [non_migratable_vm_action="{ acpi_shutdown | block }" ]
```

Replace *hypervisor-IP-address* with the AHV hypervisor IP address.

The following are optional parameters:

- `wait`
- `non_migratable_vm_action`

For example:

```
nutanix@cvm$ acli host.enter_maintenance_mode 197.116.6.79  
EnterMaintenanceMode: pending  
EnterMaintenanceMode: complete
```

Note: Never put Controller VM and AHV hosts into maintenance mode on single-node clusters. Nutanix recommends shutting down user VMs before proceeding with disruptive changes.

If a cluster has HA reservation enabled, the nodes in the cluster cannot enter maintenance mode. To check if the cluster has HA reservation enabled, go to **Settings > Manage VM High Availability**.

Do not continue if the node fails to enter the maintenance mode.

- d. Verify that the node is in the maintenance mode:

```
nutanix@cvm$ acli host.get hypervisor-IP-address
```

In the output that is displayed, ensure that **node_state** is **EnteredMaintenanceMode** and **schedulable** is **False**.

Example output.

```
nutanix@cvm$ acli host.get 197.116.6.79  
197.116.6.79 {  
  cpu_usage_ppm: 192941  
  cvm_memory_size_bytes: 21474836480  
  cvm_num_vcpus: 8  
  cvm_num_vnics: 3  
  cvm_uuid: "e96dbef0-d425-4926-9fe8-4d10b1a69902"  
  host_overhead_bytes: 4957721854  
  logical_timestamp: 23  
  max_mem_ha_reserved_bytes: 0  
  mem_assigned_bytes: 0  
  mem_usage_bytes: 26654856446  
  memory_size_bytes: 269842644992  
  node_state: "EnteredMaintenanceMode"  
  num_cpus: 32  
  pool_size_bytes: 0  
  schedulable: False  
  uuid: "cc50ea78-49ce-4767-90b3-a0a2ef891446"
```

```
}
```

3. Enable backplane network segmentation:

- a. Log on to the Prism web console, click the gear icon in the top-right corner, then click **Network Configuration** in the **Settings** page.

- b. On the **Internal Interfaces** tab, in the **Backplane LAN** row, click **Configure**.

- c. In the **Backplane LAN** dialog box, follow these steps:

1. In **Subnet IP**, specify a non-routable subnet that is different from the subnet used by the AHV host and CVMs.

The AOS CVM default route uses the CVM `eth0` interface, and no route on the backplane interface. Therefore, Nutanix recommends that you use only a non-routable subnet for the backplane network. To avoid split routing, do not use a routable subnet for the backplane network.

Make sure that the backplane subnet has a sufficient number of IP addresses. Two IP addresses are required per node. Reconfiguring the backplane to increase the size of the subnet involves cluster downtime, so ensure that the subnet can accommodate new nodes in the future.

You can also use an IP pool to configure network segmentation. For more information, see [Configuring Backplane IP Pool](#) on page 183

2. In **Netmask**, specify the network mask.
3. (Optional) To assign the interfaces on the network to a VLAN, specify the VLAN ID in the **VLAN ID** field.

Nutanix strongly recommends configuring a separate VLAN. If you do not specify a VLAN ID, AOS applies the untagged VLAN on the virtual switch.

4. In the **Virtual Switch** list, select the virtual switch that you created for the backplane traffic.

- d. Click **Verify and Save**.

If the network settings that you specified pass validation, the segmentation process creates a backplane network and the CVMs perform a reboot in a rolling fashion (one at a time), after which the services use the new backplane network. You can track the progress of this operation on the Prism Element web console Tasks page.

4. Log in to a CVM in the cluster with SSH and stop Acropolis cluster-wide:

```
nutanix@cvm$ allssh genesis stop acropolis
```

5. Restart Acropolis cluster-wide:

```
nutanix@cvm$ cluster start
```

6. Remove all nodes from maintenance mode:

- a. From any CVM in the cluster, exit the AHV host from the maintenance mode:

```
nutanix@cvm$ acli host.exit_maintenance_mode hypervisor-IP-address
```

Replace *hypervisor-IP-address* with the IP address of the node.

For example,

```
nutanix@cvm$ acli host.exit_maintenance_mode 197.116.6.79
ExitMaintenanceMode: pending
```

```
ExitMaintenanceMode: complete
```

This command migrates all the VMs that were previously running on the host back to the host.

- b. Verify that the node has exited the maintenance mode:

```
nutanix@cvm$ acli host.get hypervisor-IP-address
```

Replace *hypervisor-IP-address* with the IP address of the node.

In the output that is displayed, ensure that **node_state** is **kAcropolisNormal** or **AcropolisNormal** and **schedulable** is **True**.

Example output.

```
nutanix@cvm$ acli host.get 197.116.6.79
197.116.6.79 {
  cpu_usage_ppm: 192941
  cvm_memory_size_bytes: 21474836480
  cvm_num_vcpus: 8
  cvm_num_vnics: 3
  cvm_uuid: "e96dbef0-d425-4926-9fe8-4d10b1a69902"
  host_overhead_bytes: 4957721854
  logical_timestamp: 23
  max_mem_ha_reserved_bytes: 0
  mem_assigned_bytes: 0
  mem_usage_bytes: 26654856446
  memory_size_bytes: 269842644992
  node_state: "AcropolisNormal"
  num_cpus: 32
  pool_size_bytes: 0
  schedulable: True
  uuid: "cc50ea78-49ce-4767-90b3-a0a2ef891446"
}
```

7. Power on the guest VMs from the Prism Element web console.

Physically Isolating the Backplane Traffic on an ESXi Cluster

Configure a dedicated vSwitch, subnet, and VLAN to physically isolate backplane traffic and apply settings using the Prism Element web console for secure segmentation.

Before you begin

On the ESXi hosts, follow these steps:

1. Create a vSwitch for the backplane traffic.
2. From **vSwitch0**, remove the uplinks (physical NICs) that you plan to add to the vSwitch that you created for the backplane traffic.
3. On the backplane vSwitch, create one port group for the CVM and another for the host.

Ensure that at least one uplink is present in the Active Adaptors list for each port group if you have overridden the failover order.

You don't need to manually create the VMkernel adapter for backplane segmentation since the workflow takes care of creating it on the port group selected as the host port group.

For more information, see VMware ESXi documentation.

Before you perform the following procedure, ensure that the uplinks you added to the vSwitch are in the UP state.

About this task

To physically segment the backplane traffic, follow these steps:

Procedure

1. Log in to Prism Element as an administrator.
2. Click the gear icon in the top-right corner, then in the **Settings** page, click **Network Configuration**.
3. On the **Internal Interfaces** tab, in the **Backplane LAN** row, click **Configure**.

4. In the **Backplane LAN** dialog box, follow these steps:

- a. In **Subnet IP**, specify a non-routable subnet that is different from the subnet used by the ESXi host and CVMs.

The AOS CVM default route uses the CVM `eth0` interface, and no route on the backplane interface. Nutanix recommends that you use only a non-routable subnet for the backplane network. To avoid split routing, do not use a routable subnet for the backplane network.

Make sure that the subnet has a sufficient number of IP addresses. Each node requires two IP addresses. Reconfiguring the backplane to increase the size of the subnet involves cluster downtime, so you might also want to make sure that the subnet can accommodate new nodes in the future.

- b. In **Netmask**, specify the network mask.
- c. (Optional) To assign the interfaces on the network to a VLAN, specify the VLAN ID in the **VLAN ID** field. Nutanix strongly recommends configuring a separate VLAN. If you do not specify a VLAN ID, AOS applies the default VLAN on the virtual switch.
- d. In the **Host Port Group** list, select the port group that you created for the host.
- e. In the **CVM Port Group** list, select the port group that you created for the CVM.

A port group can be a standard vSwitch port group, a distributed vSwitch port group, or a NSX segment reflected as a distributed port group in the vCenter.

A port group can be a standard vSwitch port group, a distributed vSwitch port group, or a NSX segment reflected as distributed port group in the vCenter.

Note:

Nutanix clusters support both vSphere Standard Switches and vSphere Distributed Switches with either normal portgroups or NSX segment backed portgroups. However, you must configure only one type of virtual switches in one cluster. Configure all the backplane and management traffic in one cluster on either vSphere standard switches or vSphere distributed switches. Do not mix standard and distributed vSwitches on a single cluster. Make sure that you configure the same port group type (normal or NSX segment) on all nodes of the cluster.

5. Click **Verify and Save**.

If the network settings that you specified pass validation, the segmentation process creates the backplane network and the CVMs perform a reboot in a rolling fashion, after which the services use the new backplane network. You can track the progress of this operation in the Prism Element web console Tasks page.

Physically Isolating the Backplane Traffic on a Hyper-V Cluster

Configure a dedicated Hyper-V switch, subnet, and VLAN to physically isolate backplane traffic and apply settings using the Prism Element web console for secure segmentation.

Before you begin

On the Hyper-V hosts, perform these steps:

1. Create a Hyper-V Virtual Switch for the backplane traffic.

2. From the default External Switch, remove the uplinks (physical NICs) that you want to add to the backplane Virtual Switch you created for the backplane traffic.
3. On the backplane Virtual Switch, create a subnet and, optionally, assign a VLAN.

For more information, see Hyper-V documentation on the Microsoft portal.

Before you perform the following procedure, ensure that the uplinks that you added to the backplane Virtual Switch are in the UP state.

About this task

To physically segment the backplane traffic, follow these steps:

Procedure

1. Log in to Prism Element as an administrator.
2. Click the gear icon in the top-right corner, and then in the **Settings** page, click **Network Configuration**.
3. On the **Internal Interfaces** tab, in the **Backplane LAN** row, click **Configure**.
4. In the **Backplane LAN** dialog box, follow these steps:
 - a. In **Subnet IP**, specify a non-routable subnet that is different from the subnet used by the Hyper-V host and CVMs.

The AOS CVM default route uses the CVM `eth0` interface, and no route on the backplane interface. Nutanix recommends that you use only a non-routable subnet for the backplane network. To avoid split routing, do not use a routable subnet for the backplane network.

Make sure that the subnet has a sufficient number of IP addresses. Each node requires two IP addresses. Reconfiguring the backplane to increase the size of the subnet involves cluster downtime, so you might also want to make sure that the subnet can accommodate new nodes in the future.
 - b. In **Netmask**, specify the network mask.
 - c. (Optionally) To assign the interfaces on the network to a VLAN, specify the VLAN ID in the **VLAN ID** field. Nutanix strongly recommends that you configure a separate VLAN. If you do not specify a VLAN ID, AOS applies the default VLAN on the virtual switch.
 - d. In the **Bridge** list, select the Hyper-V switch that you created for the backplane traffic.
5. Click **Verify and Save**.

If the network settings you specified pass validation, the segmentation process creates the backplane network and the CVMs perform a reboot in a rolling fashion, after which the services use the new backplane network. You can track the progress of this operation in the Prism Element web console Tasks page.

Segmenting backplane traffic can involve up to two rolling reboots of the CVMs. The first rolling reboot moves the backplane interface (`eth2`) of the CVM to the selected port group or virtual switch. This is done only for CVMs whose backplane interface is not already connected to the selected port group or bridge virtual switch. The second rolling reboot migrates the cluster services to the newly configured backplane interface.

Enabling Physical Backplane Segmentation on Hyper-V Using CLI

Physical backplane segmentation support is now available on a cluster containing Hyper-V nodes.

About this task

To enable physical backplane segmentation on a cluster containing Hyper-V node using CLI, follow these steps:

Procedure

1. Log on to any CVM in the cluster using SSH.
2. Enable backplane segmentation on a Hyper-V node:

```
nutanix@CVM:~$ network_segmentation --backplane_network  
--ip_pool=IP-Pool-name  
--backplane_vlan=VLAN-ID  
--host_physical_network=hyperv_host_physical_network
```

- Replace *IP-Pool-name* with a user defined IP pool name
- Replace *VLAN-ID* with a backplane VLAN ID
- Replace *hyperv_host_physical_network* with the Hyper-V switch name

For example:

```
nutanix@CVM:~$ network_segmentation --backplane_network  
--ip_pool=BackplanePool  
--backplane_vlan=1234  
--host_physical_network=BackplaneSwitch
```

Enabling Backplane Network Segmentation on a Mixed Hypervisor Cluster

Enable backplane network segmentation on a mixed hypervisor cluster.

About this task

You can enable Backplane Network Segmentation on a mixed hypervisor cluster:

- ESXi and AHV storage-only nodes
- Hyper-V and AHV storage-only nodes

To enable backplane network segmentation on a mixed hypervisor cluster, follow these steps:

Procedure

1. Log in to any CVM in the cluster using SSH.
2. Enable network segmentation for backplane traffic:

- On a cluster containing ESXi and AHV storage only nodes:

```
nutanix@cvm$ network_segmentation --backplane_network  
--ip_pool=IP-pool-name  
--backplane_vlan=VLAN-ID  
[--esx_host_physical_network=ESXi-host-portgroup-name ]  
[--esx_cvm_physical_network=ESXi-cvm-portgroup-name ]  
[--ahv_host_physical_network=AHV-network-name ]
```

- On a cluster containing Hyper-V and AHV storage-only nodes:

```
nutanix@cvm$ network_segmentation --backplane_network  
--ip_pool=IP-pool-name  
--backplane_vlan=VLAN-ID  
[--hyperv_host_physical_network=HyperV-host-network-name ]  
[--ahv_host_physical_network=AHV-network-name ]
```

- Replace *IP-Pool-name* with a user defined IP pool name

- Replace *VLAN-ID* with a backplane VLAN ID
- Replace *ESXi-host-portgroup-name* with the ESXi host network name
- Replace *ESXi-cvm-portgroup-name* with the ESXi CVM network name
- Replace *AHV-network-name* with the AHV storage-only node bridge name
- Replace *HyperV-host-network-name* with the Hyper-V switch name

For example, enable network segmentation on a mixed hypervisor containing ESXi and AHV storage-only nodes:

```
nutanix@cvm$ network_segmentation --backplane_network
--ip_pool=BackplanePool
--backplane_vlan=1234
--esx_host_physical_network=host-pg
--esx_cvm_physical_network=cvm-pg
--ahv_host_physical_network=br1
```

Backplane IP Pool

Create and manage small IP pools for backplane traffic using the CLI to optimize IP usage and avoid full subnet allocation.

Network segmentation for backplane traffic previously required an entire subnet even if a cluster has a small number of nodes. Allocating an entire subnet resulted in inefficient use of IP addresses. The backplane IP address pool feature creates a small IP address pool instead of an entire subnet.

You can create an IP address pool using the `network_segmentation ip_pool` command. The named IP address pool includes one or more IP address ranges. For example, you can define one range from 172.16.1.100 to 172.16.1.105 and another range from 172.16.1.120 to 172.16.1.125, all within the same named IP address pool and subnet.

Currently, the Prism Element web console does not offer an option to create an IP address pool specifically for backplane segmentation. However, it does allow for the creation of small IP address pools for service-specific traffic, such as Nutanix Volumes and Nutanix Disaster Recovery (DR). You can use the new `network_segmentation ip_pool` CLI to create IP address pools for backplane, Nutanix Volumes, and Nutanix DR. You can also manage (edit, delete, and update) IP address pools that are created for backplane, Nutanix Volumes and Nutanix DR using the new CLI.

Configuring Backplane IP Pool

Create an IP pool for backplane interfaces using the CLI.

About this task

To configure IP address pool for backplane, perform the following steps:

Procedure

1. Log in to any CVM in the cluster using SSH.
2. Create a new IP address pool and define the IP address ranges:

```
nutanix@cvm$ network_segmentation --ip_pool_name=IP-Pool-name --
ip_pool_netmask=netmask
--ip_ranges="[('First-IP-Address', 'Last-IP-Address'), ('First-IP-Address', 'Last-IP-Address')]"
ip_pool create
```

- Replace *IP-Pool-name* with a user-defined IP address pool name
- Replace *Netmask* with a network mask in dot-decimal mask notation

- Replace *First-IP-Address* with the first IP address in the range
- Replace *Last-IP-Address* with the last IP address in the same range

For example:

```
nutanix@cvm$ network_segmentation --ip_pool_name=BackplanePool --
ip_pool_netmask=255.255.255.0
--ip_ranges="[('172.16.1.100', '172.16.1.105'), ('172.16.1.120', '172.16.1.125')]"
ip_pool create
```

Reconfiguring the Backplane Network

Backplane network reconfiguration is a CLI-driven procedure that you perform on any one of the CVMs in the cluster. The change is propagated to the remaining CVMs.

About this task

Caution: At the end of this procedure, the cluster stops and restarts, even if only the VLAN is changed, and involves cluster downtime.

To reconfigure the cluster, follow these steps:

Procedure

1. Log on to any CVM in the cluster using SSH.
2. Reconfigure the backplane network:

```
nutanix@cvm$ backplane_ip_reconfig [--backplane_vlan=vlan-id] \
[--backplane_ip_pool=ip_pool_name]
```

Replace *vlan-id* with the new VLAN ID, and *ip_pool_name* with the newly created backplane IP address pool.

For information on creating a backplane IP address pool, see [Configuring Backplane IP Pool](#) on page 183.

For example, reconfigure the backplane network to use VLAN ID 10 and newly created backplane IP pool *NewBackplanePool*:

```
nutanix@cvm$ backplane_ip_reconfig --backplane_vlan=10 \
--backplane_ip_pool=NewBackplanePool
```

Example output:

```
This operation will do a 'cluster stop', resulting in disruption of
cluster services. Do you still want to continue? (Type "yes" (without quotes)
to continue)
Type yes to confirm that you want to reconfigure the backplane network.
```

During the reconfiguration process, you might receive an error message similar to the following:

```
Failed to reach a node.
```

You can ignore this error message and therefore do not stop the script manually.

Note: The `backplane_ip_reconfig` command is not supported on ESXi clusters with vSphere Distributed Switches. To reconfigure the backplane network on a vSphere Distributed Switch setup, disable the backplane network (see [Disabling Network Segmentation on an ESXi and Hyper-V Clusters](#) on page 186) and enable again with a different subnet or VLAN.

3. Type yes to confirm.

The reconfiguration procedure takes a few minutes and includes a cluster restart. If you type anything other than yes, the system aborts the network reconfiguration.

4. After the process completes, verify that the backplane was reconfigured:

- a. Verify that the IP addresses of the `eth2` interfaces on the CVM are set correctly:

```
nutanix@cvm$ svmips -b
```

Output similar to the following is displayed:

```
172.30.25.1 172.30.25.3 172.30.25.5
```

- b. Verify that the IP addresses of the backplane interfaces of the hosts are set correctly.

```
nutanix@cvm$ hostips -b
```

Output similar to the following is displayed:

```
172.30.25.2 172.30.25.4 172.30.25.6
```

The `svmips` and `hostips` commands, when used with the option `b`, display the IP addresses assigned to the interfaces on the backplane.

Update Backplane Port Groups

Update backplane port groups on ESXi clusters without disabling segmentation.

You can update the backplane port groups that are assigned to CVM and host nodes. Previously, to change a port group that is assigned to a CVM and host, you had to disable network segmentation and re-enable it with the new port groups.

This feature is only supported on a cluster running an ESXi hypervisor.

Updating backplane port groups:

- Helps you to move from one vSphere Standard Switch (VSS) port group to another VSS port group within the same virtual standard switch
- Helps you to move from one VSS port group to another VSS port group in a different Virtual Standard Switch
- Helps you to move from a VSS port group to a vSphere Distributed Switch (VDS) port group
- Helps you to move from a VDS port group to a VSS

Note:

To rename existing VSS or VDS port groups, you need to manually perform the rename operation either through the vCenter application or by using the ESXi CLI. Run the update operation with the new port group name. This process ensures that the configuration in the Nutanix internal database is kept up to date.

Limitations of Updating Backplane Port Group

Consider the following limitations before updating backplane port groups:

- Clusters running on an AHV or Hyper-V hypervisors do not support updating backplane port groups feature.
- This feature does not support updating any other configuration such as VLAN ID and IP addresses.

Note: This feature does not perform any network validation on the new port groups. You must ensure the port group settings are accurate before proceeding with the port group update operation. If the settings are not accurate, the CVM on that node might not be able to communicate with its peers and this results in a stuck rolling reboot.

Updating Backplane Port Group

Update the backplane port groups that are assigned to CVM and host nodes.

About this task

To update the backplane port groups that are assigned to CVM and host nodes, follow these steps:

Procedure

1. Log in to any CVM in the cluster using SSH.

2. Update the CVM and host port groups:

```
nutanix@cvm$ network_segmentation --backplane_network
--host_physical_network=new-host-portgroup-name
--cvm_physical_network=new-cvm-portgroup-name
--update
```

- Replace *new-host-portgroup-name* with the new host port group name
- Replace *new-cvm-portgroup-name* with the new CVM port group name

For example:

```
nutanix@cvm$ network_segmentation --backplane_network
--host_physical_network=new-bp-host-pgroup
--cvm_physical_network=new-bp-cvm-pgroup
--update
```

For more information, see [Creating Port Groups on the Distributed Switch](#) in *vSphere Administration Guide for Acropolis*.

Disabling Network Segmentation on an ESXi and Hyper-V Clusters

Backplane network reconfiguration is a CLI-driven procedure that you perform on any one of the CVMs in the cluster. The change is propagated to the remaining CVMs.

About this task

To disable network segmentation on an ESXi and Hyper-V clusters, follow these steps:

Procedure

1. Log on to any CVM in the cluster using SSH.

2. Disable network segmentation on an ESXi and Hyper-V cluster:

```
nutanix@cvm$ network_segmentation --backplane_network --disable
```

Output similar to the following appears:

```
Operation type : Disable
Network type : kBackplane
Params : {}
Please enter [Y/y] to confirm or any other key to cancel the operation
```

Type Y or y to confirm.

If you type Y or y, network segmentation is disabled and the controller VMs (CVMs) restart in a rolling manner, one CVM at a time. If you type anything other than Y or y, network segmentation is not disabled.

This method does not involve cluster downtime.

3. Verify that you successfully disabled network segmentation.

You can verify in one of two ways:

- » Verify that the backplane is disabled.

```
nutanix@cvm$ network_segment_status
```

Output similar to the following is displayed:

```
2017-11-23 06:18:23 INFO zookeeper_session.py:110 network_segment_status is attempting to connect to Zookeeper
```

```
Network segmentation is disabled
```

- » Verify that the commands to show the backplane IP addresses of the CVMs and hosts list the same management IP addresses. Run the `svmips` and `hostips` command with and without the `-b` option, then compare the IP addresses shown in the output. For example:

```
nutanix@cvm$ svmips
192.127.3.2 192.127.3.3 192.127.3.4
nutanix@cvm$ svmips -b
192.127.3.2 192.127.3.3 192.127.3.4
nutanix@cvm$ hostips
192.127.3.5 192.127.3.6 192.127.3.7
nutanix@cvm$ hostips -b
192.127.3.5 192.127.3.6 192.127.3.7
```

In the example, the outputs of the `svmips` and `hostips` commands with and without the `-b` option are the same, indicating that the backplane network segmentation is disabled.

Disabling Network Segmentation on an AHV Cluster

Disable network segmentation on an AHV cluster.

About this task

You perform backplane network reconfiguration procedure on any one of the CVMs in the cluster. The change propagates to the remaining CVMs.

To disable network segmentation on an AHV cluster, follow these steps:

Procedure

1. Shut down all the guest VMs in the cluster from within the guest OS or use the Prism Element web console.

2. Place all nodes of a cluster into the maintenance mode:

- a. Use SSH to log on to a Controller VM in the cluster.
- b. Determine the IP address of the node you want to put into the maintenance mode:

```
nutanix@cvm$ acli host.list
```

Note the value of hypervisor IP address for the node you plan to put in the maintenance mode.

- c. Put the node into the maintenance mode:

```
nutanix@cvm$ acli host.enter_maintenance_mode hypervisor-IP-address [wait="{ true  
| false }" ] [non_migratable_vm_action="{ acpi_shutdown | block }" ]
```

Replace *host-IP-address* with either the IP address or host name of the AHV host you plan to shut down.

The following are optional parameters for running the `acli host.enter_maintenance_mode` command:

- `wait`
- `non_migratable_vm_action`

Do not continue if the host fails to enter the maintenance mode.

- d. Verify that the host is in the maintenance mode:

```
nutanix@cvm$ acli host.get host-ip
```

In the output that is displayed, ensure that **node_state** is **EnteredMaintenanceMode** and **schedulable** is **False**.

3. Disable backplane network segmentation from the Prism web console.

- a. Log in to the Prism Element web console, click the gear icon in the top-right corner, and then click **Settings>Network Configuration**.
- b. In the **Internal Interfaces** tab, in the **Backplane LAN** row, click **Disable**.
- c. Click **Yes** to disable backplane LAN.

Disabling backplane LAN involves a rolling reboot of CVMs to migrate the cluster services back to the external interface.

4. Log in to a CVM in the cluster with SSH and stop Acropolis cluster-wide:

```
nutanix@cvm$ allssh genesis stop acropolis
```

5. Restart Acropolis cluster-wide:

```
nutanix@cvm$ cluster start
```

6. Remove all nodes from maintenance mode:

- a. From any CVM in the cluster, run the following command to exit the AHV host from the maintenance mode:

```
nutanix@cvm$ acli host.exit_maintenance_mode host-ip
```

Replace *host-ip* with the new IP address of the host.

This command migrates all the VMs that were previously running on the host back to the host.

- b. Verify that the host has exited the maintenance mode:

```
nutanix@cvm$ acli host.get host-ip
```

In the output that is displayed, ensure that **node_state** equals to **kAcropolisNormal** or **AcropolisNormal** and **schedulable** equals **True**.

7. Power on the guest VMs from the Prism Element web console.

Service-Specific Traffic Isolation

Isolating the traffic associated with a specific service is a two-step process:

- Configure the networks and uplinks on each host manually. The Prism Element only creates the vNIC that the service requires and places that vNIC on the bridge or port group that you specify. You must manually create the bridge or port group on each host and add the required physical NICs as uplinks to that bridge or port group.
- Configure network segmentation for the service by using the Prism Element. Create an extra vNIC for the service, specify any additional parameters that are required (for example, IP address pools), and the bridge or port group that you plan to dedicate to the service.

Isolating Service-Specific Traffic

Isolate a service to a separate virtual network.

Before you begin

- Ensure that each host is configured as described in [Configuring the Network on Existing Nodes](#) on page 172.
- Ensure that you meet all [Prerequisites](#) on page 170.

About this task

To isolate a service to a separate virtual network, follow these steps:

Procedure

1. Log in to Prism Element as an administrator.
2. At the top-right corner of the page, click the gear icon.
3. In the left pane, click **Network Configuration**.
4. In the details pane, on the **Internal Interfaces** tab, click **Create New Interface**.
The **Create New Interface** dialog box is displayed.

5. On the **Interface Details** tab, follow these steps:

- a. Specify a descriptive name for the network segment.
- b. (Optional for AHV) Specify a **VLAN ID**.
Make sure that the VLAN ID is configured on the physical switch.
- c. In **Bridge** (on AHV) or **CVM Port Group** (on ESXi), select the bridge or port group that you created for the network segment.
- d. Specify an IP address pool for the network segment, click **Create New IP Pool**, then in the **IP Pool** dialog box, follow these steps:

- In the **Name** field, specify a name for the pool.
- In the **Netmask** field, specify the network mask for the pool.
- Click **Add an IP Range** and specify the start and end IP addresses in the **IP Range** dialog box that is displayed.
- Use **Add an IP Range** to add as many IP address ranges as you need.

Note: Add at least $n+1$ IP addresses in an IP address range considering n is the number of nodes in the cluster.

- Click **Save**.
- (Optional) Use **Add an IP Pool** to add more IP address pools. You can use only one IP address pool at any given time.
- Select the IP address pool that you plan to use and then click **Next**.
You can also use an existing unused IP address pool.

6. On the **Feature Selection** tab, follow these steps:

- a. Select the service whose traffic you plan to isolate.
- b. Configure the settings for the selected service.

The setting shown on this page vary based on the service that you select. For information about service-specific settings, see [Service-Specific Settings and Configurations](#) on page 192.

- c. Click **Save**.

You cannot enable network segmentation for multiple services at the same time. Complete the configuration for one service before you enable network segmentation for another service.

7. In the **Create Interface** dialog box, click **Save**.

The CVMs are rebooted multiple times, one after another. This procedure might trigger more tasks on the cluster.

What to do next

See [Service-Specific Settings and Configurations](#) on page 192 for any additional tasks that are required after you segment the network for a service.

Modifying Network Segmentation Configured for a Service

To modify network segmentation configured for a service, you must first disable network segmentation for that service, then create the network interface again for that service with the new IP address pool and VLAN.

About this task

For example, if the interface of the service you want to modify is `ntnx0`, after the reconfiguration, the same interface (`ntnx0`) is assigned to that service if that interface is not assigned to any other service. If `ntnx0` is assigned to another service, a new interface (for example `ntnx1`) is created and assigned to that service.

To reconfigure network segmentation configured for a service, follow these steps:

Procedure

1. Disable the network segmentation configured for a service.
For instructions, see [Disabling Network Segmentation Configured for a Service](#) on page 191.
2. Create the network interface again for that service with the new IP address pool and VLAN.
For instructions, see [Isolating Service-Specific Traffic](#) on page 189.

Disabling Network Segmentation Configured for a Service

To disable network segmentation configured for a service, you must disable the dedicated vNIC. Disabling network segmentation frees the name of the vNIC. The free name is reused in a subsequent network segmentation configuration.

About this task

To disable the network segmentation configured for a service, perform the following steps:

Procedure

1. Log in to Prism Element as an administrator.
2. At the top-right corner of the page, click the gear icon.
3. In the left pane, click **Network Configuration**.
4. On the **Internal Interfaces** tab, for the interface that you plan to disable, click **Disable**.

The defined IP address pool is available even after disabling the network segmentation.

What to do next

At the end of this procedure, the cluster performs a rolling restart. Disabling network segmentation disrupts the functioning of the associated service. To restore normal operations, you may need to perform additional tasks immediately after the rolling restart. For information about the follow-up tasks, see [Service-Specific Settings and Configurations](#) on page 192.

Deleting a vNIC Configured for a Service

If you disable network segmentation for a service, the vNIC for that service is not deleted. AOS reuses the vNIC if you enable network segmentation again. However, you can manually delete a vNIC by logging into any CVM in the cluster with SSH.

Before you begin

Ensure that you meet the following pre-requisites before you delete the vNIC configured for a service:

- Network segmentation is disabled for a service. For instructions, see [Disabling Network Segmentation Configured for a Service](#) on page 191.
- You observe the limitations specified in [Limitation for vNIC Hot-Unplugging](#) topic in *AHV Admin Guide*.

About this task

To delete a vNIC that is configured for a service, follow these steps:

Procedure

1. Log in to any CVM in the cluster with SSH.
2. Delete the vNIC:

```
nutanix@cvm$ network_segmentation --service_network --interface="interface-name" --delete
```

Replace *interface-name* with the name of the interface you plan to delete. For example, *ntnx0*.

Service-Specific Settings and Configurations

Settings required by the services that support network segmentation.

Prism Element supports network segmentation for specific services to isolate traffic and improve security. For Nutanix Volumes, you can configure multiple iSCSI networks and migrate client connections to segmented networks. Disaster recovery configurations also support segmentation, including stretched layer 2 networks. Each service requires specific parameters such as virtual IP address, client subnet, and gateway.

Nutanix Volumes

Network segmentation for Volumes also requires you to migrate iSCSI client connections to the new segmented network. If you no longer require segmentation for Volumes traffic, you must also migrate connections back to *eth0* after disabling the vNIC used for Volumes traffic.

You can create up to two different networks for Nutanix Volumes with different IP address pools, VLANs, and data services IP addresses. For example, you can create two iSCSI networks, one for production and one for non-production traffic, on the same Nutanix cluster.

Follow the instructions in [Isolating Service-Specific Traffic](#) on page 189 again to create the second network for Volumes after you create the first network.

Settings to be Specified When Configuring Traffic Isolation

When configuring the traffic isolation, specify the following settings:

- Virtual IP: This is an optional setting. Specifies a Virtual IP address for the service. If specified, select the IP address from the specified IP address pool. If you do not specify an IP address, the system selects a random IP address from the specified IP address pool.
- Client subnet: The network (in CIDR notation) that hosts iSCSI clients. Required only if clients are outside the subnet of the iSCSI segment.

You can specify multiple client subnets while configuring network segmentation for Volumes in the CLI from a CVM.

To update client subnets for network segmentation for Volumes segmentation:

```
nutanix@cvm$ network_segmentation --service_network --update  
--interface="interface_name" --service_name="kVolumes" --  
client_subnets="existing_subnet,new_client_subnet"
```

For example:

```
nutanix@cvm$ network_segmentation --service_network --update --interface="ntnx0" --  
service_name="kVolumes" --client_subnets="10.2.2.0/24,10.2.3.0/24"
```

For more information on using the CLI, run the `nutanix@cvm$ network_segmentation --help` command.

You can specify only one client subnet while configuring network segmentation for Volumes in the Prism Element web console.

- Gateway: Specifies the gateway to the subnetwork that hosts the iSCSI clients. Required if you specify the client subnet.

Migrating iSCSI Connections to the Segmented Network

After you enable network segmentation for Volumes, you must manually migrate connections from existing iSCSI clients to the newly segmented network. Even though support is available to run iSCSI traffic on both the segmented and management networks at the same time, Nutanix recommends that you move the iSCSI traffic for guest VMs to the segmented network to achieve true isolation.

Before you begin

Ensure that the task for enabling network segmentation for the service succeeds.

About this task

To migrate iSCSI connections to the segmented network, perform the following steps:

Procedure

1. Log out of all the clients connected to iSCSI targets that are using CVM `eth0` or the data service IP address.
2. (Optional) Remove all the discovery records for the data services IP address (DSIP) on `eth0` interface.
3. If the clients are allowlisted by their IP address, remove the client IP address that is on the management network:

```
nutanix@cvm$ acli vg.detach_external vg_name initiator_network_id=old_vm_IP
nutanix@cvm$ acli vg.attach_external vg_name initiator_network_id=new_vm_IP
```

Replace `vg_name` with the name of the volume group and `old_vm_IP` and `new_vm_IP` with the old and new client IP addresses, respectively.

4. Discover the virtual IP address specified for Volumes.
5. Connect to the iSCSI targets from the client.

Migrating Existing iSCSI Connections to the Management Network

If you no longer require segmentation for Volumes traffic, you must migrate connections back to the controller VM `eth0` after disabling the vNIC used for Volumes traffic.

About this task

To migrate existing iSCSI connections to `eth0`, follow these steps:

Procedure

1. Log out of all the clients connected to iSCSI targets using the CVM vNIC dedicated to Volumes.
2. Remove all the discovery records for the data service IP address (DSIP) on the new interface.
3. Discover the DSIP for `eth0`.
4. Connect the clients to the iSCSI targets.

Disaster Recovery with Protection Domains

The settings for configuring network segmentation for disaster recovery apply to all asynchronous, nearSync, and metro availability replication schedules.

You can use disaster recovery with asynchronous, nearSync, and metro availability replications only if you have configured network segmentation for both the primary site and the recovery site. Before enabling or disabling the network segmentation on a host, make sure to disable all disaster recovery replication schedules that are currently running on that host.

When configuring the traffic isolation, specify the following settings:

- **Virtual IP:** This is an optional setting. Specifies a Virtual IP address for the service. If specified, select the IP address from the specified IP address pool. If you do not specify an IP address, the system selects a random IP address from the specified IP address pool.
- **Gateway:** Gateway to the subnet.

Consider the following points when configuring network segmentation for disaster recovery:

- In the Prism Central web console, you cannot configure network segmentation for DR without a gateway. Therefore, you must use the CLI method.
- While configuring network segmentation for L2 stretched DR using the CLI, specifying a gateway is optional if you use the `stretched_metro` CLI option.
- When you configure network segmentation for a network which is intended as a stretched Layer 2 network across AOS clusters without a gateway using CLI, disaster recovery (DR) is limited to AOS clusters that are within the same Layer 2 broadcast domain.

Remote Site Configuration

After configuring network segmentation for disaster recovery, configure remote sites at both locations. You also need to reconfigure remote sites if you disable network segmentation.

When you configure a remote site within a Protection Domain, you must allow communication between all CVM IP addresses at the remote site and the VIP using IP address pools. However, after you enable network segmentation and register the remote site, the system allows only the VIP by default. As a result, communication between the primary site and remote site can fail. To fix this issue, configure firewall rules to allow communication over the required ports for all IP addresses in the network segmentation pool. For more information, see [KB-19733](#).

For information about configuring remote sites, see [Remote Site Configuration](#) in the *Protection Domain-Based Disaster Recovery with Prism Element Guide*.

Segmenting a Stretched Layer 2 Network for Disaster Recovery

A stretched layer 2 network configuration allows the source and remote metro clusters to be in the same broadcast domain and communicate without a gateway.

Before you begin

Create an IP address pool for DR.

For information on how to create an IP pool, see [Configuring Backplane IP Pool](#) on page 183.

About this task

You can enable network segmentation for disaster recovery on a stretched layer 2 network that does not have a gateway. A stretched layer 2 network is usually configured across physically remote clusters, such as a Metro Availability cluster deployment. A stretched layer 2 network allows the source and remote clusters to be configured in the same broadcast domain without the usual gateway.

See *AOS Release Notes* for minimum AOS version required to configure a stretched layer 2 network.

To configure a network segment as a stretched L2 network, perform the following steps:

Procedure

1. Log in to any CVM in the cluster using SSH.
2. Based on your hypervisor, run the following command:

- On a cluster running ESXi hypervisor:

```
nutanix@cvm$ network_segmentation --service_network --service_name=kDR --  
ip_pool=DR-ip-pool-name  
--service_vlan=DR-vlan-id --desc_name=Description --  
host_physical_network=portgroup  
--stretched_metro
```

- On a cluster running AHV hypervisor:

```
nutanix@cvm$ network_segmentation --service_network --service_name=kDR --  
ip_pool=DR-ip-pool-name  
--service_vlan=DR-vlan-id --desc_name=Description --host_virtual_switch=virtual-  
switch  
--stretched_metro
```

Replace the following:

- *DR-ip-pool-name* with the name of the **IP Pool** created for the DR service or any existing unused IP address pool.
- *DR-vlan-id* with the **VLAN ID** used for the DR service.
- *Description* with a suitable description of this stretched L2 network segment.
- *portgroup* with the details of the **CVM Port Group** used for the DR service in the ESXi hypervisor.
- *virtual-switch* with the details of the **Virtual Switch** used for the DR service in the AHV hypervisor.

The following example shows how to configure network segment as a stretched L2 network on a cluster running ESXi hypervisor:

```
nutanix@cvm$ network_segmentation --service_network --service_name=kDR --  
ip_pool=DR_pool  
--service_vlan=124 --desc_name="L2 Strech for ESXi" --  
host_physical_network=portgroup0  
--stretched_metro
```

The following example shows how to configure network segment as a stretched L2 network on a cluster running AHV hypervisor:

```
nutanix@cvm$ network_segmentation --service_network --service_name=kDR --  
ip_pool=DR_pool  
--service_vlan=124 --desc_name="L2 Strech for AHV" --host_virtual_switch=vs0  
--stretched_metro
```

For more information about the `network_segmentation` command, see the [Command Reference](#) guide.

Nutanix Disaster Recovery

The settings for configuring network segmentation for Nutanix Disaster Recovery apply to all asynchronous, nearSync, and synchronous replication schedules. For more information, see *Network Segmentation* in the *Nutanix Disaster Recovery Guide*.

When configuring the traffic isolation, specify the following settings:

- **Virtual IP:** This is an optional setting. Specifies a Virtual IP address for the service. If specified, select the IP address from the specified IP address pool. If you do not specify an IP address, the system selects a random IP address from the specified IP address pool.

Virtual IP address is different from the external IP address and the data services IP address of the cluster.

- **Gateway:** Specifies the gateway to the subnet.

Remote Direct Memory Access over Converged Ethernet

Remote Direct Memory Access (RDMA), RDMA port pass-through mechanism, and how to configure the RDMA network segmentation using Zero-Touch RoCE (ZTR) or Priority-Based Flow Control (PFC) mechanism.

RDMA Overview

Remote Direct Memory Access (RDMA) directly transfers data between multiple hosts without involving the CPU, OS, or system cache. RDMA reduces communication latency and increases the bandwidth output for data transfer. It directly uses the network adapters for data transfer and never creates a data copy between network layers.

When RDMA is enabled, the CPU resources available for other applications running in the cluster enhance the AOS data acceleration mechanism.

RDMA Port Pass-through Mechanism

The following table describes the supported RDMA port pass-through mechanisms:

Table 13: Supported RDMA Port Pass-through Mechanism

AOS release	Hypervisor	RDMA Port Pass-through Imaging
Prior to AOS 6.6	• AHV • ESXi	During Foundation imaging with RDMA pass-through enabled, the Controller VM (CVM) reserves the entire NIC for RDMA traffic. In this configuration, the host system cannot use any remaining NIC ports for other operations, nor can it change the selected port. If the NIC or port experiences a failure or connectivity issue, I/O traffic automatically falls back to TCP/IP mode.

AOS release	Hypervisor	RDMA Port Pass-through Imaging
AOS 6.6 and later	AHV	RDMA port pass-through behaviour during Foundation setup or after you complete cluster creation. During Foundation setup, when RDMA port pass#through is enabled during Foundation imaging, the CVM reserves the entire NIC for RDMA. At this stage, the user cannot select a specific port—only the pass#through option is available. The host cannot use the unused NIC port for other operations, nor change the selected ports. After cluster creation, a manageability feature allows you to select port. Administrators can choose which NIC port to use for RDMA pass#through, providing flexibility in case of port preferences or hardware considerations. Note: Ensure the port that you reserve for RDMA is not allocated to the uplink vswitch of the host. If the port that you want to reserve for RDMA is already allocated for uplink vswitch, the system displays it as disabled for selection and you need to manually remove it from the vswitch. • Failure handling: If the NIC or selected port fails, or if any connectivity issue occurs, I/O traffic automatically falls back to TCP/IP mode.
	ESXi	If RDMA pass#through is enabled during Foundation imaging, the entire NIC is reserved by CVM, system does not provide an option to change the RDMA port during imaging or after cluster creation.

For more information, see [Configuring Foundation VM by Using the Foundation GUI](#) and [RDMA Supported Platforms](#) in *NX Series Hardware Administration Guide*.

Node Upgrade Case

When you add RDMA-capable NICs to an existing node after the cluster is already created and you plan to configure RDMA, follow the instructions provided in the [KB 17964](#).

Cluster Expansion Case

When you add a new node to increase the capacity of a cluster, you cannot change the RDMA ports after the cluster is created. In this situation, the system checks whether RDMA is enabled in the cluster. If RDMA is enabled, the system provisions RDMA for the new node and selects the first available port in the NIC. To change the RDMA port, you must disable RDMA for the entire cluster, then reconfigure it with the new port selection.

ZTR Specifications

ZTR is a deployment mechanism for RDMA setup in which the Mellanox NIC firmware handles the entire configurations (card optimizations) without any user intervention or dependency on the customized switch profiles or switch compatibility. ZTR reduces RDMA deployment duration and does not require the support of PFC and end-to-end congestion notification (ECN) settings.

ZTR functionality is supported with both ESXi and AHV hypervisors. For information about how to enable ZTR for RDMA network segmentation, see [Isolating the Backplane Traffic on an Existing RDMA Cluster](#) on page 200.

Nutanix recommends that you use ZTR only if NVIDIA Mellanox Connect X-5 Ethernet adapters (Cx5 NICs) or NVIDIA Mellanox Connect X-6 Ethernet adapters (Cx6 NICs) are available in your setup. For more information about the NICs' compatibility for ZTR feature, see [NIC Compatibility Matrix for RDMA Features](#) on page 198.

Priority-Based Flow Control without Zero-Touch RoCE Specifications

The following are the RDMA switch configurations you must perform when using datacenter bridging (DCB) or priority flow control mode (PFC) without ZTR:

- Ensure that the switches are DCB capable and that DCB or PFC is enabled.
- Ensure that switch ports mapped to servers are configured with DCB or PFC.
- Ensure that when you are configuring DCB with Enhanced Transmission Selection (ETS), the sum of all traffic class (TC) bandwidth allocations equals 100%. The bandwidth percentages are assigned across TCs according to workload requirements, such as RDMA#heavy, storage#heavy, or balanced. The exact ETS bandwidth allocation for RDMA traffic is use#case dependent and must follow the switch vendor's specific recommendations.
- Connect the RDMA-supported server NIC port to the switch.
- Configure the connected switch port to allow the selected VLAN.
- Disable send and receive flow control on the switch port.
- Enable priority flow control mode (PFC) with priority 3 or priority 4.

For more information on how to configure these settings, see to the switch vendor documentation or contact the switch vendor for support.

RDM switches using PFC with ZTR do not require additional configurations as described before, because the NICs negotiate directly with each other.

For information on how to configure the RDMA network segmentation using either PFC or ZTR from Prism Element, see [Isolating the Backplane Traffic on an Existing RDMA Cluster](#) on page 200.

NIC Compatibility Matrix for RDMA Features

The following table provides the information about NIC compatibility with RDMA features, RDMA Port Pass-through and ZTR, and the required workaround:

Table 14: NIC Compatibility Matrix for RDMA Features

NIC Available at Site	RDMA Feature	Compatibility Information	Workaround
NVIDIA Mellanox ConnectX-4 Ethernet Adapter (CX-4)	ZTR	Not supported Nutanix recommends that you use ZTR only if CX-5 NICs are available in your setup.	None
	RDMA port pass-through	Not supported after cluster creation.	Perform RDMA port pass-through during Foundation setup, then upgrade to AOS 6.6 release.

NIC Available at Site	RDMA Feature	Compatibility Information	Workaround
NVIDIA Mellanox ConnectX-5 Ethernet Adapter (CX-5)	ZTR	Supported	None
	RDMA port pass-through	Supported The RDMA port pass-through is supported for the following scenarios: - During Foundation setup where the CVM reserves the entire NIC for RDMA port pass-through. In this case, the host cannot use the empty NIC port for other operations nor can it change the selected port. - After cluster creation for AHV only.	None
NVIDIA Mellanox ConnectX-6 Ethernet Adapter (CX-6)	ZTR	Supported	None
	RDMA port pass-through	Supported The RDMA port pass-through is supported for the following scenarios: - During Foundation setup where the CVM reserves the entire NIC for RDMA port pass-through. In this case, the host cannot use the empty NIC port for other operations nor can it change the selected port. - After cluster creation for AHV only.	None

You can have a mix of NIC cards across nodes in a cluster. In this case, the oldest family of cards dictates the feature supported for the cluster. For example, if any CX-4 NIC is present, the system blocks the RDMA live port pass-through (RDMA port pass-through after cluster creation) and ZTR functions, and allows only RDMA data replication.

Important: You cannot have a mix of RDMA NIC families on the same node. For example, you cannot have a combination of NVIDIA Mellanox ConnectX-4 Ethernet adapter (CX-4 NIC) and NVIDIA Mellanox ConnectX-5 Ethernet adapter (CX-5 NIC) or a combination of CX-5 NICs with different speeds such as CX-5 10Gb NIC and CX-5 25Gb NIC, on the same node. If you have any of these combination types of RDMA NIC families on the same node, the system never allows you to enable RDMA on the cluster.

Isolating the Backplane Traffic on an Existing RDMA Cluster

Configure the RDMA network segmentation settings using either PFC or ZTR from Prism Element.

About this task

The network segmentation process creates a separate network for RDMA communications on the existing default virtual switch and places the `rdma0` interface (created on the CVMs during upgrade) and the host interfaces on the newly created network. From the specified subnet, IP addresses are assigned to each new interface. Two IP addresses are therefore required per node. If you specify the optional VLAN ID, the newly created interfaces are placed on the VLAN. A separate VLAN is highly recommended for the RDMA network to achieve true segmentation.

Before you begin

Ensure that the following prerequisites are met before you proceed with RDMA network segmentation:

- You must specify a non-routable subnet. The interfaces on the backplane network are automatically assigned IP addresses from the subnet. You reserve the entire subnet for the backplane network alone.
- If you plan to specify a VLAN for the RDMA network, ensure that the VLAN is configured on the physical switch ports to which the nodes are connected.
- For information on prerequisites related to switch configuration when using datacenter bridging (DCB) or priority flow control mode (PFC) without ZTR, see [Priority-Based Flow Control without Zero-Touch RoCE Specifications](#) on page 198.
- Observe the NICs compatibility information specified in [NIC Compatibility Matrix for RDMA Features](#) on page 198.
- Ensure that the cluster does not have a mixed configuration where some nodes have RDMA port pass-through or ZTR enabled while other nodes have it disabled, such mix configuration is not supported. You must configure all nodes in the cluster uniformly with the RDMA functionality.

The RDMA configuration option is disabled if any of the following conditions exist in your setup:

- All the nodes in the cluster do not contain at least two RDMA-enabled NIC cards.
- A mix of RDMA NIC families on the same node is present. For example, a combination of NVIDIA Mellanox ConnectX-4 Ethernet Adapter (CX-4 NIC) and NVIDIA Mellanox ConnectX-5 Ethernet Adapter (CX-5 NIC) or a combination of CX-5 NICs with different speeds such as CX-5 10Gb NIC and CX-5 25Gb NIC, on the same node is present. If you have any of these combination types of RDMA NIC families on the same node, the system never allows you to enable RDMA on the cluster.

For more information, see [NIC Compatibility Matrix for RDMA Features](#) on page 198.

The following prerequisites are applicable only for ZTR:

- NVIDIA Mellanox Connect X-5 Ethernet Adapters (Cx5 NICs) or NVIDIA Mellanox Connect X-6 Ethernet Adapters (Cx6 NICs) are available. For details, see [NIC Compatibility Matrix for RDMA Features](#) on page 198.
- NICs on all nodes are running the same NIC firmware. Perform a NCC health check to verify the minimum driver version recommended and supported by Mellanox NIC running on the Nutanix platforms. For information about how to perform NCC health check, see [KB 4289](#)
- AHV or ESXi hypervisor and AOS 6.6 or later (CVM 6.6) are deployed for the cluster. For information about supported AHV or ESXi hypervisor versions, see [KB 4289](#).

Procedure

To isolate the backplane traffic on an existing RDMA cluster, follow these steps:

1. Log in to Prism Element as an administrator.
2. In the top-right corner, click the gear icon, and then click **Network Configuration** in the **Settings** page. The **Network Configuration** dialog box displays.
3. Click the **Internal Interfaces** tab. The **Internal Interfaces** tab displays.

Note: If the system detects that NVIDIA Mellanox Cx5 NIC or NVIDIA Mellanox Cx6 NIC is used, it provides you the option to configure RDMA.

4. Click **Configure** in the **RDMA** row, and set the following attributes in **RDMA** dialog box:
 - a. Under **Subnet IP**, specify a non-routable Subnet IP.
Ensure that the subnet size can accommodate cluster expansion in the future.
 - b. Under **Netmask**, specify a non-routable netmask.
 - c. Under **VLAN**, specify a VLAN ID for the RDMA LAN.
VLAN ID is optional with ZTR but Nutanix recommends that you use it for true network segmentation and enhanced security.
 - d. To enable ZTR, select either the **Use Zero Touch RoCE** checkbox or select the **PFC** value configured on the physical switch port.
When you select the Use Zero Touch RoCE checkbox:
 - The PFC field is disabled.
 - The system automatically defines the RDMA port if RDMA port pass-through is done during Foundation setup. You cannot change the RDMA port in this case.
5. Click **Verify and Save**.
The selected RDMA network segmentation settings are saved.
If you do not perform the RDMA port pass-through set-up during Foundation setup and only AHV hypervisor is deployed, the system prompts you to define the RDMA port.
6. (Optional) To define RDMA port, follow these steps:
 - a. Click **Next** and then select the RDMA port in the **Port Selection** tab.
 - b. Select the port to be reserved for RDMA on all the hosts.
 - c. Click **Save**.

Support for iSCSI Extensions for RDMA

iSCSI Extensions for RDMA (iSER) enhances traditional iSCSI by using RDMA to eliminate data copies in the data path, improving performance and reducing CPU usage.

The Internet Small Computer Systems Interface (iSCSI) is a Storage Area Networking (SAN)-based data transfer model that uses TCP/IP to transfer the data between applications to a storage array. iSER is an extension of iSCSI that uses the direct data placement technology of the RDMA protocol to eliminate the copies in the data path. The system appropriately utilizes the CPU resources and reduces latency between AHV host and CVM I/O transfer.

iSER provides the following key benefits:

- Enhances I/O flow performance between the AHV host and the CVM. iSER bypasses both the AHV host and CVM kernel spaces and enables you to achieve low latency in the I/O data transfer because the data is sent and received between AHV host and CVM without the involvement of the TCP software stack. iSER enables you to

achieve a high bandwidth in the I/O data transfer as it removes the unwanted context switches and system calls between AHV host and CVM.

- Saves platform resources due to less CPU utilization because an application can read remote memory without any intervention by the remote processor.

The following figure shows how I/O traffic flows between the AHV host and the CVM:

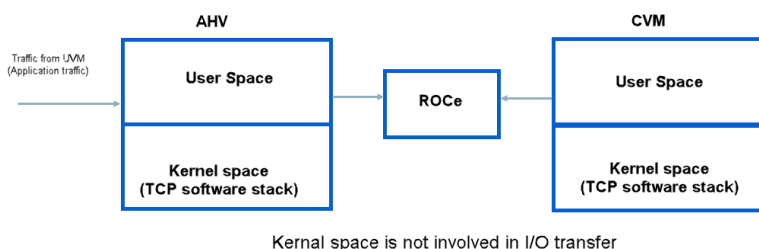


Figure 28: I/O Traffic - AHV Host and CVM

The following figure shows the detailed network architecture for iSER:

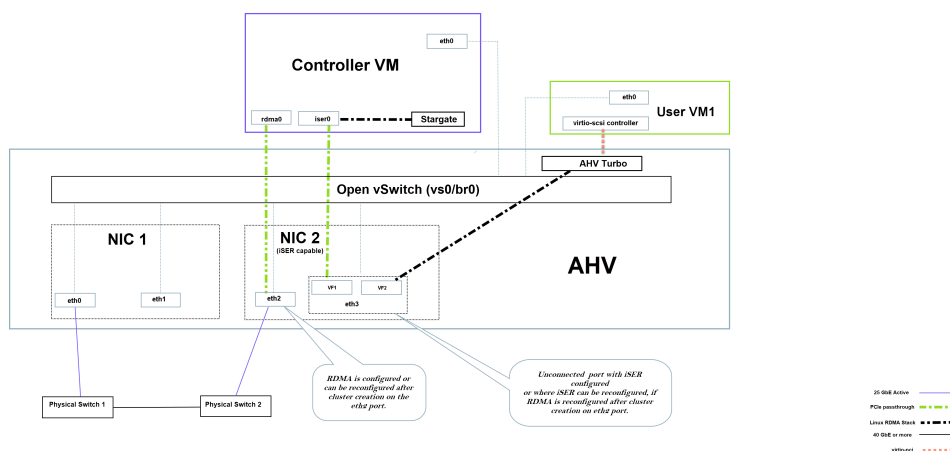


Figure 29: iSER Network Architecture

The following specifications apply to the iSER architecture:

- NIC 1 is a non-RDMA capable NIC for regular network functionality, and NIC 2 is a dual-port qualified iSER NIC for RDMA and iSER. For more information about qualified iSER NICs, see [NIC Compatibility Matrix for iSER](#) on page 204.
- The first port on NIC 2 (**eth2**) is connected to the uplink switch. The second port (**eth3**) on NIC 2 is disconnected.
- The unconnected port **eth3** can be used for iSER.
- The connected port on NIC 2 (**eth2** in the previous diagram) can be passed through to the CVM for RDMA replication.
- The VF1 interface on an iSER port is consumed by Stargate using the Linux RDMA stack.

- The VF2 interface on an iSER port is consumed by AHV Turbo using the Linux RDMA stack.
- iSER and RDMA replication can either be enabled during cluster creation or dynamically enabled or disabled on the available ports (through the RDMA manageability workflow) after cluster creation.

After cluster creation:

- You can reconfigure RDMA on the **eth2** port.
- If RDMA is reconfigured on the **eth2** port, you can reconfigure iSER on the **eth3** port.

Note: Currently, iSER uses fixed internal IPs. There is no supported workflow to change iSER IPs for the CVM or the AHV host.

The following specifications are applicable in the iSER setup:

iSER Specification	Component	Description
iSER initiator	AHV Turbo in AHV	AHV Turbo provides iSCSI packet processing.
iSER target	CVM. Stargate in CVM	Stargate in CVM acts as a transport to perform data from AHV for iSCSI.

Typically, AHV and CVM communicate with each other using the iSCSI protocol over the Open vSwitch TCP datapath. AHV forwards the I/O data that originates from the guest VM to the CVM, which processes the I/O data request. After the CVM completes the I/O data processing, it returns the response and data for the request to AHV.

If iSER is enabled, the following actions occur between AHV and the CVM to establish storage connectivity:

1. The system establishes a handshake connection between AHV and the CVM over TCP.
2. On the TCP iSCSI login, the CVM responds to AHV with the status of RDMA as enabled or disabled and returns the target IP address for iSER if RDMA is enabled.
3. If the CVM indicates that RDMA connectivity is possible, AHV closes the TCP connection and attempts to establish an RDMA connection to the CVM iSER IP.
4. If the RDMA connection and TCP iSCSI login over the RDMA connection is successful, AHV submits the guest VM SCSI requests to CVM using the RDMA connection.

In case of any failure scenarios, the system re-establishes TCP connections between AHV and the CVM through the standard vSwitch path, and the data processing is resumed between AHV and the CVM with zero disruption to the running workload.

CVM Memory Requirements for iSER

For iSER, CVM requires a minimum of 64 GiB vRAM for optimum performance. For more information, see [Controller VM \(CVM\) Field Specifications](#) in *Acropolis Advanced Admin Guide*.

iSER Port-Passthrough Mechanism

You can perform an iSER NIC port pass-through either during the Foundation setup or after the cluster creation is complete using the Prism Element web console.

The following behavior applies to iSER port pass-through:

- If you perform the iSER port pass-through during Foundation setup, the CVM reserves the whole NIC for both iSER and RDMA data replication. The CVM reserves one port on the NIC for RDMA data replication and configures the other port for iSER. For more information about Foundation setup, see [Configuring Foundation VM by Using the Foundation GUI](#) in *Field Installation Guide*.

- If you perform the iSER port pass-through after cluster creation, you can configure iSER on one of the unconnected NIC ports. You can use the other NIC port to configure RDMA if RDMA port-passthrough is not done during Foundation setup.

Note: Ensure that the port that you configure for iSER is not allocated to the uplink vswitch of the host. If the port that you plan to configure for iSER is already allocated for uplink vswitch, the system displays it as disabled for selection and you need to remove it from the vswitch manually.

Node Upgrade Case

When you add the iSER-capable NICs to a node and want to configure RDMA after the cluster is created, follow the instructions provided in the [KB 17964](#).

Cluster Expansion Case

When a new node is added to increase the capacity of a cluster, the iSER ports cannot be changed after the cluster is created. In this case, the system checks if iSER is enabled in the cluster. If enabled, the system provisions the iSER for the new node and selects the first port available in the NIC. To change the iSER port, disable the iSER for the whole cluster and reconfigure it with the new port selection.

NIC Compatibility Matrix for iSER

The following table provides the information about NIC compatibility for iSER:

Table 15: NIC Compatibility Matrix for iSER

NIC Available at Site	Number of NICs Required	iSER Port Pass-through Compatibility	Workaround
NVIDIA Mellanox ConnectX-4 Ethernet Adapter (CX-4)	Not Applicable	Not supported	None

NIC Available at Site	Number of NICs Required	iSER Port Pass-through Compatibility	Workaround
NVIDIA Mellanox ConnectX-5 Ethernet Adapter (CX-5)	Two dual-port NICs for RDMA + iSER	Supported The iSER port pass-through is supported for both the following scenarios: - During Foundation setup where the CVM reserves the whole NIC for iSER and RDMA data replication. - After cluster creation for AHV only, where the system provides you an option to configure iSER on one of the unconnected NIC ports. You can use the other NIC port to configure RDMA if RDMA port-passthrough is not configured during Foundation setup.	None
NVIDIA Mellanox ConnectX-6 Ethernet Adapter (CX-6)	Two dual-port NICs for RDMA + iSER	Supported from AOS 6.7.1 onward The iSER port pass-through is supported for the following scenarios: - During Foundation setup where the CVM reserves the whole NIC for both iSER and RDMA data replication. - After cluster creation for AHV only, where the system provides you an option to configure iSER on one of the unconnected NIC ports. You can use the other NIC port to configure RDMA if RDMA port-passthrough is not configured during Foundation setup.	None

NIC Available at Site	Number of NICs Required	iSER Port Pass-through Compatibility	Workaround
NVIDIA Mellanox ConnectX-7 Ethernet Adapter (CX-7 200G)	Two dual-port NICs to support iSER	Supported from AOS 7.3 onward Mellanox CX-7 NIC does not support Foundation passthrough for iSER. Applicable scenario: After cluster creation in case of AHV environments, you must manually configure iSER on one of the unconnected NIC ports. For more information, see Configuring iSER Manually on Mellanox CX-7 NIC on page 207.	None

You can have a mix of NIC cards across nodes in a cluster. In this case, the oldest family of cards dictates the feature supported for the cluster. For example, if any CX-4 NIC is present, the system blocks the RDMA live port pass-through (RDMA port pass-through after cluster creation) and ZTR functions, and allows only RDMA data replication.

Important: You cannot have a mix of RDMA NIC families on the same node. For example, you cannot have a combination of NVIDIA Mellanox ConnectX-4 Ethernet adapter (CX-4 NIC) and NVIDIA Mellanox ConnectX-5 Ethernet adapter (CX-5 NIC) or a combination of CX-5 NICs with different speeds such as CX-5 10Gb NIC and CX-5 25Gb NIC, on the same node. If you have any of these combination types of RDMA NIC families on the same node, the system never allows you to enable RDMA on the cluster.

iSER Limitations

The following limitations apply to iSER:

- iSER is supported with AHV only because it involves changes in I/O traffic flow between the AHV host and CVM. iSER support is not available with the ESXi hypervisor in the Nutanix environment.
- The total number of VM disks supported per node using iSER is limited to 200 because of memory limitations, and you cannot specify or configure the VM disks for iSER acceleration.

Configuring iSER Port in an Existing RDMA Cluster

Configure the iSER port in an existing RDMA cluster.

About this task

iSER is enabled by default on all nodes with Mellanox CX-5 NIC on which RDMA data replication is successful. The system reserves two IP addresses for iSER. The iSER IP addresses are fixed, and during CVM restart, the system uses these reserved IP addresses for iSER.

During AOS or AHV upgrade from previous versions, system automatically enables iSER, AHV and the CVM configures an internal IP address for iSER.

Before you begin

Ensure that the following prerequisites are met before you configure the iSER port in an existing cluster:

- AHV is upgraded to AHV-20230302.207 and AOS is upgraded to AOS 6.7 version.

- Only the qualified iSER NICs; NVIDIA Mellanox dual-port CX-5 NICs, are present in your setup, or else iSER functionality is disabled. For more information, see [NIC Compatibility Matrix for iSER](#) on page 204.
- Two dual-port qualified iSER NICs are present in your setup for RDMA and iSER. For more information about qualified iSER NICs, see [NIC Compatibility Matrix for iSER](#) on page 204.
- One non-RDMA capable NIC is present in your setup for regular network functionality.
- The OFED drivers are qualified for iSER. For more information, see *Linux Drivers* information on the [NVIDIA website](#).
- NIC firmware upgrade is done through LCM.
- RDMA for iSER must be enabled for Virtual Function (VF). For more information, see [Isolating the Backplane Traffic on an Existing RDMA Cluster](#) on page 200.
- An unconnected port is available on NIC for iSER traffic.
- You observe the limitations as specified in [iSER Limitations](#) on page 206.

Procedure

To configure iSER in an existing RDMA cluster, follow these steps:

1. Log in to Prism Element as an administrator.
2. From the main menu, click the gear icon.
3. On the left navigation bar, select **Network Configuration**.
The **Network Configuration** dialog box displays.
4. Click the **Internal Interfaces** tab.
The **Internal Interfaces** tab displays.

If the system detects the NVIDIA Mellanox Cx5 NIC, it provides you the option to configure iSER.

5. Click **Configure** in the **iSER** row, and set the port in following attributes in **Configure iSER** page.
If you have deployed only AHV and not configured iSER port pass-through during Foundation setup, the system allows you to define the iSER port.
Define the iSER port:
 - a. Select the port for iSER on all the hosts.
 - b. Click **Save**.

Configuring iSER Manually on Mellanox CX-7 NIC

By default, iSER is not enabled on any nodes with Mellanox CX-7 NICs. You must manually configure iSER on Mellanox ConnectX-7 NICs.

Before you begin

Before you configure iSER manually on Mellanox CX-7 NIC, consider the following points:

- AOS version 7.3 is the minimum supported version for manually configuring iSER.
- Prism Element does not support RDMA replication factor 2.
- Prism Element with HCI supports iSER over Mellanox ConnectX-7.

About this task

To configure iSER manually on Mellanox CX-7 NIC, follow these steps:

Procedure

1. Log in to Prism Element as an administrator.
2. From the main menu, click the gear icon.
3. On the left navigation bar, select **Network Configuration**.
4. Unassign the uplink port from the vSwitch.
For more information, see [Updating a Virtual Switch](#).
5. Select the **Internal Interfaces** tab.
6. In the iSER row, click **Configure**.
7. Select the NIC port that you freed from the virtual switch in Step 4 to configure iSER on all hosts.
8. Click **Save**.

Customizing IP Addresses for each CVM and Host

Create custom IP addresses for each CVM and host for network segmentation.

About this task

IP Address customization for each CVM and host feature enables you to allocate an IP address manually to a CVM and host and helps in maintaining similarity between external and segmented IP addresses. Prism Element supports IP Address customization feature for configuring backplane segmentation, service-specific traffic isolation for Nutanix Volumes, and service-specific traffic isolation for disaster recovery.

Procedure

1. Create a JSON file with a mapping of the CVM external IP address to the new segmented IP address.

The segmented IP addresses must belong to the same IP address pool that is created from the Prism UI or the CLI command before starting the Network Segmentation operation.

Example of JSON file format:

- JSON file format for backplane segmentation:

```
{
  "svmips": {
    `cvm_external_ip1`: `cvm_backplane_ip1`,
    `cvm_external_ip2`: `cvm_backplane_ip2`,
    `cvm_external_ip3`: `cvm_backplane_ip3`
  },
  "hostips": {
    `host_external_ip1`: `host_backplane_ip1`,
    `host_external_ip2`: `host_backplane_ip2`,
    `host_external_ip3`: `host_backplane_ip3`
  }
}
```

For example:

```
{
  "svmips": {
    "10.47.240.141": "172.16.10.141",
  }
}
```

```

    "10.47.240.142": "172.16.10.142",
    "10.47.240.143": "172.16.10.143"
  },
  "hostips": {
    "10.47.240.137": "172.16.10.137",
    "10.47.240.138": "172.16.10.138",
    "10.47.240.139": "172.16.10.139"
  }
}

```

- JSON file format for service segmentation:

```

{
  "svmips": {
    `cvm_external_ip1`: `cvm_service_ip1`,
    `cvm_external_ip2`: `cvm_service_ip2`,
    `cvm_external_ip3`: `cvm_service_ip3`
  }
}

```

For example:

```

{
  "svmips": {
    "10.47.240.141": "10.47.6.141",
    "10.47.240.142": "10.47.6.142",
    "10.47.240.143": "10.47.6.143"
  }
}

```

2. Save the JSON file in any CVM in the cluster.
3. Use SSH to log on to the CVM where the JSON file is stored.
4. Enable service-specific traffic isolation using the JSON file.

For example, use the following command to enable service-specific traffic isolation for Nutanix Volumes:

```

nutanix@CVM:~$ network_segmentation --service_network
--ip_pool=pool1 --desc_name="Volumes Seg 1"
--service_name=kVolumes
--host_physical_network=dv-volumes-network-1
--service_vlan=151
--ip_map_filepath=/home/nutanix/ip_map.json

```

Network Segmentation during Cluster Expansion

During cluster expansion, Prism Element allocates IP addresses from backplane IP address pools—two for backplane and one for service-specific isolation. Ensure sufficient IP addresses and consistent VLAN settings across nodes. If IP addresses are unavailable, expand the pool or reconfigure segmentation.

Keep the following points in mind when expanding a cluster:

- If you enable backplane network segmentation, Prism Element allocates two IP addresses for every new node from the backplane IP address pool.
- If you enable service-specific traffic isolation, Prism Element allocates one IP address for every new node from the respective (Volumes or DR) IP address pools.
- If enough IP addresses are not available in the specified network, the Prism Element web console displays a failure message in the tasks page. To add more IP address ranges to the IP address pool, see [Configuring Backplane IP Pool](#) on page 183.

- If you cannot add more IP addresses to the IP address pool, reconfigure that specific network segmentation. For more information, see [Reconfiguring the Backplane Network](#) on page 184.
- The network settings on the physical switch to which the new nodes are connected must be identical to those of other nodes in the cluster. New nodes communicate with current nodes using the same VLAN ID for segmented networks. Otherwise, the cluster expansion task fails in the network validation stage.
- After fulfilling the earlier points, you can add nodes to the cluster. For instructions about how to add nodes to your Nutanix cluster, see [Expanding a Cluster](#) in the *Prism Element Web Console Guide*.

Network Segmentation-Related Changes During an AOS Upgrade

During an AOS upgrade to a version supporting network segmentation, Prism Element creates `eth2` interface automatically on each CVM, but services continue using `eth0` until you configure segmentation.

When you upgrade from an AOS version that does not support network segmentation to an AOS version that does, the `eth2` interface (used to segregate backplane traffic) is automatically created on each CVM. However, the network remains unsegmented, and the cluster services on the CVM continue to use `eth0` until you configure network segmentation.

Prism Element does not create vNICs such as `ntnx0`, `ntnx1`, and so on during an upgrade to a AOS release that supports service-specific traffic isolation. Prism Element creates the vNICs when you configure traffic isolation for a service.

Note:

Do not delete the `eth2` interface that is created on the Controller VMs, even if you are not using the network segmentation feature.

ACCESSING A LIST OF OPEN SOURCE SOFTWARE RUNNING ON A CLUSTER

As an admin user, you can access a text file that lists all of the open source software running on a cluster.

About this task

To access a list of the open source software running on a cluster, follow these steps:

Procedure

1. SSH into any of the Controller VMs (CVMs) as an admin user:

```
$ ssh admin@cvm_ip_address
```

2. Access the text file by running the following command:

```
admin@cvm$ less /usr/local/nutanix/license/nutanix_bom.txt
```

COPYRIGHT

Copyright 2026 Nutanix, Inc.

Nutanix, Inc.

1740 Technology Drive, Suite 150

San Jose, CA 95110

All rights reserved. This product is protected by U.S. and international copyright and intellectual property laws. Nutanix and the Nutanix logo are registered trademarks of Nutanix, Inc. in the United States and/or other jurisdictions. All other brand and product names mentioned herein are for identification purposes only and may be trademarks of their respective holders.